Basic Concepts

In electrical circuits we are dealing with different physical quantities, as charge, current, voltage, energy and power.

Charge: is the electrical property of the atomic particle of which matter is built, measured in Coulomb (C).

The smallest charge in nature is the charge of the electron, $e = -1.602x10^{-19} C$. one Coulomb is a large quantity of charge and is equal to the charge of

$$\frac{1}{1.602x10^{-19}} = \frac{10^{19}}{1.602} = \frac{10x10^{18}}{1.602} = 6.24\,x10^{18} \text{ electrons.}$$

Note that the charge is always an integer multiple of the charge of the electron, as a fractional electron doesn't exist.

Note also that the principle of charge conservation is always applied, this means that the charge can't be created nor destroyed. The charge can only move from one place to another.

Current: is the time rate of change of charge. measured in Amperes (A), or:

$i = \frac{dq}{dt}$, and $q = \int i dt$

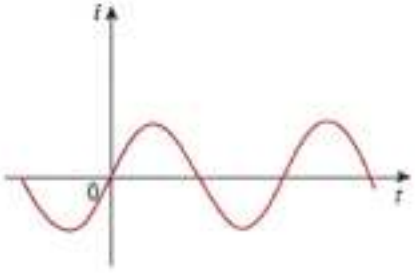
A current through a given element can be direct current (DC) or alternating current (AC).

A current is said to be direct current (DC) if it remains constant with time. And the current is said to be an alternating current (AC) if it is changing with time.

The following figure shows a DC and an AC currents as functions of time.



A DC current



An AC current

Note that capital letters are usually used to denote constant quantity, while small letters are used to denote varying quantities

Note that the direction of the current is the direction of movement of positive charges. which is the opposite to the direction of movement of eletrons

Example 1:

The total charge entering a terminal is given by $q = (10 - 10e^{-2t})mC$. Calculate the current at $t = 0.5s$.

Solution:

$$i = \frac{dq}{dt} = \frac{d}{dt}(10 - 10e^{-2t})x10^{-3} = 20x10^{-3}e^{-2t}$$

$$i(05) = 20x10^{-3} e^{-2.x05} = 7.36x10^{-3} A = 7.36 \ \ mA$$

Example 2:

The current flowing through an element is given by:

$$i = \begin{cases} 2 \ \ A, & 0 < t < 1 \\ 2t^2 \ A, & t > 1 \end{cases}$$

Calculate the charge entering the element from $t = 0$ to $t = 2$ s.

Solution:

$$q = \int i dt = \int_0^1 2dt + \int_1^2 2t^2 dt = 2t \Big|_0^1 + \frac{2}{3}t^3 \Big|_1^2 = 2 + \frac{2}{3}(7) = 6.67 \ \ C$$

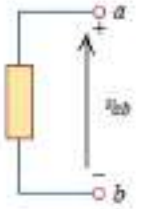
Voltage:

The voltage, or potential difference between two points, is equal to the energy required to move a unit charge between these two points. In other words, if the energy required to move charge q between points a and b is w thus, the voltage between these two points is:

$$v_{ab} = \frac{dw}{dq}$$

Note that w is measured in Jules (J) and q is measured in Coulombs (C), while v is measured in Volts. Or the voltage v_{ab}, in volts, is the work required to move 1 C between points a and b.

In electrical circuits the is directed toward the point of higher potential, as shown below.



Note in this figure that point a is at a higher potential than point b.

Note also that:

$$v_{ab} = -v_{ba}$$

Power & Energy

We know that power absorbed by a load is the rate of change of the energy absorbed by this load and in the same way, the power generated by a generator is the rate of change of the energy generated by thus generator. Or:

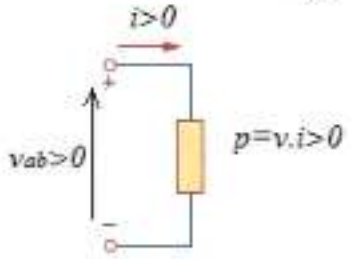
$$p = \frac{dw}{dt}$$

In terms of voltage and current, the power can be expressed as:

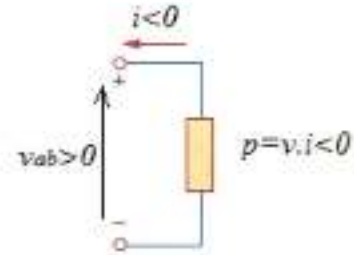
$$p = \frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt} = v.i$$

Or the power is the product of the voltage and the current.

For a given element, the power absorbed by the element is assumed positive, while the power consumed by the element is assumed negative.
Note that the positive voltage is toward the terminal with a higher potential and the positive current is entering from the terminal of higher potential towards the terminal of lower potential, thus the power of a given element is positive (consumed) if its current is entering the positive terminal toward the negative terminal as shown below.



In the same meaning, if the current is supplied from the terminal of higher potential, the power consumed by the element is negative, or the power is being generated by the element, as below.



Note that the law of conservation of energy is applied in all electrical circuits. Consequently, the algebraic sum of power in a circuit, at any instant of time, must be zero, or:

$$\sum p = 0$$

Or, in other words, the total power supplied to the circuit is to the total power absorbed.
Note also the energy supplied to or absorbed from an element can be found as:

$$w = \int p.dt = \int v.i dt$$

Note also, that the electric power utility companies measure energy in watt-hours (Wh), where:

$$1 \ \ Wh = 3600 \ \ J$$

Example:
Find the power delivered to an element at $t = 5ms$ if the current entering its positive terminal is $i = 5\cos(60\pi) \ \ A$, and the voltage is:

a) $v = 2i \ \ V$

b) $v = \left(10 + 5\int_0^t i.dt\right) \ \ V$

Solution:
a)

$$v = 2i = 10\cos(60\pi) \ \ V$$

$$p(t) = v.i = 50\cos^2(60\pi) \ \ W$$

$$p(5ms) = 50\cos^2\left(60\pi x5x10^{-3}\right) = 50\cos^2(0.3\pi) = 17.27 \quad W$$

b)

$$v = 10 + 5\int_0^t i\,dt = 10 + 5\int_0^t [5\cos(60\pi t)]dt = 10 + 25\int_0^t [\cos(60\pi t)]dt$$

$$= 10 + \frac{25}{60\pi}\sin(60\pi t)$$

$$p(t) = v.i = [5\cos(60\pi t)]\left[10 + \frac{25}{60\pi}\sin(60\pi t)\right]$$

$$p(5ms) = \left[5\cos\left(60\pi x5x10^{-3}\right)\right]\left[10 + \frac{25}{60\pi}\sin\left(60\pi x5x10^{-3}\right)\right] = 29.7W$$

Circuit Elements:

Circuit elements are the building blocks of the electrical circuits, and the electrical circuit can be defined as an interconnection of circuit element.
In general, circuit elements can be classified as:

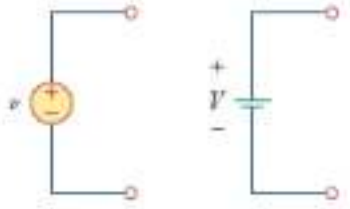
- passive elements: elements which are not producing energy, as resistors, capacitors and coils.
- active elements: are the elements which are capable of producing energy, as electric generators and batteries.

Active elements can also be divided into voltage and current sources. A voltage source is an element that produces a voltage between its terminals irrespective of the current supplied by it, while a current source is an elements that supplies a current to the circuit irrespective of the voltage between its terminals.

Active elements can also be classified as independent (uncontrolled) sources and dependent (controlled) sources. Independent or uncontrolled source produces fixed or constant values (voltage between its terminals, or current supplied from it). While, dependent or controlled elements produces values which are functions of other values in the circuit.
Different types of dependent (controlled) sources are existing in electrical circuits, these are:

a) voltage-controlled voltage sources (VCVS).
b) current-controlled voltage sources (CCVS).
c) voltage-controlled current sources (VCCS).
d) current-controlled current sources (CCCS).

The figure below shows the symbols of the different active elements in the electrical circuits.



independent voltage sources



independent current source



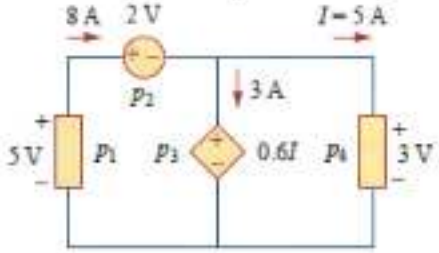
dependent voltage source



dependent current source

Example:

Calculate the power absorbed or supplied by each component of the following circuit.



Solution:

- element p_1: the current is out of the positive terminal, or this element is producing (or generating) power, and $p_1 = (5V)(-8A) = -40W$

- element p_2: the current is entering the positive terminal, or this element is consuming power, and $p_2 = (2V)(8A) = 16W$

- element p_3: the current is entering the positive terminal, or this element is consuming power, and $p_3 = (0.6I)(3A) = (0.6x5)(3A) = 9W$

- element p_4: the current is entering the positive terminal, or this element is consuming power, and $p_4 = (3V)(5A) = 15W$

Note that: $\sum p_n = p_1 + p_2 + p_3 + p_4 = 0$

Elements of network topology

An electrical circuit, in general, consists of elements connected together. Some are passive and others are active elements.

The following definitions of network topology are usually used.

- A network is an interconnection of elements or devices, whereas a circuit is a network with at least one closed paths.

- A branch represents any two-terminal element.

- A node is the point of connection between two or more branches. Note that, if a short circuit (a connecting wire) connects two nodes, the two nodes constitute a single node.

- A loop is any closed path in a circuit.

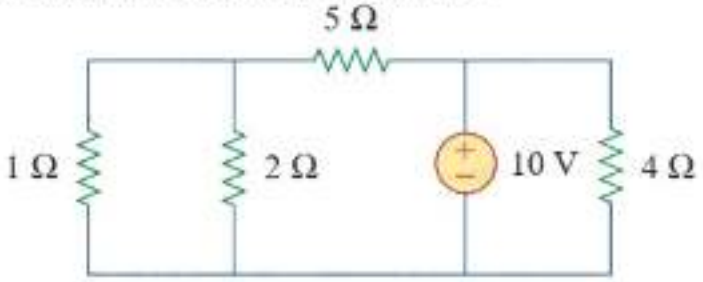
- Note that a network with b branches, n nodes, and l independent loops will satisfy the fundamental theorem of network topology:

$b = l + n - 1$

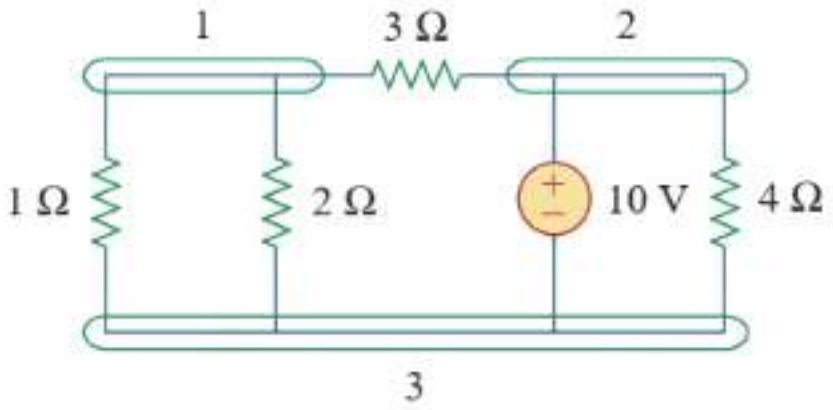
- Two or more elements are in series if they carry the same current.

- Two or more elements are in parallel if they have the same voltage across them.

Example:
How many branches, nodes and loops does the circuit in the following figure have. ? Identify the elements that are in series and in parallel.



Solution:
- the circuit has 5 branches: 1Ω, 2Ω, 5Ω, 4Ω resistors and the $10V$ voltage source. Or. $b = 5$.
- the circuit has 3 nodes, or $n = 3$, as in the following figure.



- the circuit has 3 loops, $l = 3$

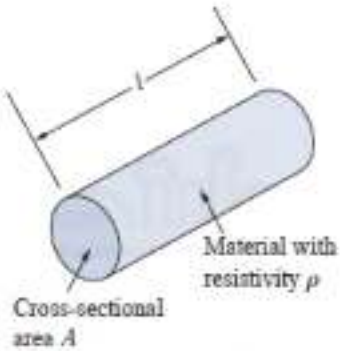
Note that $b = l + n - 1$ is fulfilled.
- the 1Ω and the 2Ω resistors are in parallel. Also the 4Ω resistor and the voltage source are in parallel.

Basic Laws

Ohm's Law:

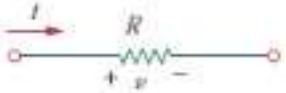
This physical property, or ability to resist current, for a given element is known as resistance and is represented by the symbol R.
The resistance of any material with a uniform cross-sectional area, with cross-sectional area A and length l, as shown in the figure below, is:

$$R = \rho \frac{l}{A}$$



ρ is known as the resistivity of the material in ohm-meters. Good conductors, such as copper and aluminum, have low resistivities, while insulators, such as mica and paper, have high resistivities.
As representatives of conductors, for copper, $\rho = 1.72x10^{-8}\Omega m$ and for aluminum $\rho = 2.8x10^{-8}\Omega m$.

As representatives of insulators, for paper $\rho = 10^{10}\Omega m$ and for glass $\rho = 10^{12}\Omega m$.
The symbol for the resistor, which is the simplest passive element, is as shown in the following figure.



The relationship between the current through a resistor and the voltage across its terminal is linear and is known as Ohm's law, and it is:

$v = Ri$

Note that the resistance is measured in Ohms $[\Omega]$

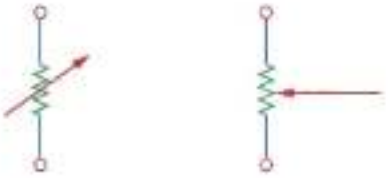
From the above law, we find that:

$$R = \frac{v}{i}$$

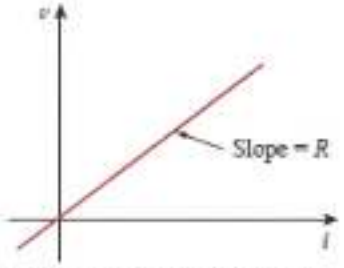
The values of the resistor can range from zero to infinity. If $R = 0$, for a given element, then $v = Ri = 0$ and the element is called a short-circuit. Or, for short-circuit element $v = 0$ irrespective of the value of the current.

If R approaching infinity, for a given element, then $i = \frac{v}{R}$ is zero irrespective of the value of the voltage and the element is called an open-circuit.

A resistor is either fixed or variable. For fixed resistors, the resistance remains constant. For variable resistors, the resistance can be adjusted or changed. The symbols of a variable resistor are as follows.

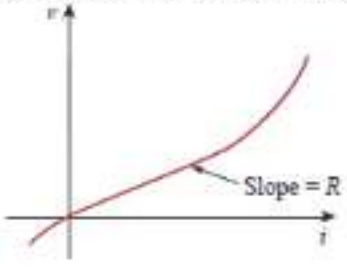


Note that, a resistor that obeys Ohm's law is known as a linear resistor. It has a constant resistance and thus its current-voltage characteristic is as illustrated in the figure below.



Note that the i-v graph of the resistor is a straight line passing through the origin.

A nonlinear resistor does not obey Ohm's law. Its resistance varies with current and its i-v characteristic is typically shown in the figure below.



Note that, the reciprocal of resistance R is known as conductance and is denoted by G, or:

$$G = \frac{1}{R} = \frac{i}{v}$$

The unit of conductance is the mho $\mho$, or Siemens, S

The power dissipated by a resistor can be expressed in terms of R, or:

$$p = v.i = (Ri).i = Ri^2$$

Or:

$$p = v.i = v.\left(\frac{v}{R}\right) = \frac{v^2}{R}$$

Also the power dissipated by a resistor can be expressed in terms of G. or:

$$p = v.i = v.(Gv) = Gv^2$$

Or:

$$p = v.i = \left(\frac{i}{G}\right).i = \frac{i^2}{G}$$

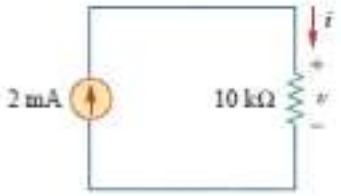
Note that:

a) The power dissipated in a resistor is a nonlinear function of either current or voltage.

b) Since R and G are positive quantities, the power dissipated in a resistor is always positive. Thus, a resistor always absorbs power from the circuit. Or the resistor is a passive element.

Example:

For the circuit shown in in the figure, calculate the voltage v, the conductance G and the power p.



Solution:

$$v = Ri = (10x10^3)(2x10^{-3}) = 20V$$

$$G = \frac{1}{R} = \frac{1}{10x10^3} = 0.1x10^{-3}S = 0.1mS = 100\mu S$$

$$p = Ri^2 = (10x10^3)(2x10^{-3})^2 = 40x10^{-3}W = 40mW$$

Example:

A resistor absorbs an instantaneous power of $20\cos^2 t \ \ mW$ when connected to a voltage source $10\cos t \ \ V$. Find i and R.

Solution:

$p = v.i$, or:

$$i = \frac{p}{v} = \frac{(20\cos^2 t)x10^{-3}}{10\cos t} = 2x10^{-3}\cos t \quad A = 2\cos t \quad mA$$

$$R = \frac{v}{i} = \frac{10\cos t}{(2\cos t)x10^{-3}} = 5x10^{3} \quad \Omega = 5 \quad k\Omega$$

Kirchhoff's Laws

There are 2 Kirchhoff's laws. these are Kirchhoff's current law (KCL) and Kirchhoff's voltage law (KVL).

Kirchhoff's current law (KCL):

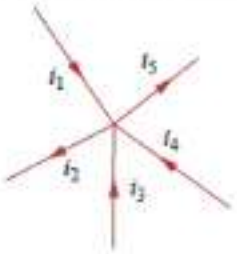
This law states that the algebraic sum of currents entering a node (or a closed boundary) is zero. Or,

$$\sum_{n=1}^{N} i_n = 0$$

where N is the number of branches connected to the node and i_n is the nth current entering (or leaving) the node. By this law, currents entering a node may be regarded as positive, while currents leaving the node may be taken as negative or vice versa.

This law is based on the law of conservation of charge, which requires that the algebraic sum of charges within a system cannot change.

Assume a set of currents flow into a node, as shown in the figure below.



The law of conservation of electric charge requires that the algebraic sum of electric charges at the node must not change. Or the sum of the charges entering the node is equal to the sum of the charges leaving the node. Or:

$$q_1 - q_2 + q_3 = q_4 + q_5$$

Or:

$$\int i_1\,dt + \int i_3\,dt - \int i_4\,dt = \int i_2\,dt + \int i_5\,dt$$

$$\int (i_1 + i_3 + i_4)\,dt = \int (i_2 + i_5)\,dt$$

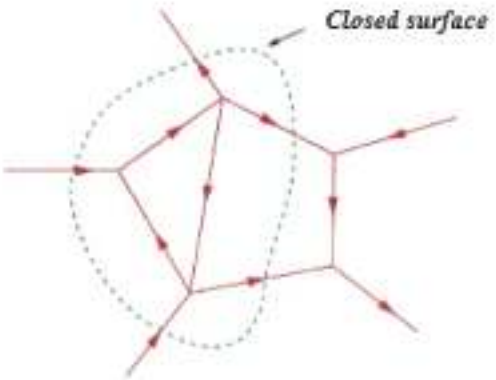
$$i_1 + i_3 + i_4 = i_2 + i_5$$

$$i_1 + (-i_2) + i_3 + i_4 + (-i_5) = 0$$

Or:

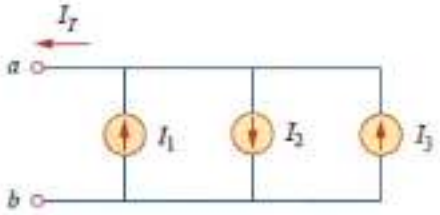
$$\sum_{n=1}^{5} i_n = 0$$

It has to be noted that Kirchhof's current low can be generalized to a closed surface. Consider the following figure.



In this figure, the total current entering the closed surface is equal to the total current leaving the surface.

A simple application of KCL is combining current sources in parallel. Consider the circuit in the figure below.



For this circuit:

$I_T + I_3 = I_1 + I_2$

Or:

$I_T - I_1 + I_3 - I_2$

Kirchhoff's voltage law (KVL):

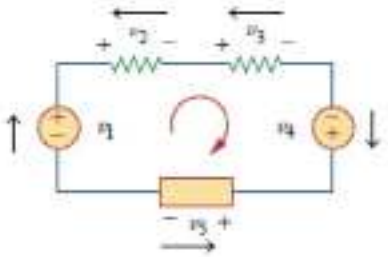
This law states that the algebraic sum of all voltages around a closed path (or loop) is zero. Or,

$$\sum_{m=1}^{M} v_m = 0$$

where M is the number of voltages in the loop. And v_m is the mth voltage.

This law is based on the law of conservation of energy in the electric field, which requires that the energy required to move a charge through a closed path is zero.

Consider the circuit in the figure below.



For this circuit, and going around the loop in the clockwise direction:

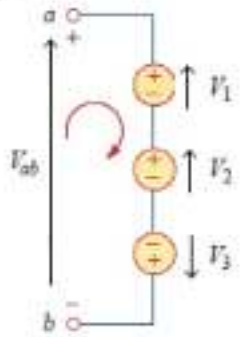
$v_1 - v_2 - v_3 + v_4 - v_5 = 0$

Or:

$v_1 + v_4 = v_2 + v_3 + v_5$

The same result is found if we go through the loop in the counter-clockwise direction.

When voltage sources are connected in series, KVL can be applied to obtain the total voltage. Consider the circuit below.



Going in the clockwise direction:

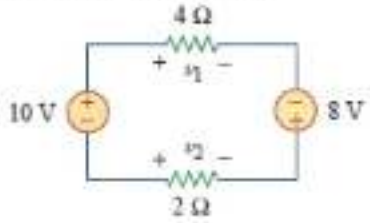
$V_{ab} - V_1 - V_2 + V_3 = 0$

Or:

$V_{ab} = V_1 + V_2 - V_3$

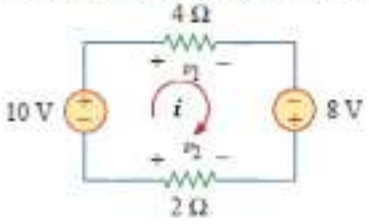
Example 1:

Find v_1 and v_2 in the circuit in the figure below.



Solution:

Assuming the current in the clockwise direction, as below.



$v_1 = 4i$

$v_2 = -2i$

Applying KVL:

$10 - v_1 + 8 + v_2 = 0$

Or:

$10 - 4i + 8 - 2i = 0$

From which:

$i = 3A$

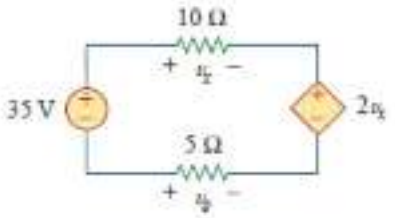
And:

$v_1 = 4i = 12V$

$v_2 = -2i = -6V$

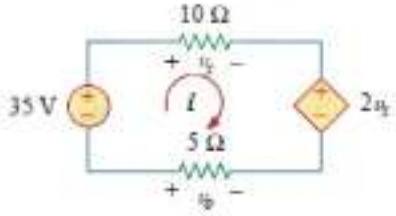
Example 2:

Find v_x and v_0 in the circuit below.



Solution:

Assuming the current in the clockwise direction, as below.



$35 - 10i - 2v_x + 5i = 0 \qquad (1)$

And:

$v_x = 10i \qquad (2)$

Substituting (2) in (1), we find that:

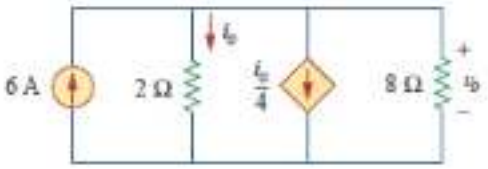
$35 - 10i - 20i - 5i = 0$

From this equation:

$i = 1A$, $v_x = 10i = 10V$ and $v_0 = -5i = -5V$

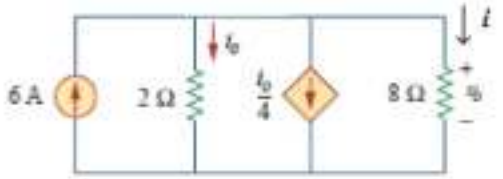
Example 3:

Find v_0 and i_0 in the circuit below.



Solution:

From the circuit below.



$$i_o + \frac{i_o}{4} + i = 6 \qquad (1)$$

Where:

$i = \frac{v_o}{8}$ and $v_o = 2i_o$, or:

$$i = \frac{2i_o}{8} = \frac{i_o}{4} \qquad (2)$$

Substituting (2) in (1):

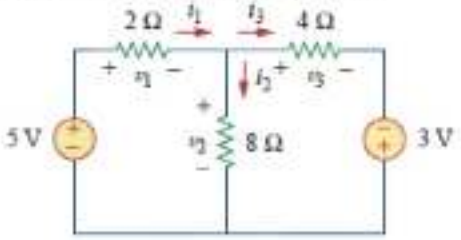
$$i_o + \frac{i_o}{4} + \frac{i_o}{4} = 6$$

Or:

$i_o = 4A$ and $v_o = 2i_o = 8V$

Example 4:

Find the currents and voltages in the following circuit.



Solution:

For the node, by KCL:

$$i_1 = i_2 + i_3 \qquad (1)$$

For the first loop, by KVL and going clockwise direction:

$$5 - 2i_1 - 8i_2 = 0 \qquad (2)$$

For the second loop, by KVL and going clockwise direction:

$$3 + 8i_2 - 4i_3 = 0 \qquad (3)$$

Substituting (1) in (2),

$$5 - 2(i_2 + i_3) - 8i_2 = 0$$

Or:

$$10i_2 - 2i_3 = 5 \qquad (4)$$

From equation (3) and (4), we find:

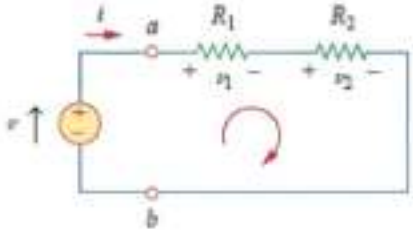
$i_2 = 0.25A$, $i_3 = 1.25A$

From (1), $i_1 = i_2 + i_3 = 1.5A$

$v_1 = 2i_1 = 3V$, $v_2 = 8i_2 = 2V$ and $v_3 = 4i_3 = 5V$

Series Resistors and Voltage Division

Resistors are in series if they have the same current. Consider the circuit shown below.



For this circuit:

$v_1 = R_1 i$ (1)

$v_2 = R_2 i$ (2)

And:

$v - v_1 - v_2 = 0$

Or:

$v = v_1 + v_2$ (3)

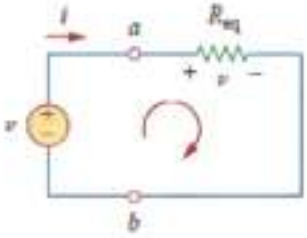
Substituting (1) and (2) in (3):

$v = R_1 i + R_2 i = (R_1 + R_2) i$

Or:

$$i = \frac{v}{R_1 + R_2} \quad (4)$$

Comparing this result with the current of the equivalent circuit below, in which:



$$i = \frac{v}{R_{eq}} \quad (5)$$

From equations (4) and (5), we conclude that:

$R_{eq} = R_1 + R_2$ (6)

Or, for the two series resistors, the equivalent resistance is the sum of their resistances.

In general, the equivalent resistance of resistors connected in series is equal to the sum of their resistances. Or, for n resistors in series,

$$R_{eq} = R_1 + R_2 + \ldots + R_n = \sum_{i=1}^{n} R_i \quad (7)$$

For n equal serially connected resistors:

$$R_{eq} = \sum_{i=1}^{n} R_i = n.R$$

Substituting (4) in (1) and (2), we find:

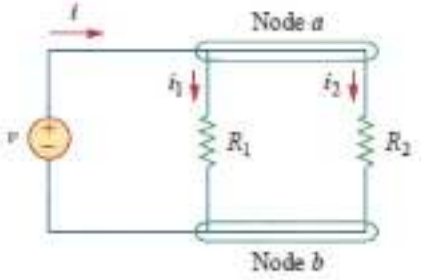
$$v_1 = R_1 i = \frac{R_1}{R_1 + R_2} v \qquad (8)$$

$$v_2 = R_2 i = \frac{R_2}{R_1 + R_2} v \qquad (9)$$

Or the voltage v has been divided into voltages v_1 and v_2 as defined by (8) and (9) and resistors R_1 and R_2 are said to form a voltage divider. ∝

Parallel Resistors and Current Division

Resistors are in parallel if same voltage is applied to them. Consider the circuit shown below.



For this circuit:

$$i_1 = \frac{v}{R_1} \qquad (1)$$

$$i_2 = \frac{v}{R_2} \qquad (2)$$

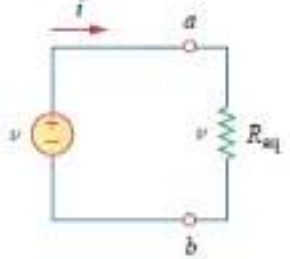
And:

$$i = i_1 + i_2 \qquad (3)$$

Substituting (1) and (2) in (3):

$$i = \frac{v}{R_1} + \frac{v}{R_2} = \left(\frac{1}{R_1} + \frac{1}{R_2}\right) v \qquad (4)$$

Comparing this result with the current of the equivalent circuit below, in which:



$$i = \frac{v}{R_{eq}} \qquad (5)$$

From equations (4) and (5), we conclude that:

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{R_1 + R_2}{R_1 . R_2} \qquad (6)$$

In general, the equivalent resistance of resistors connected in parallel can be found by the formula:

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots + \frac{1}{R_n} = \sum_{i=1}^{n} \frac{1}{R_i} \qquad (7)$$

For n equal parallel connected resistors, each with resistance R:

$$\frac{1}{R_{eq}} = \sum_{i=1}^{n} \frac{1}{R_i} = \frac{n}{R}$$

Or:

$$R_{eq} = \frac{R}{n}$$

From (4).

$$i = \left(\frac{1}{R_1} + \frac{1}{R_2} \right) v = \frac{R_1 + R_2}{R_1 . R_2} v$$

Or:

$$v = \frac{R_1 . R_2}{R_1 + R_2} i \qquad (8)$$

Substituting (8) in (1) and (2), we find:

$$i_1 = \frac{v}{R_1} = \frac{R_2}{R_1 + R_2} i \qquad (9)$$

$$i_2 = \frac{v}{R_2} = \frac{R_1}{R_1 + R_2} i \qquad (10)$$

Or the current i has been divided into currents i_1 and i_2 as defined by (9) and (10) and the circuit is said to be a current divider.

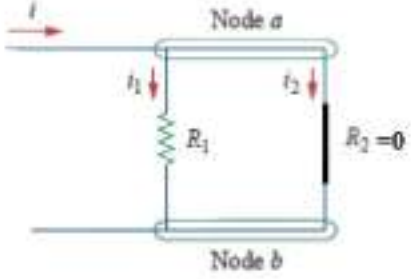
Some important notes:

1) equation (7) can be written using conductances as:

$$G_{eq} = G_1 + G_2 + \ldots + G_n = \sum_{i=1}^{n} G_i$$

Or the equivalent conductance of parallel resistors is equal to the sum of their conductances.

2) for two parallel resistors, if one is short-circuited, as shown below,



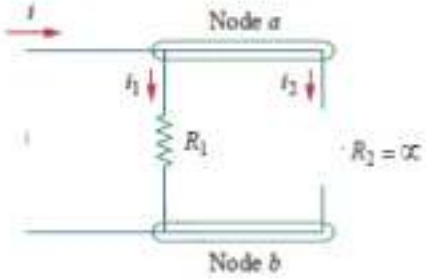
Using equations (9) and (10):

$$i_1 = \frac{R_2}{R_1 + R_2} i = 0$$

$$i_2 = \frac{R_1}{R_1 + R_2} i = i$$

Or all the supply current will pass through the short-circuited path.

3) for two parallel resistors, if one is open-circuited, as shown below,



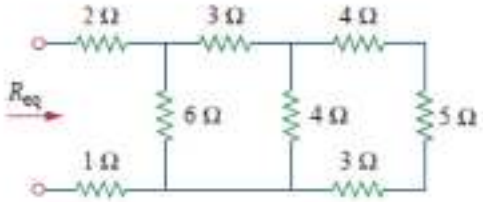
$$i_1 = \frac{R_2}{R_1 + R_2} i = \frac{\frac{R_2}{R_2}}{\frac{R_1}{R_2} + \frac{R_2}{R_2}} i = \frac{1}{0 + 1} i = i$$

$$i_2 = \frac{R_1}{R_1 + R_2} i = 0$$

Or all the supply current will pass through the non-open-circuited path.

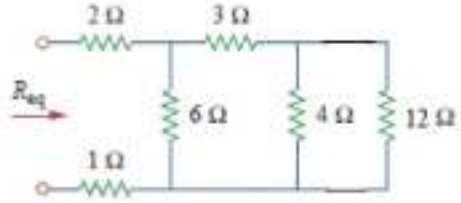
Example 1:

Find R_{eq} in the following circuit.

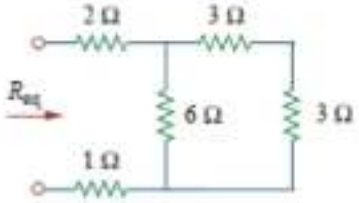


Solution:

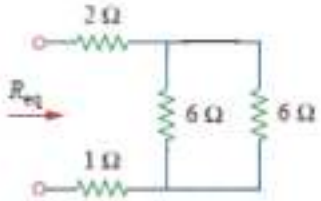
The 4Ω, 5Ω and 3Ω are in series, their equivalent is $4+5+3=12\Omega$, the circuit is reduced to the one below.



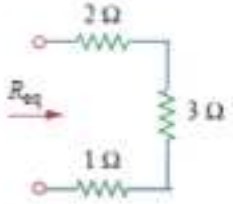
The 12Ω and the 4Ω are in //, their equivalent is $\frac{4x12}{4-12}=3\Omega$, the circuit is reduced to the one below.



The two 3Ω resistors are in series and their equivalent is $3+3=6\Omega$, the circuit is reduced to the one below.



The two 6Ω resistors are in // and their equivalent is $\frac{6}{2}=3\Omega$, the circuit is reduced to the one below.

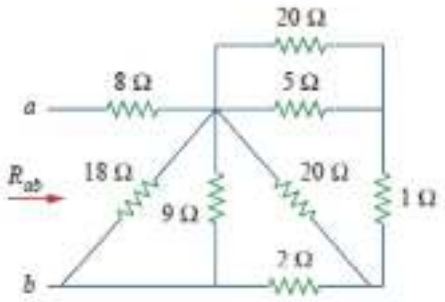


These 3 resistors are in series, and:

$R_{eq}=2+3+1=6\Omega$

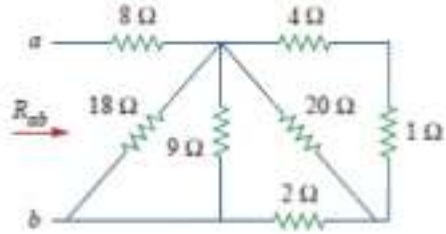
Example 2:

Find R_{ab} in the following circuit.

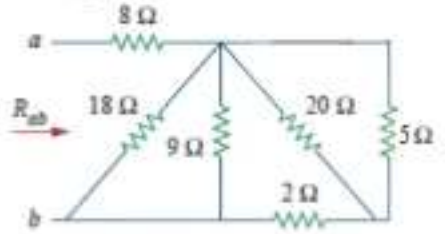


Solution:

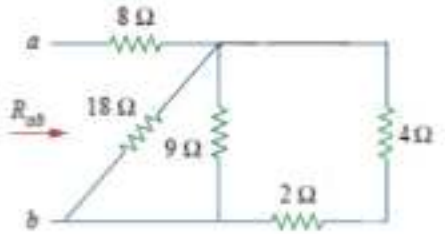
The 20Ω and the 5Ω resistors are in // and their equivalent is $\frac{20x5}{20+5} = 4\Omega$ and the circuit is reduced to the following one.



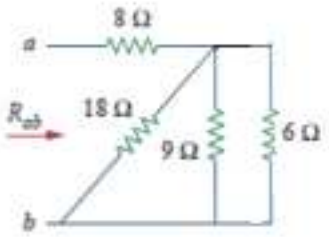
The 4Ω and the 1Ω resistors are in series and their equivalent is $4+1=5\Omega$ and the circuit is reduced to the following one.



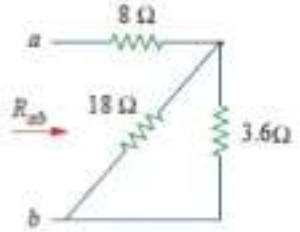
The 20Ω and the 5Ω resistors are in // and their equivalent is $\frac{20x5}{20+5} = 4\Omega$ and the circuit is reduced to the following one.



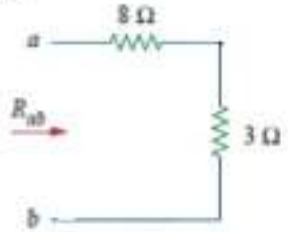
The 4Ω and the 2Ω resistors are in series and their equivalent is $4+2=6\Omega$ and the circuit is reduced to the following one.



The 6Ω and the 9Ω resistors are in // and their equivalent is $\frac{6x9}{6+9} = 3.6\Omega$ and the circuit is reduced to the following one.



The 3.6Ω and the 18Ω resistors are in // and their equivalent is $\frac{3.6x18}{3.6-18} = 3\Omega$ and the circuit is reduced to the following one.

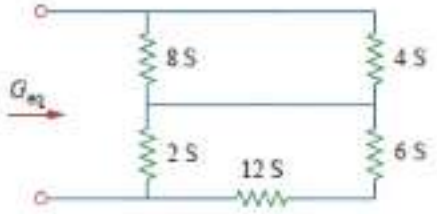


From this circuit:

$R_{ab} = 8+3 = 11\Omega$

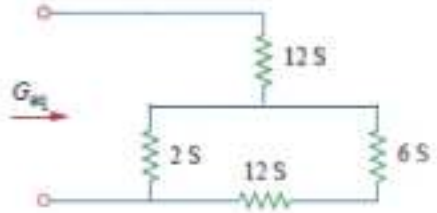
Example 3:

Find G_{eq} in the following circuit.

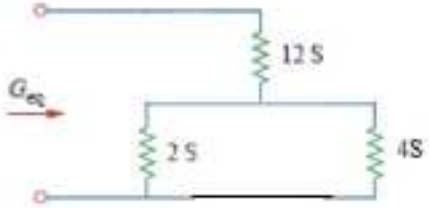


Solution:

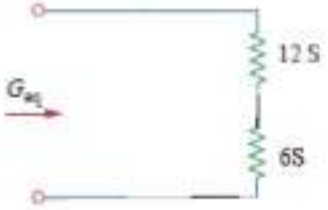
The $8S$ and $4S$ admittances are in // and their equivalent is $8+4 = 12S$, and the circuit is reduced to the following one.



The $6S$ and $12S$ admittances are in series and their equivalent is $\frac{6x12}{6+12}=4S$, and the circuit is reduced to the following one.



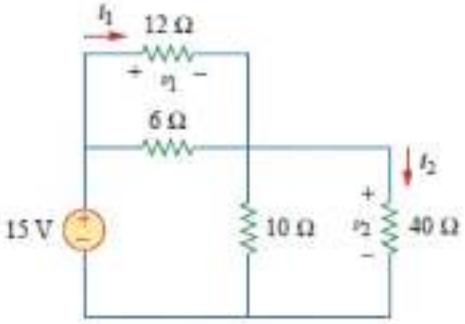
The $4S$ and $2S$ admittances are in // and their equivalent is $2+4=6S$, and the circuit is reduced to the following one.



The $6S$ and $12S$ admittances are in series and their equivalent is $G_{eq}=\frac{6x12}{6+12}=4S$

Example 4:

Find v_1, v_2, i_1, i_2 and the powers dissipated in the 12Ω and the 40Ω resistors in the following circuit.



Solution:

Resistors 10Ω and 40Ω are in // and their equivalent is in series to the parallel resistors 12Ω and 6Ω. Thus the equivalent resistance supplied by the voltage source is:

$$R_{eq}=\frac{10x40}{10+40}+\frac{6x120}{6+12}=12\Omega$$

The current out of the source is:

$$i=\frac{V}{R_{eq}}=\frac{15}{12}=1.25A$$

This current will pass in the parallel 12Ω and 6Ω resistors, thus:

$$v_1=i(12\Omega\,//\,6\Omega)=1.25\left(\frac{12x6}{12+6}\right)=5V$$

The same current will also pass in the parallel 10Ω and 40Ω resistors, thus:

$$v_2 = i(10\Omega // 40\Omega) = 1.25\left(\frac{10x40}{10+40}\right) = 10V$$

Note that $v_1 + v_2 = 5 + 10 = 15V$

Also:

$$i_1 = \frac{v_1}{12\Omega} = \frac{5}{12} = 0.4167A = 416.7mA$$

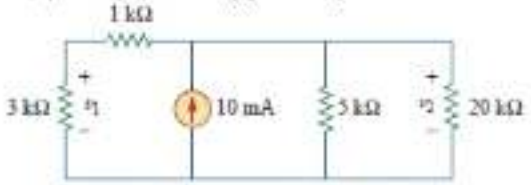
$$i_2 = \frac{v_2}{40\Omega} = \frac{10}{40} = 0.25A = 250mA$$

The power in the 12Ω resistor is: $R(i_1)^2 = 12(0.4167)^2 = 2.083W$

The power in the 40Ω resistor is: $R(i_2)^2 = 40(0.25)^2 = 2.5W$

Example 5:

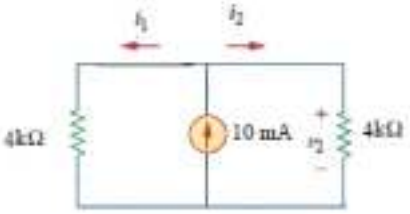
For the circuit shown below, find: (a) v_1 and v_2 (b) the power dissipated in the $3k\Omega$ and $20k\Omega$ resistors, and (c) the power supplied by the current source.



Solution:

The equivalent of the parallel $5k\Omega$ and the $20k\Omega$ resistors is $\frac{5x20}{5+20} = 4k\Omega$

The equivalent of the series $1k\Omega$ and the $3k\Omega$ resistors is $1+3=4k\Omega$, the circuit is reduced to the one below.



From this circuit:

$$i_1 = i_2 = 10x\frac{4}{4+4} = 5mA$$

$$v_1 = i_1(3k\Omega) = (5x10^{-3})(3x10^3) = 15V$$

$$v_2 = i_2(4k\Omega) = (5x10^{-3})(4x10^3) = 20V$$

the power dissipated in the $3k\Omega$ is:

$$p_3 = (3k\Omega)(i_1)^2 = (3x10^3)(5x10^{-3})^2 = 75mW$$

the power dissipated in the $20k\Omega$ is:

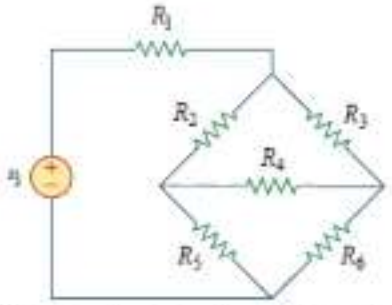
$$p_4 = \frac{(v_2)^2}{20x10^3} = \frac{20^2}{20x10^3} = 20mW$$

the power supplied by the current source is:

$$p = v_2 i = 20x10x10^{-3} = 200x10^{-3}W = 200mW$$

Wye-Delta Transformations

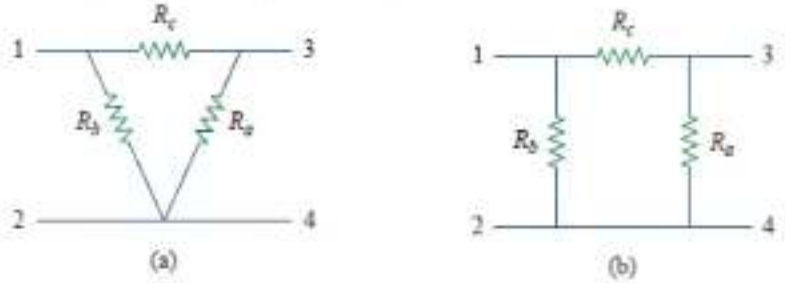
Elements in some circuits are neither in parallel nor in series, as in the bridge circuit shown below.



Circuits of this type can be simplified using $Y-\Delta$ and $\Delta-Y$ transformations.
Note that Y and Δ circuits are three-terminal circuits. The Y circuit is sometimes called T circuit is shown below.



The Δ circuit, also is called Π circuit. is shown below.



Delta to Wye Conversion

The figure below shows equivalent circuits, one is Δ connected and the other is Y connected.



These two circuits to be equivalent, the equivalent resistance between each 2 terminals in the 2 circuits must be equal.

For example in the Δ circuit, the equivalent resistance between terminals 1 and 2 is:

$$R_{12} = \frac{R_b(R_a + R_c)}{R_a + R_b + R_c} \qquad (1)$$

And in the equivalent Y circuit:

$$R_{12} = R_1 + R_2 \qquad (2)$$

Or these 2 circuits to be equivalent, thus:

$$R_1 + R_2 = \frac{R_b(R_a - R_c)}{R_a + R_b + R_c} \qquad (3)$$

In the same way, for the equivalent resistance between terminals 2 and 3, we find:

$$R_2 + R_3 = \frac{R_a(R_b + R_c)}{R_a + R_b + R_c} \qquad (4)$$

Also for the equivalent resistance between terminals 1 and 3, we find:

$$R_1 + R_3 = \frac{R_c(R_a + R_b)}{R_a + R_b + R_c} \qquad (5)$$

Adding equations (3) and (5) and subtract equation (4), we find:

$$2R_1 + R_2 + R_3 - (R_2 + R_3) = \frac{R_b(R_a + R_c)}{R_a + R_b + R_c} - \frac{R_c(R_a + R_b)}{R_a + R_b + R_c} - \frac{R_a(R_b - R_c)}{R_a + R_b + R_c}$$

Or:

$$2R_1 = \frac{R_a R_b + R_b R_c + R_a R_c + R_b R_c - R_a R_b - R_a R_c}{R_a - R_b + R_c} = \frac{2R_b R_c}{R_a + R_b + R_c}$$

Or:

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c} \qquad (6)$$

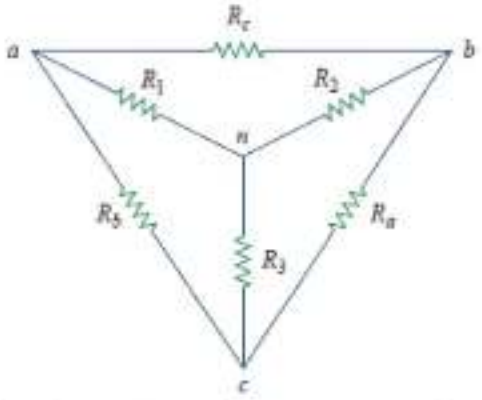
In the same way, adding equations (4) and (5) and subtract equation (3), we find:

$$R_2 = \frac{R_a R_c}{R_a + R_b + R_c} \qquad (7)$$

And by adding equations (3) and (4) and subtract equation (5), we find:

$$R_3 = \frac{R_a R_b}{R_a + R_b - R_c} \qquad (8)$$

Note that to transform a Δ network to equivalent Y, we create an extra node n, as shown in the figure below and follow the following conversion rule:



Each resistor in the Y network is the product of the resistors in the two adjacent Δ branches, divided by the sum of the three Δ resistors.

Wye to Delta Conversion

From equations (6), (7) and (8) re-written below, we find:

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c} \qquad (6)$$

$$R_2 = \frac{R_a R_c}{R_a + R_b + R_c} \qquad (7)$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c} \qquad (8)$$

$$R_1 R_2 + R_2 R_3 + R_1 R_3 = \frac{R_a R_b (R_c)^2 + (R_a)^2 R_b R_c + R_a (R_b)^2 R_c}{(R_a + R_b + R_c)^2} = \frac{R_a R_b R_c (R_a + R_b + R_c)}{(R_a + R_b + R_c)^2} = \frac{R_a R_b R_c}{R_a + R_b + R_c} \qquad (9)$$

Dividing (9) by (6), we find:

$$\frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_1} = \frac{\frac{R_a R_b R_c}{R_a + R_b + R_c}}{\frac{R_b R_c}{R_a + R_b + R_c}}$$

And from which:

$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_1} \qquad (10)$$

Dividing (9) by (7), we find:

$$R_b = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_2}$$

And dividing (9) by (8), we find:

$$R_c = \frac{R_1R_2 + R_2R_3 + R_1R_3}{R_3}$$

Or, rule for Y to Δ conversion is as follows:
Each resistor in the Δ network is the sum of all possible products of Y resistors taken two at a time, divided by the opposite Y resistor.

Note that, the Y and Δ networks are said to be balanced when:

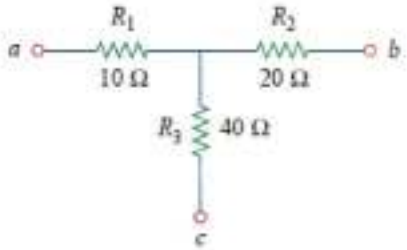
$R_1 = R_2 = R_3 = R_Y$, $\quad R_a = R_b = R_c = R_\Delta$

Under these conditions:

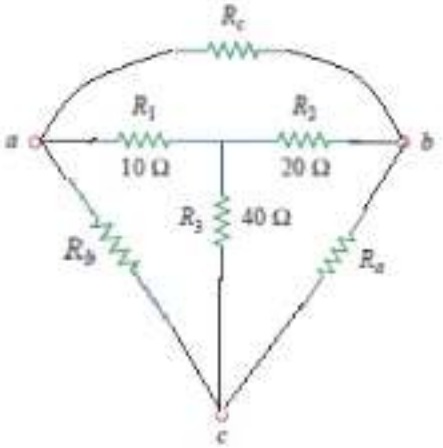
$$R_Y = \frac{R_\Delta}{3} \quad \text{or} \quad R_\Delta = 3R_Y$$

Example 1:

Find the equivalent Δ network to the network shown below.



Solution:



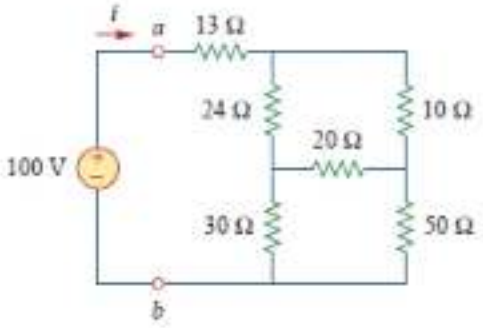
$$R_a = \frac{R_1R_2 - R_2R_3 + R_1R_3}{R_2} = \frac{10x20 + 20x40 + 10x40}{10} = 140\Omega$$

$$R_b = \frac{R_1R_2 + R_2R_3 + R_1R_3}{R_2} = \frac{10x20 + 20x40 + 10x40}{20} = 70\Omega$$

$$R_c = \frac{R_1R_2 + R_2R_3 + R_1R_3}{R_3} = \frac{10x20 + 20x40 + 10x40}{40} = 35\Omega$$

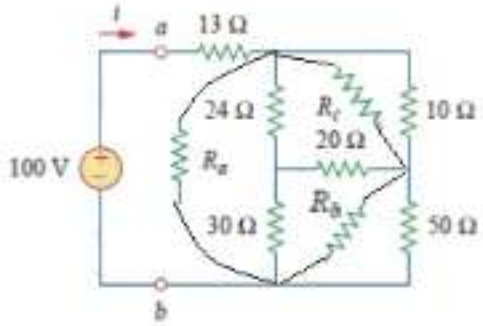
Example 2:

For the bridge network shown, find R_{ab} and i.



Solution:

The Y connected 20Ω, 24Ω and 30Ω resistors are converted to their Δ equivalent, as below.



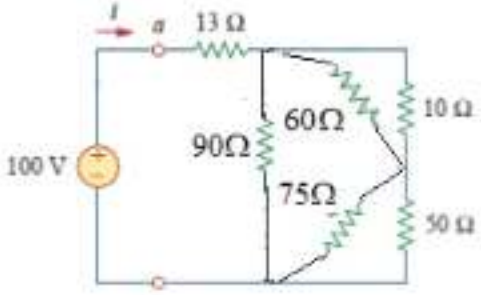
For this circuit:

$$R_a = \frac{24x30 + 24x20 + 20x30}{20} = 90\Omega$$

$$R_b = \frac{24x30 + 24x20 + 20x30}{24} = 75\Omega$$

$$R_c = \frac{24x30 + 24x20 + 20x30}{30} = 60\Omega$$

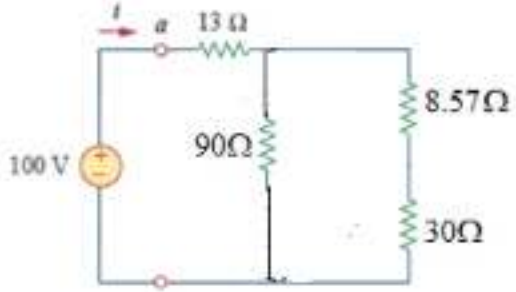
And the circuit becomes as below.



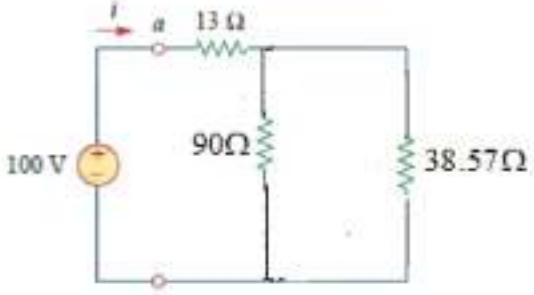
The 60Ω and the 10Ω resistors are in //, and their equivalent is: $\frac{60x10}{60-10} = 8.57\Omega$

The 75Ω and the 50Ω resistors are in //, and their equivalent is: $\frac{75x50}{75+50} = 30\Omega$

And the circuit is reduced to the one shown below.

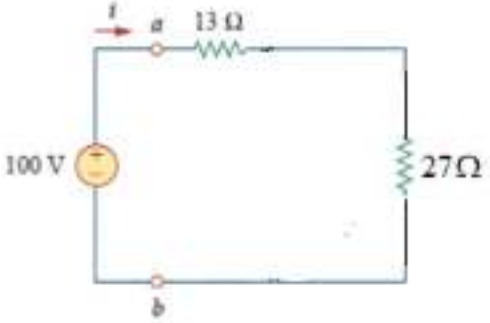


The 8.57Ω and the 50Ω resistors are in series, and their equivalent is: $8.57+30=38.57\Omega$
And the circuit is reduced to the one shown below.



The 38.57Ω and the 90Ω resistors are in //, and their equivalent is: $\frac{90x38.57}{90+38.57}=27\Omega$

And the circuit is reduced to the one shown below.



From this circuit:

$R_{ab}=27+13=40\Omega$

And:

$$i=\frac{100}{40}=2.5A$$

Methods of Analysis

The two mostly used techniques for circuit analysis are:
1) nodal analysis, which is based on Kirchhoff's current law (KCL).
2) mesh analysis, which is based on Kirchhoff's voltage law (KVL).

Nodal Analysis:

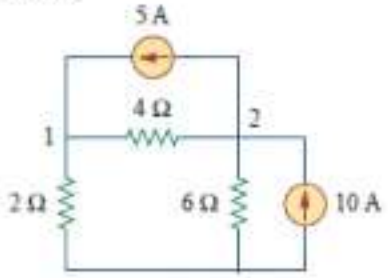
This method is based on Kirchhoff's current low (KCL). In this method one node is selected as a reference (with zero potential). KCL is applied to all other nodes. Remember that the current from node i to node j is:

$$i = \frac{v_i - v_j}{R_{ij}}$$

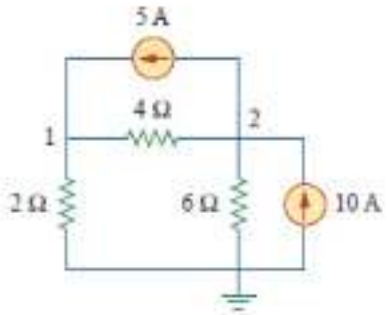
After writing equations using KCL for all non-reference nodes, these simultaneous equations are solved using Gaussian elimination, substitution, Cramer's rule or any computer based method.

Consider first circuits that don't have voltage sources.

Example 1:
For the circuit shown, using nodal analysis, calculate:
a) the currents through each of the resistances
b) the voltages on each of the resistances.



Solution:
Select the reference node (the node with zero potential), as below.



For each of nodes 1 and 2, we write the equations using KCL.
- for node 1:

$$\frac{v_1 - 0}{2} + \frac{v_1 - v_2}{4} = 5$$

From which:

$$2v_1 + v_1 - v_2 = 20$$

Or:

$$3v_1 - v_2 = 20 \qquad (1)$$

- for node 2:

$$\frac{v_2 - 0}{6} + \frac{v_2 - v_1}{4} + 5 = 10$$

From which:

$$\frac{v_2 - 0}{6} + \frac{v_2 - v_1}{4} = 5$$

$$2v_2 + 3v_2 - 3v_1 = 60$$

Or:

$$-3v_1 - 5v_2 = 60 \qquad (2)$$

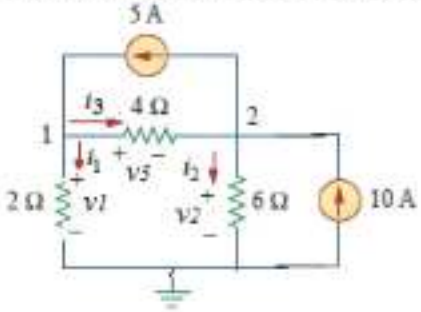
Equations (1) and (2) can be solved using different methods to find v_1 and v_2.

Adding these 2 equations, we find:

$4v_2 = 80$ or $v_2 = 20V$ and from (1):

$3v_1 = 20 + v_2 = 40$, or: $v_1 = 13.33V$

Now the currents through the resistors can be calculated and are as follows:



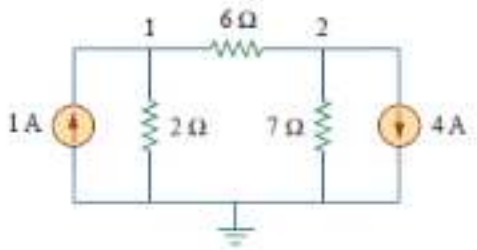
$$i_1 = \frac{v_1}{2} = \frac{13.33}{2} = 6.67A, \quad i_2 = \frac{v_2}{6} = \frac{20}{6} = 3.33A \text{ and } i_3 = \frac{v_1 - v_2}{4} = \frac{13.33 - 20}{4} = -1.67A$$

And the voltages on the resistors are:

$v_1 = 13.33V$, $v_2 = 20V$ and $v_3 = 4i_3 = -6.68V$

Example 2:

Obtain the node voltages in the circuit of the following figure.



Solution:

- For node 1: $\frac{v_1}{2} + \frac{v_1 - v_2}{6} = 1$

Or:

$4v_1 - v_2 = 6 \qquad (1)$

- For node 2: $\frac{v_2 - v_1}{6} + \frac{v_2}{7} + 4 = 0$

Or:

$-7v_1 + 13v_2 = -168 \qquad (2)$

These two equations can be solved using Cramer's rule as follows:

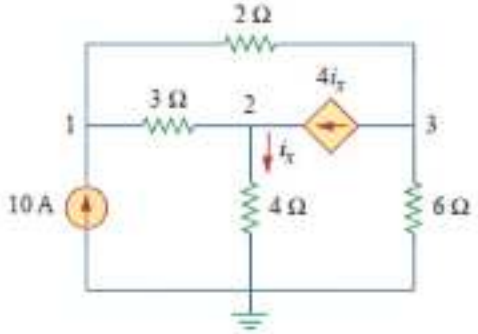
$$\Delta = \begin{vmatrix} 4 & -1 \\ -7 & 13 \end{vmatrix} = 45, \quad \Delta_1 = \begin{vmatrix} 6 & -1 \\ -168 & 13 \end{vmatrix} = -90, \quad \Delta_2 = \begin{vmatrix} 4 & 6 \\ -7 & -168 \end{vmatrix} = -630$$

Thus:

$$v_1 = \frac{\Delta_1}{\Delta} = -2V, \qquad v_2 = \frac{\Delta_2}{\Delta} = -14V$$

Example 3:

Find the voltages at the three nonreference nodes in the following circuit.



Solution:

For node 1:

$\frac{v_1 - v_2}{3} + \frac{v_1 - v_3}{2} = 10$, or: $2(v_1 - v_2) + 3(v_1 - v_3) = 60$, or:

$5v_1 - 2v_2 - 3v_3 = 60 \qquad (1)$

For node 2:

$\frac{v_2 - v_1}{3} + \frac{v_2}{4} = 4i_x$, but $i_x = \frac{v_2}{4}$, thus: $\frac{v_2 - v_1}{3} + \frac{v_2}{4} = v_2$, or: $4(v_2 - v_1) + 3v_2 = 12v_2$, or:

$4v_1 + 5v_2 = 0$ (2)

For node 3:

$\frac{v_3 - v_1}{2} + \frac{v_3}{6} + 4i_x = 0$ but $i_x = \frac{v_2}{4}$, thus: $\frac{v_3 - v_1}{2} + \frac{v_3}{6} + v_2 = 0$, $3(v_3 - v_1) + v_3 + 6v_2 = 0$, or:

$-3v_1 + 6v_2 - 4v_3 = 0$ (3)

Equations (1), (2) and (3) can be solved using Cramer's rule as follows:

$$\Delta = \begin{vmatrix} 5 & -2 & -3 \\ 4 & 5 & 0 \\ -3 & 6 & 4 \end{vmatrix} = 100 - 72 - 45 + 32 = 15$$

$$\Delta_1 = \begin{vmatrix} 60 & -2 & -3 \\ 0 & 5 & 0 \\ 0 & 6 & 4 \end{vmatrix} = 1200$$

$$\Delta_2 = \begin{vmatrix} 5 & 60 & -3 \\ 4 & 0 & 0 \\ -3 & 0 & 4 \end{vmatrix} = -960$$

$$\Delta_3 = \begin{vmatrix} 5 & -2 & 60 \\ 4 & 5 & 0 \\ -3 & 6 & 0 \end{vmatrix} = 1440 + 900 = 2340$$

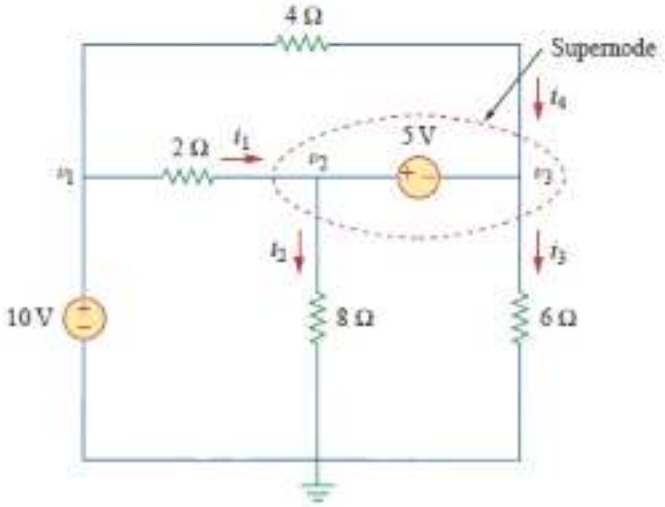
And:

$v_1 = \frac{\Delta_1}{\Delta} = \frac{1200}{15} = 80V$, $v_2 = \frac{\Delta_2}{\Delta} = -\frac{960}{15} = -64V$ and $v_3 = \frac{\Delta_3}{\Delta} = \frac{2340}{15} = 156V$

Nodal Analysis with Voltage Sources

Consider the following two cases:

CASE 1: The voltage source is connected between the reference node and a nonreference node. In this case, the voltage at the non-reference node is equal to the voltage of the voltage source. See the figure below.



In this figure:

$v_1 = 10V$

Note that, the analysis is simplified by this knowledge of the voltage at this nod.

CASE 2: The voltage source (dependent or independent) is connected between two nonreference nodes. The two nonreference nodes form a generalized node or super-node. See the previous figure. A super-node includes nodes v_2 and v_3, with $v_2 - v_3 = 5V$.

In this case, we apply both KCL and KVL to determine the node voltages.

Note that, a super-node is formed by enclosing a (dependent or independent) voltage source connected between two nonreference nodes and any elements connected in parallel with it.
Note also that, there is no way of knowing the current through the voltage source of the super node, in advance. For example, the is no way to know the current flowing through the 5V voltage source previously.

For this super node:

$i_1 + i_4 = i_2 + i_3$

Or:

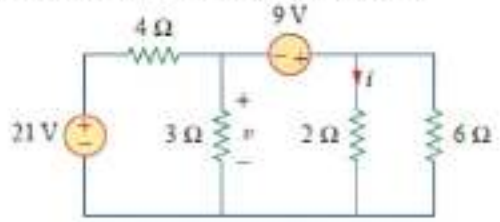
$$\frac{v_1 - v_2}{2} + \frac{v_1 - v_3}{4} = \frac{v_2}{8} + \frac{v_3}{6}$$

And:

$v_2 - v_3 = 5V$

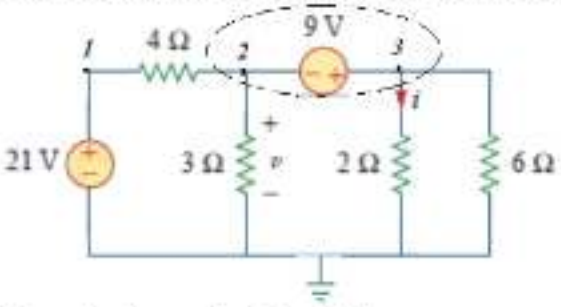
Example:

Find v and i in the circuit below.



Solution:

The reference node is chosen and the nodes are numbered as shown below.



Note also that nodes 2 and 3 form a super node.

From this circuit:

$v_1 = 21 \qquad (1)$

For the super node:

$$\frac{v_2 - v_1}{4} + \frac{v_2}{3} + \frac{v_3}{2} + \frac{v_3}{6} = 0$$

Or:

$$-3v_1 + 7v_2 + 8v_3 = 0 \quad (2)$$

And:

$$v_3 - v_2 = 9 \qquad (3)$$

Substituting (1) in (2), we find:

$$7v_2 + 8v_3 = 63 \qquad (4)$$

From equations (3) and (4), we find:

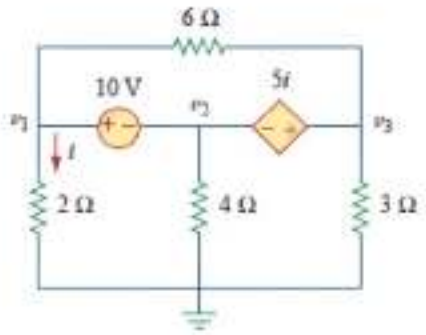
$$v_2 = -0.6V, \quad v_3 = \frac{126}{15} = 8.4V$$

Thus:

$$v = v_2 = -0.6V \text{ and } i = \frac{v_3}{2} = 4.2A$$

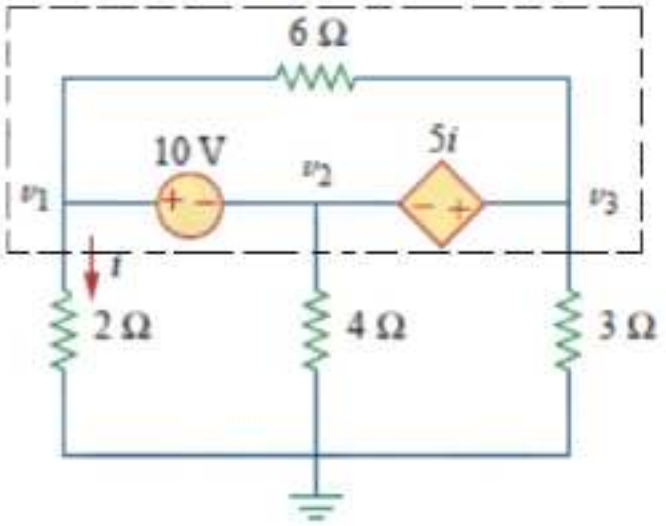
Example:

Find v_1, v_2 and v_3 in the circuit below, using nodal analysis.



Solution:

The super node contains nodes 1, 2 and 3 and the 6Ω resistor , as below.



For this circuit:

$$\frac{v_1}{2}+\frac{v_2}{4}+\frac{v_3}{3}=0$$

Or:

$$6v_1+3v_2+4v_3=0 \qquad (1)$$

Also:

$$v_1-v_2=10 \qquad (2)$$

And:

$$v_3-v_2=5i \qquad (3)$$

Where:

$$i=\frac{v_1}{2} \qquad (4)$$

Substituting (4) in (3), we find:

$$v_3-v_2=\frac{5v_1}{2}$$

Or:

$5v_1 + 2v_2 - 2v_3 = 0 \qquad (5)$

Solving equations (1), (2) and (5), using Cramer's rule, we find:

$6v_1 + 3v_2 + 4v_3 = 0 \qquad (1)$

$v_1 - v_2 = 10 \qquad (2)$

$5v_1 + 2v_2 - 2v_3 = 0 \qquad (5)$

$$\Delta = \begin{vmatrix} 6 & 3 & 4 \\ 1 & -1 & 0 \\ 5 & 2 & -2 \end{vmatrix} = 12 + 8 + 20 + 6 = 46$$

$$\Delta_1 = \begin{vmatrix} 0 & 3 & 4 \\ 10 & -1 & 0 \\ 0 & 2 & -2 \end{vmatrix} = 80 + 60 = 140$$

$$\Delta_2 = \begin{vmatrix} 6 & 0 & 4 \\ 1 & 10 & 0 \\ 5 & 0 & -2 \end{vmatrix} = -120 - 200 = -320$$

$$\Delta_3 = \begin{vmatrix} 6 & 3 & 0 \\ 1 & -1 & 10 \\ 5 & 2 & 0 \end{vmatrix} = 150 - 120 = 30$$

Thus:

$$v_1 = \frac{\Delta_1}{\Delta} = \frac{140}{46} = 3.043V$$

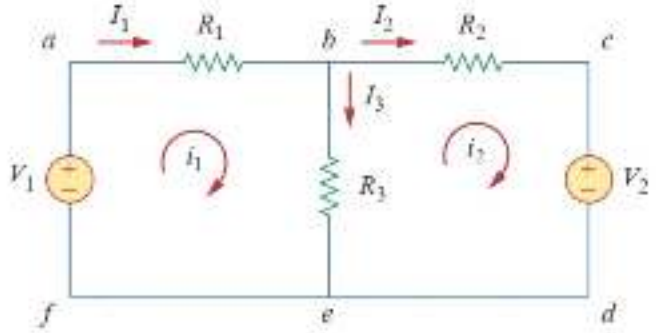
$$v_2 = \frac{\Delta_2}{\Delta} = \frac{-320}{46} = -6.956V$$

$$v_3 = \frac{\Delta_3}{\Delta} = \frac{30}{46} = 0.652V$$

Mesh Analysis

In contrary to nodal analysis, which uses Kirchhoff's current law (KCL), mesh analysis is based on Kirchhoff's Voltage Law (KVL).

Recall that a loop is a closed path with no node passed more than once. A mesh is a loop that does not contain any other loop within it.



Note that the closed paths *abefa* and *bcdeb* form meshes, while the closed path *abcdefa* doesn't form a mesh.

Note that for a mesh, the algebraic sum of the voltage drops on the resistors is equal to the algebraic sum of the voltages of the voltage sources taken in the same direction.

For example, for the above circuit, currents i_1 and i_2 are the mesh currents for meshes 1 & 2, respectively. These mesh currents circulate the two meshes in the clockwise direction.

Applying KVL for the mesh 1:

$$R_1 i_1 + R_3 (i_1 - i_2) = V_1$$

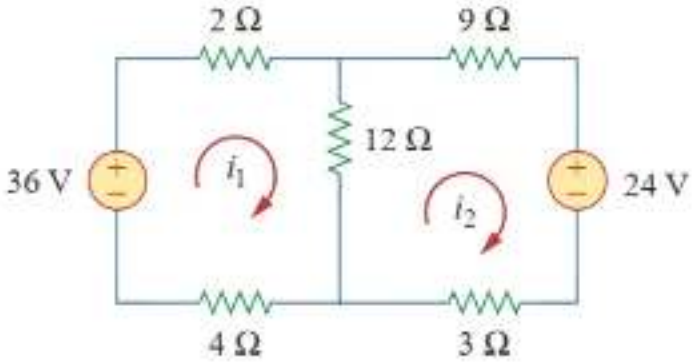
For mesh 2:

$$R_2 i_2 + R_3 (i_2 - i_1) = -V_2$$

These simultaneous equations are solved using any of the known methods.

Example 1:

Calculate the mesh currents i_1 and i_2 in the following circuit.



Solution:

For mesh 1:

$2i_1 + 12(i_1 - i_2) + 4i_1 = 36$

Or:

$18i_1 - 12i_2 = 36$

Or:

$3i_1 - 2i_2 = 6$ (1)

For mesh 2:

$9i_2 + 3i_2 + 12(i_2 - i_1) = -24$

Or:

$-12i_1 + 24i_2 = -24$

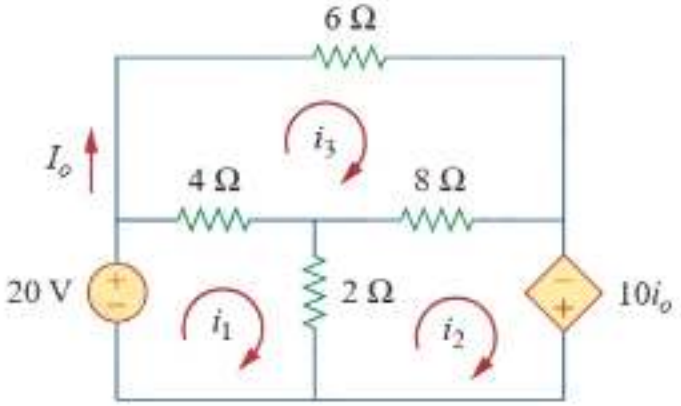
Or:

$i_1 - 2i_2 = 2$ (2)

These two equation can be solved by elimination or substitution or by Cramer's rule to find: $i_1 = 2A$ and $i_2 = 0A$.

Example 2:

Using mesh analysis, find I_o in the following circuit.



Solution:

For mesh 1:

$4(i_1 - i_3) + 2(i_1 - i_2) = 20$

Or:

$6i_1 - 2i_2 - 4i_3 = 20$

Or:

$3i_1 - i_2 - 2i_3 = 10$ (1)

For mesh 2:

$8(i_2 - i_3) + 2(i_2 - i_1) = 10i_o$

But:

$i_o = i_3$

Thus:

$$8(i_2 - i_3) + 2(i_2 - i_1) = 10i_2$$

Or:

$$-2i_1 + 10i_2 - 18i_3 = 0$$

Or:

$$i_1 - 5i_2 + 9i_3 = 0 \qquad (2)$$

For mesh 3:

$$6i_3 + 8(i_3 - i_2) + 4(i_3 - i_1) = 0$$

Or:

$$-4i_1 - 8i_2 + 18i_3 = 0$$

Or:

$$2i_1 + 4i_2 - 9i_3 = 0 \qquad (3)$$

Solving equations (1), (2) and (3) by Cramer's rule, we find:

$$3i_1 - i_2 - 2i_3 = 10 \qquad (1)$$

$$i_1 - 5i_2 + 9i_3 = 0 \qquad (2)$$

$$2i_1 + 4i_2 - 9i_3 = 0 \qquad (3)$$

$$\Delta = \begin{vmatrix} 3 & -1 & -2 \\ 1 & -5 & 9 \\ 2 & 4 & -9 \end{vmatrix} = 135 - 18 - 8 - 20 - 108 - 9 = -28$$

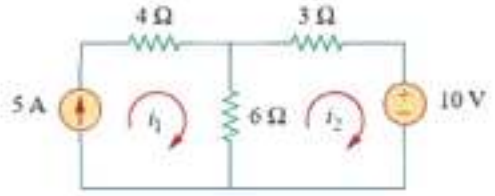
$$\Delta_2 = \begin{vmatrix} 3 & -1 & 10 \\ 1 & -5 & 0 \\ 2 & 4 & 0 \end{vmatrix} = 40 + 100 = 140$$

$$I_0 = \frac{\Delta_2}{\Delta} = -\frac{140}{28} = -5A$$

Mesh Analysis with Current Sources

Two cases can be distinguished:

1) a current source exists only in one mesh, as in the following figure.



In this case:

$i_1 = 5A$ (1)

And for mesh 2:

$3i_2 + 6(i_2 - i_1) = -10$

Or:

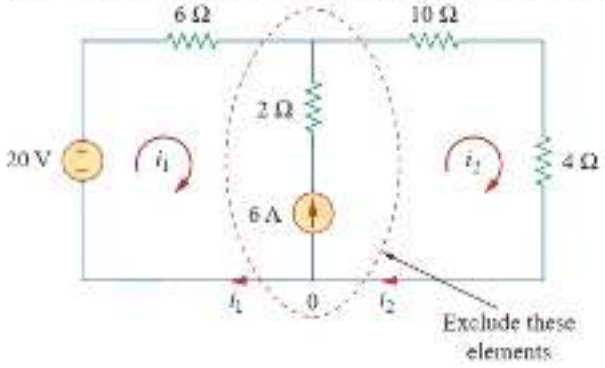
$-6i_1 + 9i_2 = -10$ (2)

Substituting the value of i_1 from (1) in equation (2), we find:

$9i_2 = 20$ or $i_2 = \frac{20}{9} A$

Note that as the value of i_1 is known, thus only one equation, equation for mesh 2 is written.

2) a current source is existing between two meshes, as in the following figure.



Here, We create a *supermesh* by excluding the current source and any elements connected in series with it. Or the supermesh will be the periphery of the two meshes, and KVL is written for this supermesh as:

$6i_1 + 10i_2 + 4i_2 = 20$ (1)

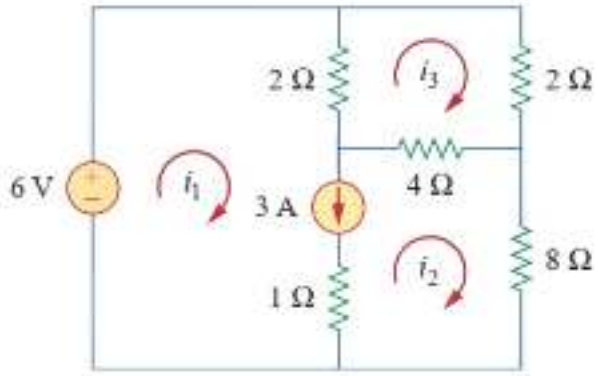
For the branch, where the two meshes intersect, we write:

$i_2 - i_1 = 6$ (2)

And these two equations are solved to find i_1 and i_2.

Example:

Use mesh analysis to find i_1, i_2 and i_3 in the following circuit.



Solution:

Meshes 1 and 2. form a super mesh as the 3A current source is common for them, thus:

$$2(i_1 - i_3) + 4(i_2 - i_3) + 8i_2 = 6$$

Or:

$$2i_1 + 12i_2 - 6i_3 = 6$$

Or:

$$i_1 + 6i_2 - 3i_3 = 3 \qquad (1)$$

Also:

$$i_1 - i_2 = 3 \qquad (2)$$

For mesh (3):

$$2(i_3 - i_1) + 2i_3 + 4(i_3 - i_2) = 0$$

Or:

$$-2i_1 - 4i_2 + 8i_3 = 0$$

Or:

$$i_1 + 2i_2 - 4i_3 = 0 \qquad (3)$$

Equations (1), (2) and (3). Re-written below are solved using Cramer's rule.

$$i_1 + 6i_2 - 3i_3 = 3 \qquad (1)$$

$$i_1 - i_2 = 3 \qquad (2)$$

$$i_1 + 2i_2 - 4i_3 = 0 \qquad (3)$$

For these equations:

$$\Delta = \begin{vmatrix} 1 & 6 & -3 \\ 1 & -1 & 0 \\ 1 & 2 & -4 \end{vmatrix} = 4 - 6 - 3 + 24 = 19$$

$$\Delta_1 = \begin{vmatrix} 3 & 6 & -3 \\ 3 & -1 & 0 \\ 0 & 2 & -4 \end{vmatrix} = 12 - 18 + 72 = 66$$

$$\Delta_2 = \begin{vmatrix} 1 & 3 & -3 \\ 1 & 3 & 0 \\ 1 & 0 & -4 \end{vmatrix} = -12 + 9 + 12 = 9$$

$$\Delta_3 = \begin{vmatrix} 1 & 6 & 3 \\ 1 & -1 & 3 \\ 1 & 2 & 0 \end{vmatrix} = 18 + 6 + 3 - 6 = 21$$

Thus:

$$i_1 = \frac{\Delta_1}{\Delta} = \frac{66}{19} = 3.473A$$

$$i_2 = \frac{\Delta_2}{\Delta} = \frac{9}{19} = 0.473A$$

$$i_3 = \frac{\Delta_3}{\Delta} = \frac{21}{19} = 1.105A$$

Circuit Theorems

Linearity Property:

Linearity is the property of an element describing a linear relationship between cause and effect.

Linearity property is a combination of both the homogeneity (scaling) property and the additivity property.

The homogeneity property requires that if the input (also called the excitation) is multiplied by a constant, then the output (also called the response) is multiplied by the same constant.

For a resistor, for example, Ohm's law relates the input i to the output v, as:
$Ri = v$
If the input (current) is increased by a constant k, then the output (voltage) increases correspondingly by k; that is,
$R(ki) = k(Ri) = kv$

The additivity property requires that the response to a sum of inputs is the sum of the responses to each input applied separately. Using the voltage-current relationship of a resistor, if
$Ri_1 = v_1$ and $Ri_2 = v_2$, thus:
$R(i_1 + i_2) = Ri_1 + Ri_2 = v_1 + v_2$

We say that a resistor is a linear element because the voltage-current relationship satisfies both the homogeneity and the additivity properties.

In general, a circuit is linear if it is both additive and homogeneous. A linear circuit consists of only linear elements, linear dependent sources, and independent sources.

Note that since $p = Ri^2 = \frac{v^2}{R}$ (making it a quadratic function rather than a linear one), the relationship between power and voltage (or current) is nonlinear.

Consider for example, the circuit below.

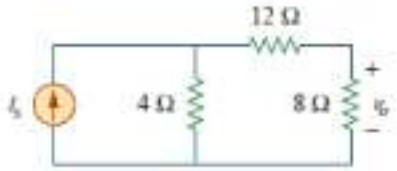


As this circuit is linear, thus the relation between v_s and i is linear, or $i = k.v_s$.
For example, if when $v_s = 10V$, $i = 2A$, then $k = 0.2$.

And if v_s is doubled then i will double, and in general if v_s is multiplied by any constant, then i will be multiplied by the same constant.

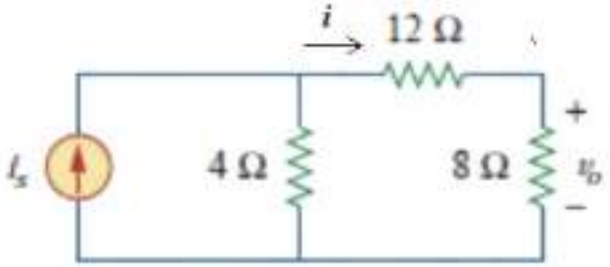
Example1:

For the circuit in the following figure, find v_o when $i_s = 15A$ and $i_s = 30A$



Solution:

From the figure, and applying the current divider rule,



$$i = i_s \frac{4}{4-(12+8)} = \frac{1}{6} i_s$$

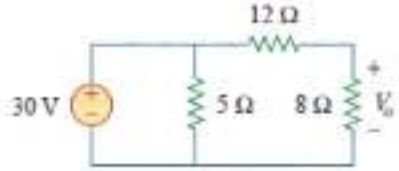
And:

$v_o = 8i = \frac{8}{6} i_s$ and is a linear relation.

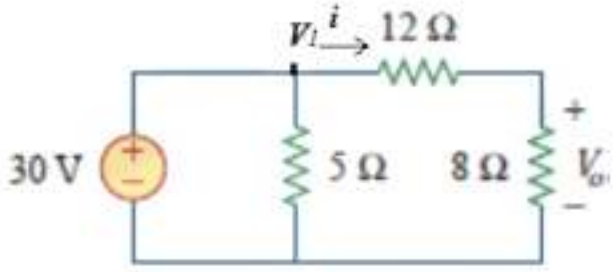
Thus, for $i_s = 15A$ $v_o = \frac{8}{6}(15) = 20V$ and for $i_s = 30A$ $v_o = \frac{8}{6}(30) = 40V$

Example2:

Assume that $v_o = 1V$ and use linearity to calculate the actual value of V_s in the following circuit.



Solution:



From the circuit, when $v_o = 1V$ the current $i = \frac{v_o}{8} = \frac{1}{8}A$ and the voltage $V_1 = i(12+8) = 2.5V$, or $v_o = \frac{1}{2.5}V_1 = 0.4V_1$, thus when $V_1 = 30V$, $v_o = 0.4V_1 = 0.4(30) = 12V$.

Superposition:

If a circuit has two or more independent sources, one way to determine the value of a specific variable (voltage or current) is to determine the contribution of each independent source to the variable and then add them up. is known as the superposition, it is based on the linearity property.

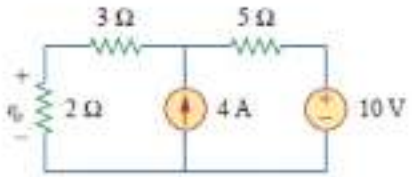
Note that a voltage source is removed by making its terminal voltage equal to zero, or its terminals are short-circuited. In contrary; a current source is removed by making its current equals to zero, or its terminals are open-circuited.

Note also that superposition does help reduce a complex circuit to simpler circuits through replacement of voltage sources by short circuits and of current sources by open circuits.

Keep in mind that superposition is based on linearity. For this reason, it is not applicable to the effect on power due to each source, because the power absorbed by a resistor depends on the square of the voltage or current. If the power value is needed, the current through (or voltage across) the element must be calculated first using superposition.

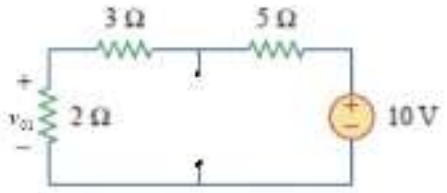
Example1:

Using the superposition theorem, find v_o in the circuit of the following figure.



Solution:

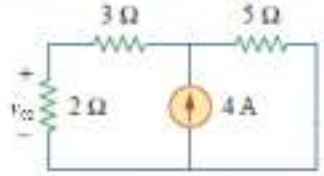
Assume v_{o1} is part of v_o generated by the voltage source (current source is open-circuited), as below.



From this circuit, and using the voltage divider principle:

$$v_{o1} = \frac{10}{3+2+5}(2) = 2V$$

v_{o2} is part of v_o generated by the current source (voltage source is short-circuited), as below.



From this circuit, and using the current divider principle:

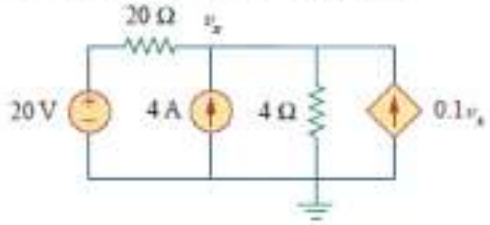
$$v_{o2} = \frac{4(5)}{3+2+5}(2) = 4V$$

Thus:

$$v_o = v_{o1} + v_{o2} = 2 + 4 = 6V$$

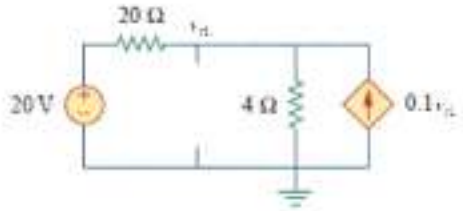
Example2:

Use superposition to find v_x in the circuit of following figure.



Solution:

Assume v_{x1} is part of v_x generated by the voltage source (current source is open-circuited), as below.



Using nodal analysis:

$$\frac{v_{x1} - 20}{20} + \frac{v_{x1}}{4} = 0.1v_{x1}$$

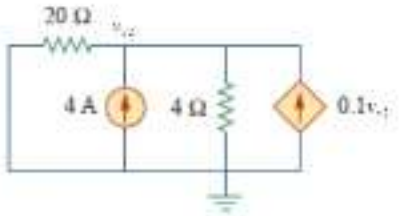
Or:

$$v_{x1} - 20 + 5v_{x1} = 2v_{x1}$$

From which:

$$v_{x1} = \frac{20}{4} = 5V$$

Assume v_{x2} is part of v_x generated by the current source (voltage source is short-circuited), as below.



Using nodal analysis:

$$\frac{v_{x2}}{20} + \frac{v_{x2}}{4} = 4 + 0.1v_x$$

Or:

$$v_{x2} + 5v_{x2} = 80 + 2v_{x2}$$

From which:

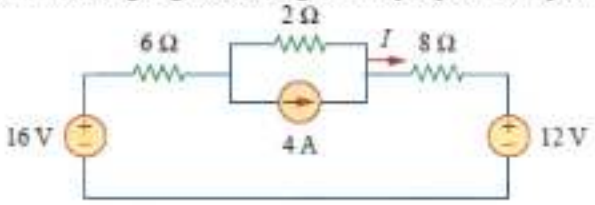
$$v_{x2} = 20V$$

Thus:

$$v_x = v_{x1} + v_{x2} = 25V$$

Example3:

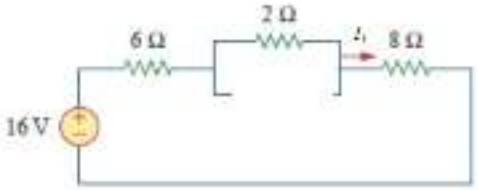
Find I in the circuit of the following figure, using the superposition principle.



Solution:

Let $I = I_1 + I_2 + I_3$, where I_1 is caused by the 16V source, I_2 is caused by the 12V source and I_3 is caused by the 4A source.

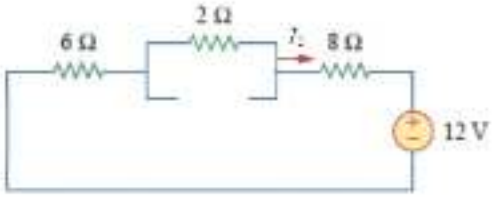
Consider the circuit below.



From this circuit:

$$I_1 = \frac{16}{6+2+8} = 1A$$

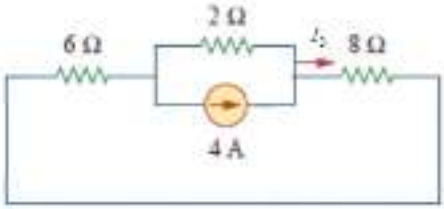
The current component I_2 is found from the following circuit.



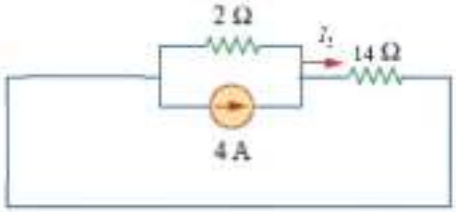
From this circuit:

$$I_2 = -\frac{12}{6+2-8} = -0.75A$$

The current component I_3 is found from the following circuit.



This circuit can be simplified as:



The current component I_3 can be found using the current division principle, as:

$$I_3 = \frac{4(2)}{14+2} = 0.5A$$

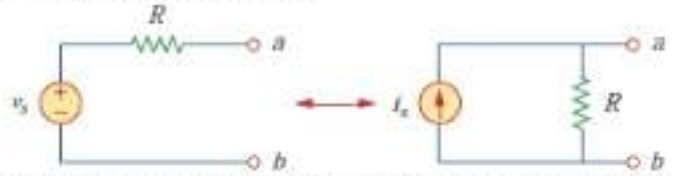
Thus:

$$I = I_1 + I_2 + I_3 = 1 - 0.75 + 0.5 = 0.75A$$

Circuit Theorems

Source Transformation

In source transformation, we transform a voltage source with a series resistor to its equivalent current source with parallel resistor, as below.



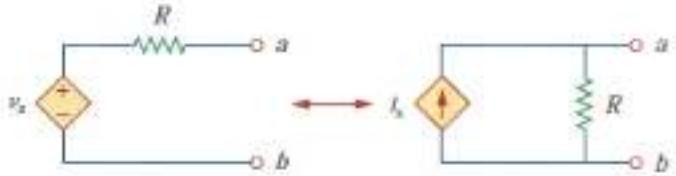
Note that equivalent circuits are the ones whose v-i characteristics are identical.
For the above two equivalent circuits, their open-circuit voltage must be equal, or:

$v_s = RI_s$

Also, their short-circuit currents must be equal, or:

$$I_s = \frac{v_s}{R}$$

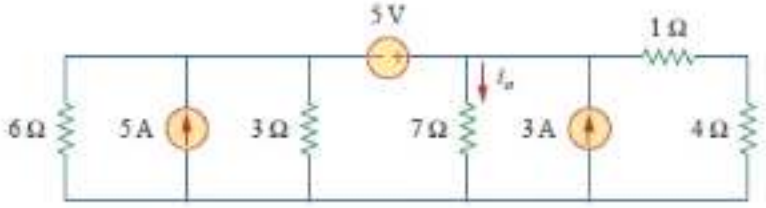
Source transformation also applies to dependent sources, as shown below.



A dependent voltage source in series with a resistor can be transformed to a dependent current source in parallel with the resistor or vice versa where and the above equations for independent sources are used.

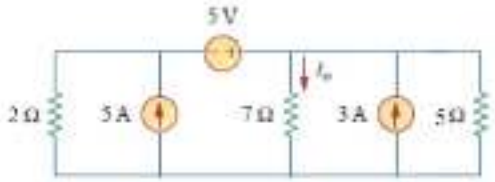
Example:

Find i_o in the following circuit using source transformation.

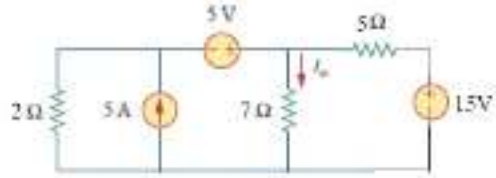


Solution:

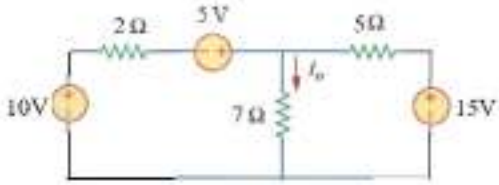
1Ω and 4Ω resistors are in series, their equivalent is a 5Ω resistor.
Also, 6Ω and 3Ω resistors are in parallel, their equivalent is a 2Ω resistor.
The circuit becomes as follows.



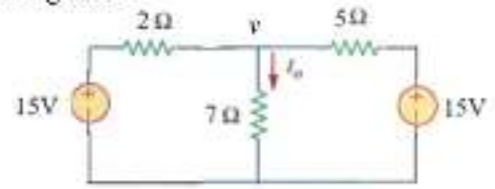
The 3A current source and the 5Ω resistor are replaced by their equivalent voltage source, $v_s = 3x5 = 15V$, in series with a resistor of 5Ω and the circuit is reduced to the following one.



In the same way, the 5A current source and the 2Ω resistor are replaced by their equivalent voltage source, $v_s = 5x2 = 10V$, in series with a resistor of 2Ω and the circuit is reduced to the following one.



The 10V and the 5V voltage sources are in series and their equivalent is a 15V source, and the circuit is reduced to the following one.



By nodal method:

$$\frac{v-15}{2}+\frac{v-15}{5}+\frac{v}{7}=0$$

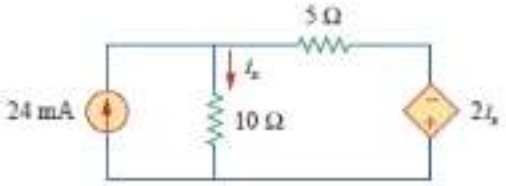
Or:

$$35(v-15)+14(v-15)+10v=0$$

$59v = 735$ from which $v = \frac{735}{59}V$ and $i_o = \frac{v}{7} = 1.78V$

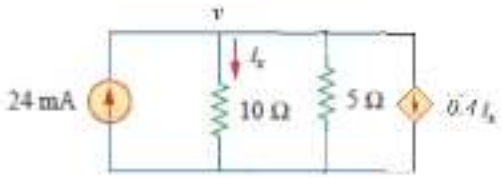
Example:

Find i_x in the following circuit using source transformation.



Solution:

The controlled voltage source in series with the 5Ω resistor is transformed to the equivalent current source $\frac{2i_x}{5} = 0.4i_x$ in parallel with the resistor, as below.



Using nodal analysis:

$$\frac{v}{10} + \frac{v}{5} + 0.4i_x = 24x10^{-3}$$

But:

$$i_x = \frac{v}{10}$$

Thus:

$$\frac{v}{10} + \frac{v}{5} + 0.04v = 24x10^{-3}$$

Or:

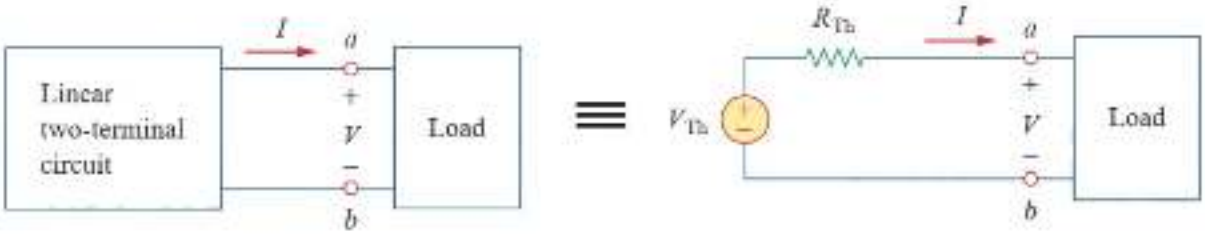
$$10v + 20v + 4v = 24x10^{-1}$$

$$v = \frac{24x10^{-1}}{34} \text{ and } i_x = \frac{v}{10} = \frac{24x10^{-2}}{34} = 7.059mA$$

Circuit Theorems

Thevenin's Theorem

According to Thevenin's theorem, a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a voltage source V_{Th} in series with a resistor R_{Th}, where V_{Th} is the open-circuit voltage at the terminals and R_{Th} is the input or equivalent resistance at the terminals when the independent sources are turned off. See the figure below.



Note that when the circuit is not loaded, the voltage between terminal a and b is the open-circuit voltage between these terminals, while the voltage between these terminals in the equivalent circuit is V_{Th}

, or V_{Th} is equal to the open-circuit voltage, as said before.

Also if load is disconnected and V_{Th} is removed from the equivalent circuit, the resistance between terminals a and b in this equivalent circuit will be R_{Th}. As the two circuits are equivalent, this R_{Th} must be the same as the resistance between terminal a and b with load is disconnected and independent sources are removed.

To find the Thevenin's equivalent of a given circuit, two cases can be discussed:

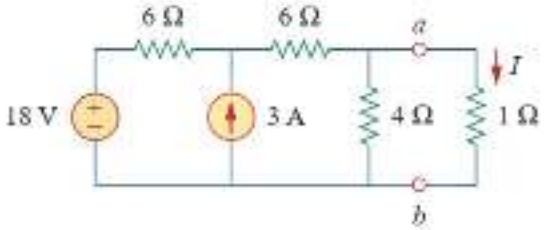
a) the circuit has no dependent sources: in this case we turn off all independent sources and R_{Th} is the input resistance of the network looking between terminals a and b.

b) If the circuit has dependent sources, we turn off all independent sources and apply a voltage source v_0 at terminals a and b, and determine the resulting current i_0, then $R_{Th} = \frac{v_0}{i_0}$.

Alternatively, we may insert a current source i_0 at terminals a-b and find the terminal voltage v_0. Again $R_{Th} = \frac{v_0}{i_0}$.

Example 1:

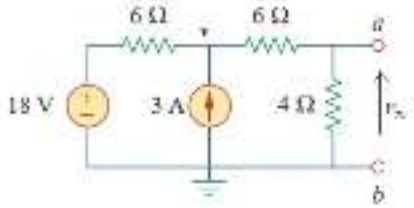
Find the Thevenin's equivalent circuit to the left of the terminals $a-b$ of the circuit shown below. Find also the current I.



Solution:

V_{Th} is the open-circuit voltage (load disconnected) between terminals a and b.

From the following circuit and using nodal analysis:



$$\frac{v-18}{6}+\frac{v}{6+4}=3$$

Or:

$$5(v-18)+3v=90$$

Or:

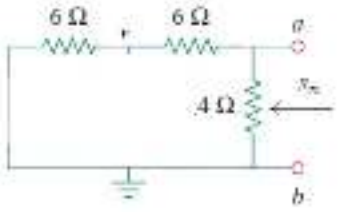
$$v=22.5V$$

Using the voltage division principle:

$$V_{Th}=(v)\frac{4}{4+6}=0.4v=0.4(22.5)=9V$$

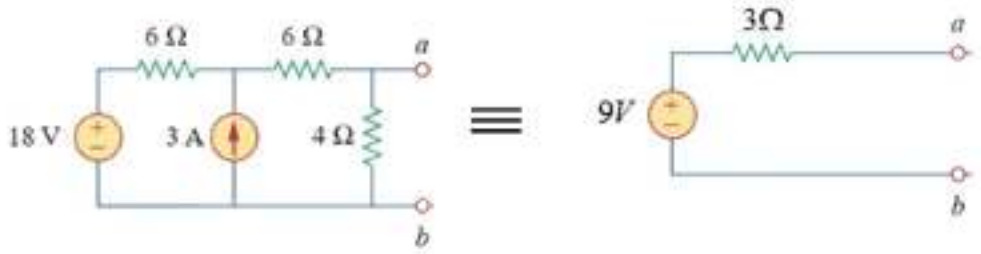
R_{Th} is the resistance seen between terminals a and b, when the sources are removed.

From the following circuit:



$$R_{Th}=\frac{4(12)}{4+12}=3\Omega$$

Or Thevenin's equivalent to the above circuit is as follows.

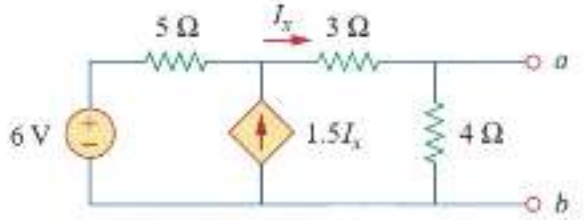


When a 1Ω is connected between terminals, the current I becomes:

$$I = \frac{9}{3+1} = 2.25A$$

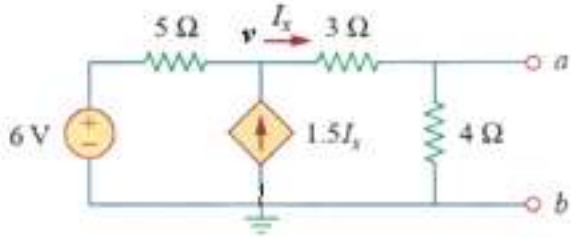
Example 2:

Find the Thevenin's equivalent circuit to the left of the terminals $a-b$ of the circuit shown below.



Solution:

Using nodal analysis, for the circuit below:



$$\frac{v-6}{5}+\frac{v}{7}=1.5I_x$$

And:

$$I_x=\frac{v}{7}$$

From these 2 equations:

$$\frac{v-6}{5}+\frac{v}{7}=\frac{1.5v}{7}$$

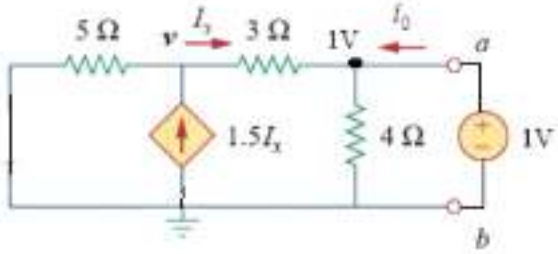
$$7(v-6)+5v=7.5v$$

$$v=\frac{42}{4.5}=9.333V$$

Using the voltage division principle:

$$V_{Th} = V_{ab} = v\frac{4}{4+3} = (9.333)\left(\frac{4}{7}\right) = 5.333V$$

To find R_{Th}, the independent source is removed and a voltage source $v_o = 1$ is applied at terminals a and b, as below.



Using nodal analysis:

$\frac{v}{5} + \frac{v-1}{3} = 1.5I_x$ and $I_x = \frac{v-1}{3}$, thus:

$$\frac{v}{5} + \frac{v-1}{3} = (1.5)\frac{v-1}{3}$$

Or:

$$\frac{v}{5} - (0.5)\frac{v-1}{3} = 0$$

$$6v - 5(v-1) = 0$$

Or:

$$v = -5V$$

And:

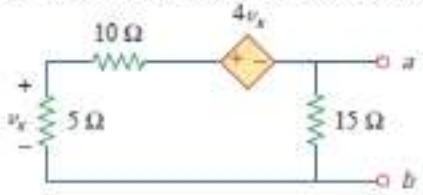
$$i_o = \frac{1}{4} + \frac{1-v}{3} = \frac{1}{4} - \frac{1+5}{3} = 2.25A$$

Or:

$$R_{Th} = \frac{v_o}{i_o} = \frac{1}{2.25} = 0.44\Omega$$

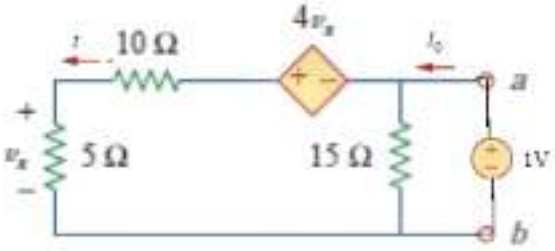
Example 3:

Obtain the Thevenin equivalent of the circuit in the following figure.



Solution:

This circuit has no independent sources. A voltage source $v_o = 1$ is applied at terminals a and b, as below.



For the outer loop:

$1+4v_x-10i-5i=0$

And:

$v_x=5i$, thus:

$1+20i-10i-5i=0$; or:

$i=-0.2A$

$$i_o=i+\frac{1}{15}=-0.2+\frac{1}{15}=-0.133A$$

And

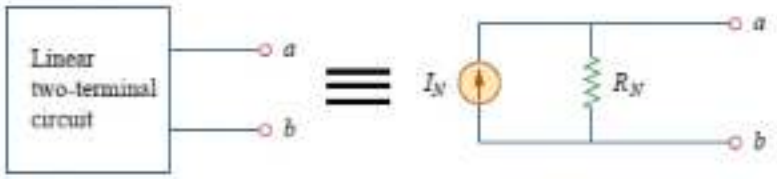
$$R_{Th}=-\frac{1}{0.133}=-7.5\Omega \text{ (supplying power)}$$

$V_{Th}=0V$

Circuit Theorems

Norton's Theorem

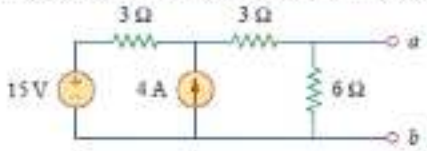
Norton's theorem states that: a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a current source I_N in parallel with a resistor R_N, where I_N is the short-circuit current through the terminals and R_N is the input or equivalent resistance at the terminals when the independent sources are turned off, see the figure below.



Note that Norton's equivalent circuit can be found from Thevenin's equivalent circuit, and vice versa, using source transformation. Or, $I_N = \frac{V_{Th}}{R_{Th}}$ and $R_N = R_{Th}$.

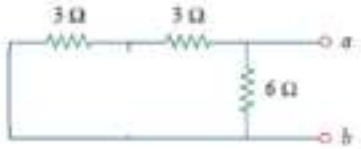
Example 1:

Find the Norton equivalent circuit for the circuit in the figure below, at terminals a-b.



Solution:

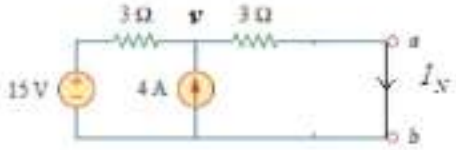
R_N is found as the equivalent resistance between terminals a-b after sources are removed, as below.



From this figure:

$$R_N = \frac{6(3+3)}{6+(3+3)} = 3\Omega$$

I_N is found after terminals a-b are short-circuited, as below.



Note that the 6Ω resister is removed from this circuit as it is bridged by the short-circuit at a-b terminals.

Using nodal analysis:

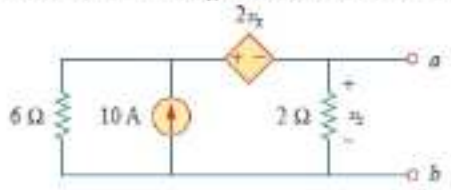
$$\frac{v-15}{3}+\frac{v}{3}=4$$

From which:

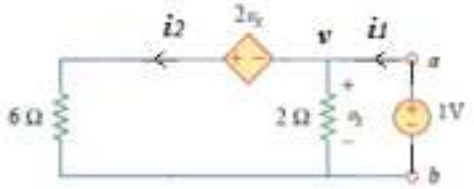
$$v-15+v=12 \text{ or } v=13.5V \text{ and } I_x=\frac{v}{3}=\frac{13.5}{3}=4.5A$$

Example 2:

Find the Norton equivalent circuit for the circuit in the figure below, at terminals a-b.



R_N is found as the equivalent resistance between terminals a-b after the independent source is removed , as below.



For this circuit:

$v=v_x=1V$

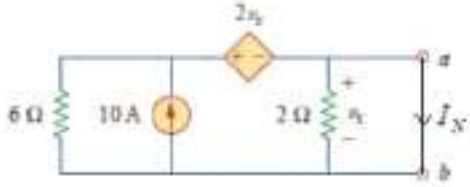
$6i_2=1+2v_x$ or $6i_2=3$ and $i_2=0.5A$

$$i_1=i_2+\frac{v}{2}=1A$$

Thus:

$$R_N=\frac{v}{i_1}=1\Omega$$

I_N is found after terminals a-b are short-circuited, as below.



From this circuit:

$v_x=0$ and the voltage between the terminals of the dependent source is also zero, or this source is short-circuited and the circuit becomes as shown.



From this circuit:

$I_x = 10A$

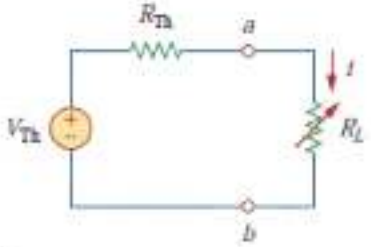
Circuit Theorems

Maximum Power Transfer

Assume a linear circuit is loaded by a load having resistance R_L. The question is to find the value of R_L, such that the power delivered on it is maximum.



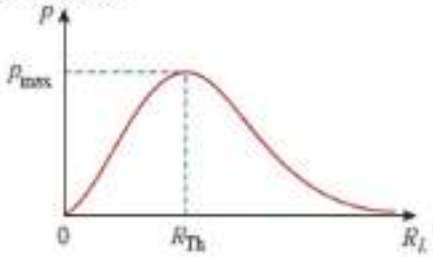
The Thevenin equivalent is useful in solving this problem. The Thevenin's equivalent of this circuit is shown below.



The power delivered to the load is:

$$p = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 .R_L \qquad (1)$$

The plot of this relation is shown below.



From equation (1), we find:

$$\frac{dp}{dR_L} = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 + 2R_L \left(\frac{V_{Th}}{R_{Th} + R_L} \right) \left[\frac{-V_{Th}}{(R_{Th} + R_L)^2} \right]$$

The maximum value of p is found when this derivative is equal to zero, or:

$$\frac{dp}{dR_L} = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 + 2R_L \left(\frac{V_{Th}}{R_{Th} + R_L} \right) \left[\frac{-V_{Th}}{(R_{Th} + R_L)^2} \right] = 0$$

$$\left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 + 2R_L \left(\frac{V_{Th}}{R_{Th} + R_L} \right) \left[\frac{-V_{Th}}{(R_{Th} + R_L)^2} \right] = 0$$

$$\left(\frac{V_{Th}}{R_{Th}+R_L}\right)-2R_L\left[\frac{-V_{Th}}{(R_{Th}+R_L)^2}\right]=0$$

$$\frac{V_{Th}(R_{Th}+R_L)}{(R_{Th}+R_L)^2}-\frac{2V_{Th}R_L}{(R_{Th}+R_L)^2}=0$$

$$\frac{V_{Th}(R_{Th}+R_L)-2V_{Th}R_L}{(R_{Th}+R_L)^2}=0$$

$$V_{Th}(R_{Th}+R_L)-2V_{Th}R_L=0$$

$$V_{Th}[(R_{Th}+R_L)-2R_L]=0$$

$$(R_{Th}+R_L)-2R_L=0$$

Or:

$$R_L=R_{Th}$$

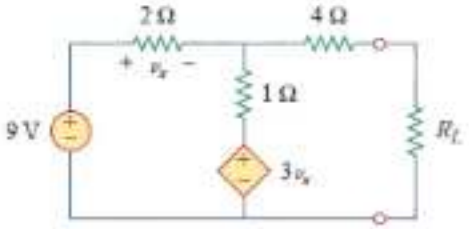
Or, the maximum power transfer takes place when the load resistance R_L equals the Thevenin resistance R_{Th}.

The value of the maximum power delivered is found by substituting this condition in equation (1), or:

$$P_{max}=\left(\frac{V_{Th}}{R_{Th}+R_{Th}}\right)^2.R_{Th}=\frac{(V_{Th})^2}{4R_{Th}}$$

Example:

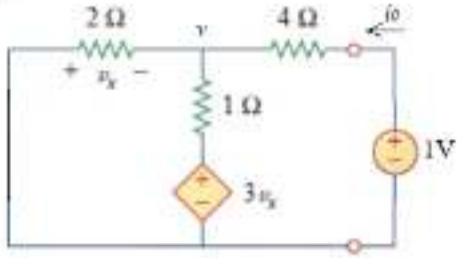
Find the value of R_L that will draw the maximum power from the rest of the circuit in the figure below. Calculate the maximum power.



Solution:

We find the Thevenin's equivalent to this circuit as:

R_{Th} is found from the following circuit after removing the independent voltage source and exciting the circuit by $v_0=1V$, as below.



Using nodal analysis:

$$\frac{v}{2}+\frac{v-3v_x}{1}+\frac{v-1}{4}=0$$

And:

$$v_x=-v$$

From these two equations:

$$\frac{v}{2}+\frac{v+3v}{1}+\frac{v-1}{4}=0$$

$$2v+4(v+3v)+v-1=0$$

From which:

$$v=\frac{1}{19}V$$

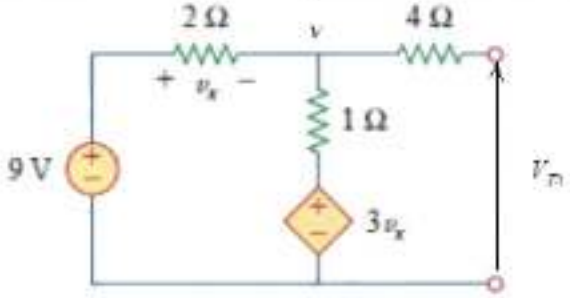
And:

$$i_o=\frac{1-v}{4}=\frac{1-\frac{1}{19}}{4}=\frac{\frac{18}{19}}{4}=\frac{18}{4x19}=\frac{9}{38}A$$

And:

$$R_{Th}=\frac{1}{i_o}=\frac{38}{9}=4.222\Omega$$

V_{Th} is found from the open-circuit after removing the load, as below.



Using nodal analysis:

$$\frac{v-9}{2}+\frac{v-3v_x}{1}=0$$

$$v-9+2(v-3v_x)=0$$

$$3v-6v_x-9=0$$

And:

$$v_x=9-v$$

From these two equations:

$$3v-6(9-v)-9=0$$

$$3v-63+6v=0$$

Or

$$V_{Th}=v=7V$$

And:

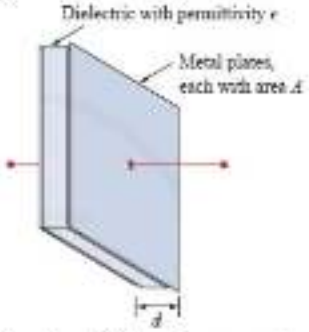
$$p_{max} = \frac{(V_{Th})^2}{4R_{Th}} = \frac{7^2}{4x4.222} = 2.901W$$

Capacitors and Inductors

Unlike resistors, which dissipate energy, capacitors and inductors do not dissipate but store energy, which can be retrieved at a later time. For this reason, capacitors and inductors are called storage elements.

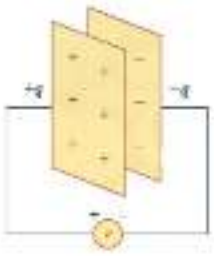
Capacitors

A capacitor is a passive element designed to store energy in its electric field. A capacitor consists of two conducting plates separated by an insulator (or dielectric), as shown below:



The plates may be aluminum foil while the dielectric may be air, ceramic, paper, or mica.

When a voltage source is connected to the capacitor, as in the figure below, the source deposits a positive charge q on one plate and a negative charge $-q$ on the other. The capacitor is said to store the electric charge. The amount of charge stored, represented by q, is directly proportional to the applied voltage so that:



$$q = Cv \qquad (1)$$

where C, the constant of proportionality, is known as the capacitance of the capacitor. The unit of capacitance is the farad (F).

From the above equation, we find:

$$C = \frac{q}{v} \qquad (2)$$

Or, the capacitance is the ratio of the charge on one plate of a capacitor to the voltage difference between the two plates, measured in farads (F).

For example, for the parallel-plate capacitor shown above, the capacitance is given by:

$$C = \frac{\varepsilon A}{d} \qquad (3)$$

where A is the surface area of each plate, d d is the distance between the plates, and ε is the permittivity of the dielectric material between the plates.

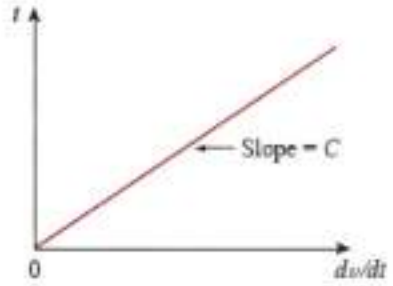
As mentioned before, the relationship between the charge on the capacitor plate and the voltage between its two plates is:

$q = Cv$

From this, we find the current-voltage relationship of the capacitor:

$$i = \frac{dq}{dt} = \frac{d}{dt}(Cv) = C\frac{dv}{dt} \qquad (4)$$

Capacitors satisfying this relation are said to be linear. Their current-voltage relationship is shown below.



Integrating (4), we find:

$i = C\frac{dv}{dt}$, or $\frac{dv}{dt} = \frac{1}{C}i$, or

$$v = \frac{1}{C}\int_{t_0}^{t} i\,dt + v(t_0) \qquad (5)$$

where $v(t_0)$ is the voltage across the capacitor at time t_0.

Equation (5) shows that capacitor voltage depends on the past history of the capacitor current. Hence, the capacitor has a memory.

The instantaneous power delivered to the capacitor is:

$$p = vi = Cv\frac{dv}{dt}$$

The energy stored in the capacitor is:

$$w = \int_{-\infty}^{t} p\,dt = C\int_{-\infty}^{t} v\frac{dv}{dt}dt = C\int_{v(-\infty)}^{v(t)} v\,dv = \frac{1}{2}Cv^2\Big|_{v(-\infty)}^{v(t)}$$

Note that $v(-\infty) = 0$ because the capacitor was uncharged at $t = -\infty$, thus:

$$w = \frac{1}{2}Cv^2 \qquad (6)$$

As:

$$v = \frac{q}{C} \qquad (7)$$

Thus, we find also that:

$$w = \frac{1}{2}C\left(\frac{q}{C}\right)^2 = \frac{q^2}{2C} \qquad (8)$$

Equations (6) and (8) represent the energy stored in the electric field that exists between the plates of the capacitor. This energy can be retrieved, since an ideal capacitor cannot dissipate energy.

Example 1:

What is the voltage across a $3\mu F$ capacitor if the charge on one plate is $0.12mC$? How much energy is stored?

Solution:

$$v = \frac{q}{C} = \frac{0.12x10^{-3}}{3x10^{-6}} = 40V$$

$$w = \frac{1}{2}Cv^2 = \frac{1}{2}\left(3x10^{-6}\right)(40)^2 = 2.4mJ$$

Example 2:

If a $10\mu F$ capacitor is connected to a voltage source with $v(t) = 50\sin 2000t$, determine the current through the capacitor.

Solution:

$$i = C\frac{dv}{dt} = 10x10^{-6}\left(50x2000\cos 2000t\right) = \cos 2000t \quad A$$

Example 3:

The current through a $100\mu F$ capacitor is $i(t) = 50\sin\left(120\pi t\right)mA$. Calculate the voltage across it at $t = 1ms$ and $t = 5ms$, Take $v(0) = 0$.

Solution:

$$v(t) = \frac{1}{C}\int_0^t i\,dt + v(0) = \frac{1}{100x10^{-6}}\int_0^t 50x10^{-3}\sin\left(120\pi t\right)dt = \frac{50x10^{-3}}{100x10^{-6}}\int_0^t \sin\left(120\pi t\right)dt$$

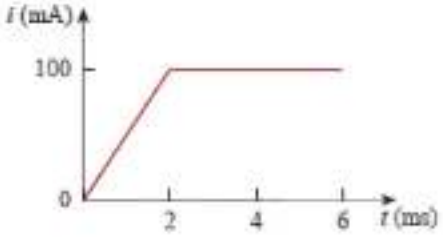
$$= 500\int_0^t \sin\left(120\pi t\right)dt = \frac{500}{120\pi}\cos(120\pi t)\Big|_0^t = \frac{500}{120\pi}\left[\cos(120\pi t) - 1\right]$$

$$v(1ms) = \frac{500}{120\pi}\left[\cos\left(120\pi x1x10^{-3}\right) - 1\right] = 1.326\left[\cos(0.377) - 1\right] = -0.09312V = -93.12mV$$

$$v(5ms) = \frac{500}{120\pi}\left[\cos\left(120\pi x5x10^{-3}\right) - 1\right] = 1.326\left[\cos(1.885) - 1\right] = -1.736V$$

Example 4:

An initially uncharged 1-mF capacitor has the current shown in the following figure across it. Calculate the voltage across it at $t = 2ms$ and $t = 5ms$.



Solution:

$$i(t) = \begin{cases} 50t & 0 < t < 2x10^{-3}S \\ 0.1 & t > 2x10^{-3}S \end{cases}$$

$$v(t) = \frac{1}{C}\int_0^t i\,dt + v(0)$$

Thus:

$$v(t) = \frac{1}{1x10^{-3}}\int_0^t 50t\,dt = \frac{25}{1x10^{-3}}t^2 \quad \text{for } 0 < t < 2x10^{-3}S$$

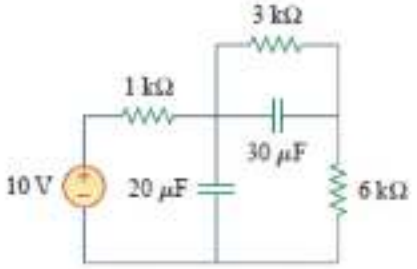
$$v(2ms) = \frac{25}{1x10^{-3}}\left(2x10^{-3}\right)^2 = 25x10^3\,x4x10^{-6} = 0.1V = 100mV$$

$$v(5ms) = \frac{1}{1x10^{-3}}\int_{2x10^{-3}}^{5x10^{-3}} 0.1\,dt + v(2ms) = 1000x0.1t\Big|_{2x10^{-3}}^{5x10^{-3}} + 0.1$$

$$= 100\left(3x10^{-3}\right) + 0.1 = 0.4V = 400mV$$

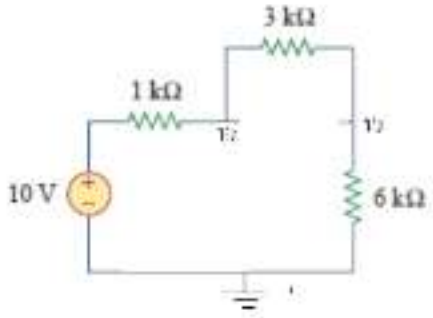
Example 5:

Under dc conditions, find the energy stored in the capacitors in the following figure.



Solution:

Under dc conditions, the capacitors are open-circuited and the circuit become as follows.



The current in the circuit is:

$$i = \frac{10}{1+3-6} = 1mA$$

Thus:

$$v_2 = (6x10^3)(1x10^{-3}) = 6V$$

$$v_1 = (6x10^3 + 3x10^3)(1x10^{-3}) = 9V$$

The voltage on the $20\mu F$ capacitor is v_1 and the energy stored in it is:

$$w_1 = \frac{1}{2}Cv_1^2 = \frac{1}{2}(20x10^{-6})(9^2) = 810\mu J$$

The voltage on the $30\mu F$ capacitor is $v_1 - v2 = 3V$ and the energy stored in it is:

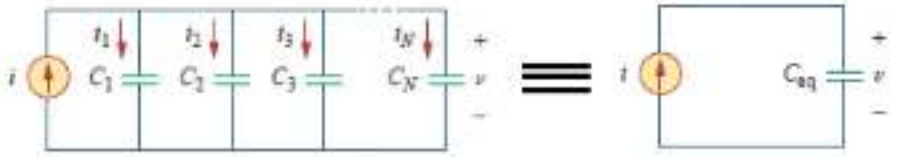
$$w_2 = \frac{1}{2}C(v_1 - v_2)^2 = \frac{1}{2}(30x10^{-6})(3^2) = 135\mu J$$

Capacitors and Inductors

Series and Parallel Capacitors

Parallel Capacitors:

The figure shows n parallel capacitors and their equivalent C_{eq}.



For these circuits:

$i = i_1 + i_2 + ... + i_N \qquad (1)$

From this circuit:

$v_1 = v_2 = ... = v_N = v$

$i_1 = C_1 \dfrac{dv_1}{dt} = C_1 \dfrac{dv}{dt}$

$i_2 = C_2 \dfrac{dv}{dt}$

.

.

$i_N = C_N \dfrac{dv}{dt}$

From the equivalent circuit:

$i = C_{eq} \dfrac{dv}{dt}$

Substituting all these equations in equation (1), we find:

$C_{eq} \dfrac{dv}{dt} = C_1 \dfrac{dv}{dt} + C_2 \dfrac{dv}{dt} + ... + C_N \dfrac{dv}{dt}$

Or:

$C_{eq} \dfrac{dv}{dt} = (C_1 + C_2 + ... + C_N) \dfrac{dv}{dt}$

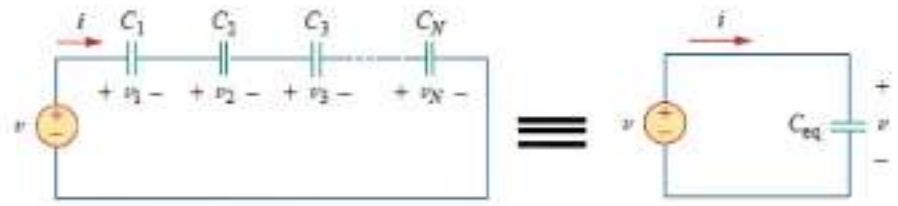
From which:

$C_{eq} = C_1 + C_2 + ... + C_N$

Or the equivalent capacitance for parallel capacitors is equal to the sum of their capacitances.

Series Capacitors

The figure shows n series capacitors and their equivalent C_{eq}.



For this circuit:

$$v = v_1 + v_2 + \ldots - v_N \qquad (2)$$

From this circuit:

$$v_1 = \frac{1}{C_1}\int_{t_0}^{t} i\,dt + v_1(t_0)$$

$$v_2 = \frac{1}{C_2}\int_{t_0}^{t} i\,dt + v_2(t_0)$$

.

.

$$v_N = \frac{1}{C_N}\int_{t_0}^{t} i\,dt + v_N(t_0)$$

From the equivalent circuit:

$$v = \frac{1}{C_{eq}}\int_{t_0}^{t} i\,dt + v(t_0)$$

Substituting these equations in equation (2), we find:

$$\frac{1}{C_{eq}}\int_{t_0}^{t} i\,dt + v(t_0) = \frac{1}{C_1}\int_{t_0}^{t} i\,dt + v_1(t_0) + \frac{1}{C_2}\int_{t_0}^{t} i\,dt + v_2(t_0) + \ldots + \frac{1}{C_N}\int_{t_0}^{t} i\,dt + v_N(t_0)$$

Or:

$$\frac{1}{C_{eq}}\int_{t_0}^{t} i\,dt + v(t_0) = \left(\frac{1}{C_1} + \frac{1}{C_2} + \ldots + \frac{1}{C_N}\right)\int_{t_0}^{t} i\,dt + \left[v_1(t_0) + v_2(t_0) + \ldots + v_N(t_0)\right] \qquad (3)$$

But, initially:

$$v(t_0) = v_1(t_0) + v_2(t_0) + \ldots + v_N(t_0)$$

Tus:

$$\frac{1}{C_{eq}}\int_{t_0}^{t} i\,dt + v(t_0) = \left(\frac{1}{C_1} + \frac{1}{C_2} + \ldots + \frac{1}{C_N}\right)\int_{t_0}^{t} i\,dt + \left[v_1(t_0) + v_2(t_0) + \ldots + v_N(t_0)\right]$$

Or:

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \ldots + \frac{1}{C_N}$$

Or:

The equivalent capacitance of series-connected capacitors is the reciprocal of the sum of the reciprocals of the individual capacitances.

For 2 capacitors in series:

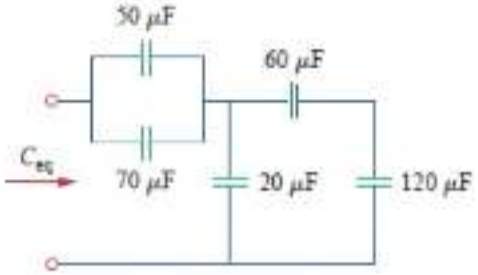
$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{C_1 + C_2}{C_1 C_2}$$

Or:

$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$$

Example 1:

Find the equivalent capacitance seen at the terminals of the circuit in the following figure:

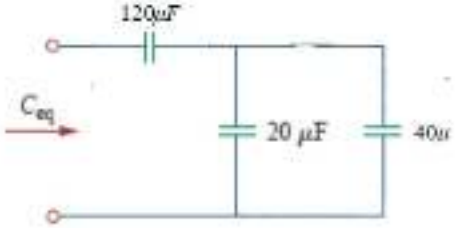


Solution:

The equivalent of the parallel $50\mu F$ and $70\mu F$ capacitors is $50 + 70 = 120\mu F$.

The equivalent of the series $60\mu F$ and $120\mu F$ capacitors is $\frac{60x120}{60+120} = 40\mu$.

The equivalent circuit is reduced to the one below.



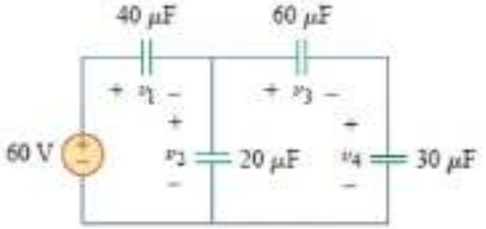
The equivalent of the parallel $20\mu F$ and $40\mu F$ capacitors is $20 + 40 = 60\mu F$.

Now:

$$C_{eq} = \frac{60x120}{60+120} = 40\mu F$$

Example 2:

Find the voltage across each of the capacitors in following figure.



Solution:

The equivalent of the series $60\mu F$ and $30\mu F$ capacitors is $\frac{60x30}{60+30}=20\mu$.

The equivalent of these 2 capacitors in parallel with the $20\mu F$ capacitor is $20+20=40\mu$

Thus $C_{eq}=\frac{40x40}{40-40}=20\mu$

Thus Q passing to charge the capacitors is:

$Q=C_{eq}V=20x10^{-6}(60)=1200\mu C$

The voltage $v_1=\frac{Q}{C_1}=\frac{1200x10^{-6}}{40x10^{-6}}=30V$

$v_2=60-v_1=60-30=30V$

The voltage v_2 will be divided between the $60\mu F$ and the $30\mu F$ capacitors.

Their equivalent capacitance is $\frac{60x30}{60+30}=20\mu$ and the amount of charge on each of them is:

$Q_1=CV=20x10^{-6}(30)=600\mu C$

Thus:

$v_3=\frac{Q_1}{C_3}=\frac{600x10^{-6}}{60x10^{-6}}=10V$

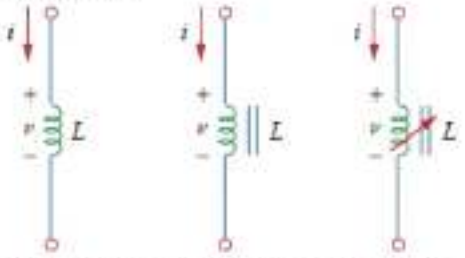
And:

$v_4=\frac{Q_1}{C_4}=\frac{600x10^{-6}}{30x10^{-6}}=20V$

Inductors

An inductor is a passive element designed to store energy in its magnetic field. An inductor consists of a coil of conducting wire.

The symbols of a conductor are as below.

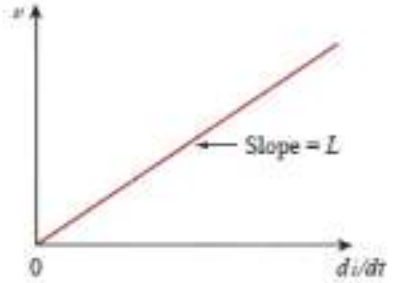


Circuit symbols for inductors: (a) air-core, (b) iron-core, (c) variable iron-core.

The relation between the voltage and the current for the inductor is:

$$v = L\frac{di}{dt} \qquad (1)$$

where L is the constant of proportionality called the inductance of the inductor. The unit of inductance is the henry (H). This relationship is shown below.



From equation (1):

$$\frac{di}{dt} = \frac{1}{L}v$$

$$i = \frac{1}{L}\int_{-\infty}^{t} v(t)dt = \frac{1}{L}\int_{t_0}^{t} v(t)dt - i(t_0)$$

The inductor is designed to store energy in its magnetic field. The energy stored can be as follows.

The power delivered to the inductor is:

$$p(t) = v(t).i(t) = Li\frac{di}{dt}$$

And the energy stored in the inductor is:

$$w = \int_{-\infty}^{t} p(t)dt = \int_{-\infty}^{t} Li\frac{di}{dt}dt = L\int_{-\infty}^{i} idi = \frac{1}{2}Li^2(t) - \frac{1}{2}Li^2(-\infty)$$

Since:

$\frac{1}{2}Li^2(-\infty)$ 0

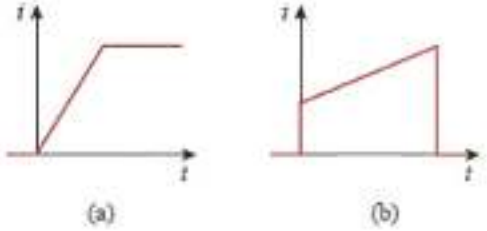
Thus:

$w = \frac{1}{2}Li^2(t)$

The following are important notes about the inductor:

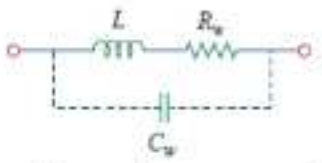
1) as for the inductor $v = L\frac{di}{dt}$, thus for constant current, the voltage is zero. or An inductor acts like a short circuit to dc.

2) as for the inductor $v = L\frac{di}{dt}$, thus the current can't change instantaneously, as in this case $\frac{di}{dt}$ becomes infinity. Or for instantaneous change of a current through an inductor, the voltage value must be infinity, which is not practically possible.
For example, the current through an inductor may take the form shown in figure (a), whereas the inductor current cannot take the form shown in figure (b).



3) Like the ideal capacitor, the ideal inductor does not dissipate energy. The energy stored in it can be retrieved at a later time. The inductor takes power from the circuit when storing energy and delivers power to the circuit when returning previously stored energy.

4) A practical, nonideal inductor has a significant resistive component, as shown in the figure below. This is due to the fact that the inductor is made of a conducting material such as copper, which has some resistance. This resistance is called the winding resistance R_w, and it appears in series with the inductance of the inductor. The presence of R_w makes it both an energy storage device and an energy dissipation device. Since R_w is usually very small, it is ignored in most cases. The nonideal inductor also has a winding capacitance C_w due to the capacitive coupling between the conducting coils.



C_w is very small and can be ignored in most cases, except at high frequencies.

Example 1:

If the current through a 1-mH inductor is $i(t) = 20\cos 100t \quad mA$. find the terminal voltage and the energy stored.

Solution:

$$v = L\frac{di}{dt} = 1x10^{-3}\left(-20x100\sin 100tx10^{-3}\right) = -2x10^{-3}\sin 100t \quad V = -2\sin 100t \quad mV$$

$$w = \frac{1}{2}Li^2 = \frac{1}{2}\left(1x10^{-3}\right)\left(20x10^{-3}\cos 100t\right)^2 = 0.2x10^{-6}\cos^2 100t \quad J = 0.2\cos^2 100t \quad \mu J$$

Example 2:

The terminal voltage of a 2-H inductor is $v = 10(1-t)V$. Find the current flowing through it at $t = 4s$ and the energy stored in it at $t = 4s$. Assume $i(0) = 2A$.

Solution:

$$i = \frac{1}{L}\int_0^t v(t)dt - i(0) = \frac{1}{2}\int_0^t 10(1-t)dt + 2 = -\frac{5}{2}(1-t)^2\Big|_0^t + 2$$

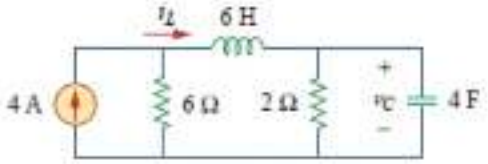
$$= -\frac{5}{2}(1-t)^2 + \frac{5}{2} - 2 = \frac{9}{2} - \frac{5}{2}(1-t)^2$$

$$i(4) = \frac{9}{2} - \frac{5}{2}(1-4)^2 = \frac{9}{2} - \frac{5}{2}(9) = -18A$$

$$w = \frac{1}{2}Li^2 = \frac{1}{2}(2)(-18)^2 = 324J$$

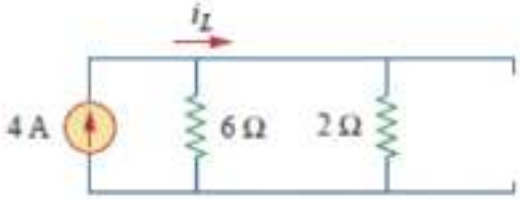
Example 3:

Determine v_c, i_L and the energy stored in the capacitor and inductor in the circuit of the following figure under dc conditions.



Solution:

The circuit under dc conditions is as follows:



The circuit forms a current divider, thus:

$$i_L = (4)\frac{6}{6+2} = 3A$$

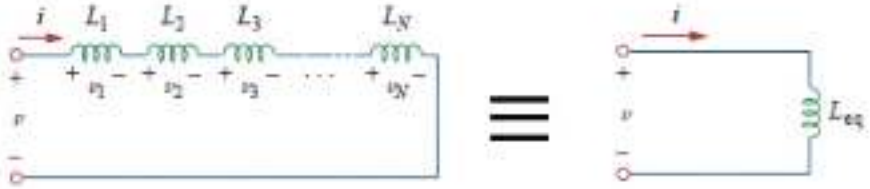
$$v_C = (2)(3) = 6V$$

$$w_C = \frac{1}{2}Cv^2 = \frac{1}{2}(4)(6^2) = 72J$$

$$w_L = \frac{1}{2}Li^2 = \frac{1}{2}(6)(3^2) = 27J$$

Series and Parallel Inductors

Consider a circuit of serially connected inductors seen below.



For this circuit:

$$v = v_1 + v_2 + ... + v_N$$

$$= L_1 \frac{di}{dt} + L_2 \frac{di}{dt} + ... + L_N \frac{di}{dt} = (L_1 + L_2 + ... + L_N)\frac{di}{dt}$$

For the equivalent circuit:

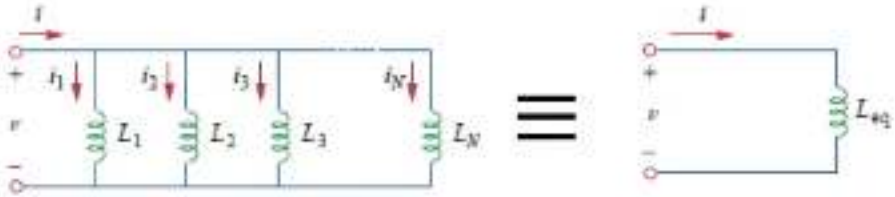
$$v = L_{eq} \frac{di}{dt}$$

Comparing these two equations, we conclude that:

$$L_{eq} = L_1 + L_2 + ... + L_N$$

Or, the equivalent inductance of series-connected inductors is the sum of the individual inductances.

Consider a circuit of parallel connected inductors seen below.



For this circuit:

$$i = i_1 + i_2 - ... + i_N$$

$$= \frac{1}{L_1}\int_{t_0}^{t} v\,dt + i_1(t_0) + \frac{1}{L_2}\int_{t_0}^{t} v\,dt + i_2(t_0) + ... + \frac{1}{L_N}\int_{t_0}^{t} v\,dt + i_N(t_0)$$

$$= \left(\frac{1}{L_1} + \frac{1}{L_2} + ... + \frac{1}{L_N}\right)\int_{t_0}^{t} v\,dt + i_1(t_0) + i_2(t_0) - ... + i_N(t_0)$$

$$= \left(\frac{1}{L_1} + \frac{1}{L_2} + ... + \frac{1}{L_N}\right)\int_{t_0}^{t} v\,dt + i(t_0)$$

For the equivalent circuit:

$$i = \frac{1}{L_{eq}} \int_{t_0}^{t} v\,dt + i(t_0)$$

Comparing these two equations, we conclude that:

$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \ldots + \frac{1}{L_N}$$

Or, the equivalent inductance of parallel inductors is the reciprocal of the sum of the reciprocals of the individual inductances.

For two inductors in parallel:

$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} = \frac{L_1 + L_2}{L_1 L_2}$$

Or:

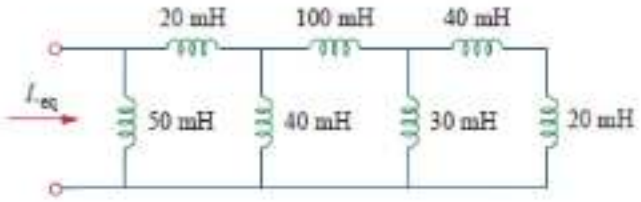
$$L_{eq} = \frac{L_1 L_2}{L_1 + L_2}$$

Now we have discussed the three passive elements used in the electrical circuits, the following table, summarizes, their characteristics:

Relation	Resistor (R)	Capacitor (C)	Inductor (L)
$v - i$	$v = Ri$	$v = \frac{1}{C}\int_{t_0}^{t} i\,dt + v(t_0)$	$v = L\frac{di}{dt}$
$i - v$	$i = \frac{v}{R}$	$i = C\frac{dv}{dt}$	$i = \frac{1}{L}\int_{t_0}^{t} v\,dt + i(t_0)$
p or w	$p = Ri^2 = \frac{v^2}{R}$	$w = \frac{1}{2}Cv^2$	$w = \frac{1}{2}Li^2$
series	$R_{eq} = R_1 - R_2 + \ldots + R_N$	$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \ldots + \frac{1}{C_N}$	$L_{eq} = L_1 + L_2 + \ldots + L_N$
parallel	$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots + \frac{1}{R_N}$	$C_{eq} = C_1 + C_2 + \ldots + C_N$	$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \ldots + \frac{1}{L_N}$
At DC	same	Open-circuit	Short-circuit

Example 1:

Calculate the equivalent inductance for the inductive ladder network in the figure below.



Solution:

The $40mH$ and the $20mH$ inductors are in series and their equivalent is $40 + 20 = 60mH$

The equivalent $60mH$ and the $30mH$ inductors are in parallel and their equivalent is

$$\frac{60x30}{60 + 30} = 20mH$$

The equivalent $20mH$ and the $100mH$ inductors are in series and their equivalent is

$20 + 100 = 120mH$

The equivalent $120mH$ and the $40mH$ inductors are in parallel and their equivalent is

$$\frac{120x40}{120 + 40} = 30mH$$

The equivalent $30mH$ and the $20mH$ inductors are in series and their equivalent is

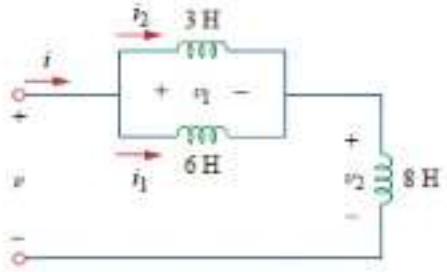
$30 + 20 = 50mH$

The equivalent $50mH$ and the $50mH$ inductors are in parallel and their equivalent is

$$\frac{50x50}{50 + 50} = 25mH$$

Example 2:

In the circuit of the following figure, $i_1(t) = 0.6e^{-2t}A$. If $i(0) = 1.4A$ find:



(a) $i_2(0)$

(b) $i_1(t)$ and $i(t)$

(c) $v_1(t)$, $v_2(t)$ and $v(t)$

Solution:

a) $i_2(0) = i(0) - i_1(0) = 1.4 - 0.6e^{-2(0)} = 0.8A$

b) $v_1(t) = L_1 \frac{di_1}{dt} = (6)\frac{d}{dt}\left(0.6e^{-2t}\right) = -7.2e^{-2t}$

$$i_2(t) = \frac{1}{L_2}\int_0^t v_1(t)dt + i_2(0) = -\frac{7.2}{3}\int_0^t e^{-2t}dt + 0.8 = 1.2e^{-2t}\Big|_0^t + 0.8$$

$$= 1.2e^{-2t} - 1.2 + 0.8 = 1.2e^{-2t} - 0.4 \quad A$$

$$i(t) = i_1(t) + i_2(t) = 0.6e^{-2t} + 1.2e^{-2t} - 0.4 = 1.8e^{-2t} - 0.4 \quad A$$

c) $$v_1(t) = L_1\frac{di_1}{dt} = (6)\frac{d}{dt}\left(0.6e^{-2t}\right) = -7.2e^{-2t}$$

$$v_2(t) = L_2\frac{di}{dt} = (8)\frac{d}{dt}\left(1.8e^{-2t} - 0.4\right) = 8(1.8)(-2)e^{-2t} = -28.8e^{-2t} \quad V$$

$$v(t) = v_1(t) + v_2(t) = -7.2e^{-2t} - 28.8e^{-2t} = -36e^{-2t}$$

First-Order Circuits

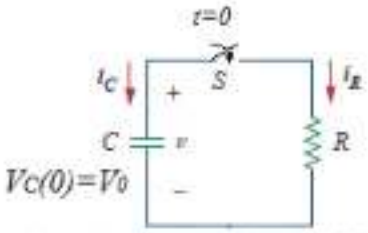
First-order circuits are electrical circuits, the operation of which is described by a first-order differential equations. These circuits are circuits comprising a resistor and a capacitor (RC circuit) or a circuit comprising a resistor and an inductor (RL circuit).
According to the excitation method, these circuits are:

- circuits excited by the initial condition of the storage elements. These circuits are called source-free circuits. The energy initially stored in the capacitive or inductive element causes current to flow in the circuit and is gradually dissipated in the resistors. These circuits have no independent sources but may have dependent sources.

- circuits excited by independent sources (voltage or current sources).

The Source-Free RC Circuit

A source-free RC circuit occurs when the energy stored in a capacitor is released to the resistors. Consider the following circuit.



In this circuit, when the switch S is closed at $t = 0$, the initially charged capacitor, having $V_C(0) = V_0$ starts to discharge through the resistor R.

For this circuit:

$$i_C = C\frac{dv}{dt} \qquad (1)$$

$$i_R = \frac{v}{R} \qquad (2)$$

And:

$$i_C = -i_R \qquad (3)$$

Substituting (1) and (2) in (3), we find:

$$C\frac{dv}{dt} = -\frac{v}{R}$$

This is a first-order differential equation (variables separable type). This equation can be written in the form:

$$\frac{dv}{v} = -\frac{dt}{RC} \qquad (4)$$

Integrating both sides:

$$\ln v = -\frac{1}{RC}t + C_1$$

Or:

$$v(t) = e^{-\frac{1}{RC}t+C_1} = e^{-\frac{1}{RC}t}.e^{C_1} = C.e^{-\frac{1}{RC}t} \quad (5)$$

Where:

$C = e^{C_1}$

As :

$v(0) = V_0$ (6)

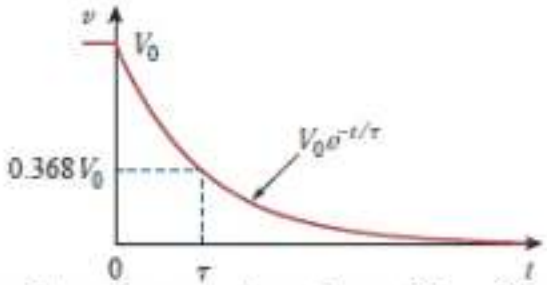
Thus from (5) and (6), we find:

$V_0 = C$ (7)

Substituting (7) in (5), we find:

$$v(t) = V_0.e^{-\frac{1}{RC}t} \quad (8)$$

This shows that the voltage response of the RC circuit is an exponential decay of the initial voltage. Since the response is due to the initial energy stored and the physical characteristics of the circuit and not due to some external voltage or current source, it is called the natural response of the circuit, and is shown in the following figure.



Note that, as t increases, the voltage decreases toward zero. The rapidity with which the voltage decreases is expressed in terms of the time constant, denoted by τ.

The time constant of a circuit is defined as the time required for the response to decay to a factor of $\frac{1}{e}$ or 36.8 percent of its initial value. Or from (8):

$\frac{1}{e} = e^{-1} = e^{-\frac{1}{RC}\tau}$, or $\tau = RC$

Thus equation (8) can be written as:

$$v(t) = V_0.e^{-\frac{t}{\tau}} \quad (9)$$

Energy conservation:

From (8), the voltage on the capacitor is:

$$v(t) = V_0 e^{-\frac{t}{RC}}$$

The current through the resistor is:

$$i(t) = \frac{v(t)}{R} = \frac{V_0}{R} e^{-\frac{t}{RC}}$$

The power dissipated on the resistor is:

$$p(t) = Ri^2 = \frac{(V_0)^2}{R} e^{-\frac{2t}{RC}}$$

The energy dissipated on the resistor is:

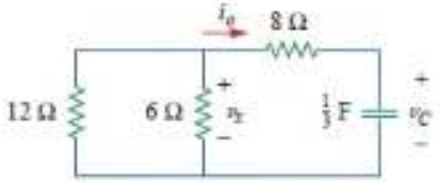
$$w(t) = \int_0^t p(t)dt = \int_0^t \frac{(V_0)^2}{R} e^{-\frac{2t}{RC}} dt = \frac{(V_0)^2}{R} \int_0^t e^{-\frac{2t}{RC}} dt = -\frac{(V_0)^2 RC}{2R} \int_0^t e^{-\frac{2t}{RC}} d\left(\frac{-2t}{RC}\right)$$

$$= -\frac{(V_0)^2 RC}{2R} e^{-\frac{2t}{RC}}\Big|_0^t = -\frac{(V_0)^2 C}{2}\left(e^{-\frac{2t}{RC}} - 1\right)$$

When $t \to \infty$ $w = \frac{1}{2}C(V_0)^2$ which is the energy previously stored in the capacitor.

Example 1:

For the circuit in the following figure, if $v_c(0) = 45V$ find v_c, v_x and i_0 for $t \geq 0$



Solution:

The equivalent resistance in parallel with the capacitor is:

$$R_e = \frac{12x6}{12+6} + 8 = 12\Omega$$

The circuit time constant is: $\tau = R_e C = 12x\frac{1}{3} = 4s$

$$v_c(t) = v_c(0)e^{-\frac{t}{\tau}} = 45e^{-\frac{t}{4}} = 45e^{-0.25t}$$

The equivalent of the parallel 12Ω and 6Ω resistors (4Ω resistor) with the 8Ω resistor are forming a voltage divider, thus:

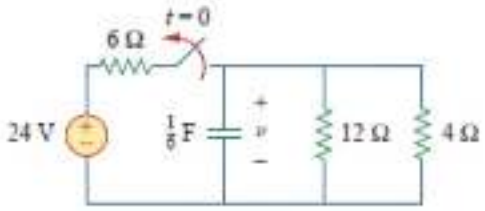
$$v_x(t) = \frac{4}{8+4} v_c(t) = \left(\frac{1}{3}\right)45e^{-0.25t} = 15e^{-0.25t}$$

The current i_0 is the current through the capacitor, or:

$$i_0(t) = C\frac{dv_C}{dt} = \frac{1}{3}\frac{d}{dt}\left(45e^{-0.25t}\right) = 15(-0.25)e^{-0.25t} = -3.75e^{-0.25t}\,A$$

Example 2:

For the circuit in the following figure, if the switch opens at $t = 0$ find $v(t)$ for $t > 0$ and $w_c(0)$.



Solution:

As the switch was closed for long time, the capacitor represents an open circuit and the equivalent of the the parallel 12Ω and 4Ω resistors (3Ω resistor) with the 6Ω resistor are forming a voltage divider, thus:

$$v(0) = \frac{3}{6+3}(24) = 8V$$

$\tau = R_e C$, where:

$$R_e = \frac{12x4}{12+4} = 3\Omega$$

Thus:

$$\tau = R_e C = 3\left(\frac{1}{6}\right) = \frac{1}{2}s$$

And:

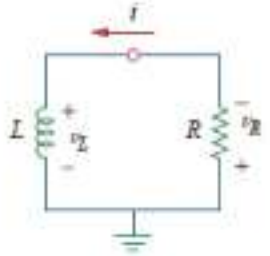
$$v(t) = v(0)e^{-\frac{t}{\tau}} = 8e^{-\frac{t}{0.5}} = 8e^{-2t}V$$

$$w_c(0) = \frac{1}{2}C[v(0)]^2 = \frac{1}{2}\left(\frac{1}{6}\right)(8^2) = \frac{64}{12} = 5.333J$$

First-Order Circuits

The Source-Free RL Circuit

Consider the series connection of a resistor and an inductor, as shown in the figure below.



we assume that the inductor has an initial current I_0 or $i_0 = I_0$.

For this circuit:

$v_L + v_R = 0$

Or:

$v_L = -v_R$

Or:

$$L\frac{di}{dt} = -Ri \qquad (1)$$

Or:

$$\frac{di}{dt} = -\frac{R}{L}i$$

$$\frac{di}{i} = -\frac{R}{L}dt$$

Integrating both sides:

$$\ln i = -\frac{R}{L}t + C_1$$

Or:

$$i(t) = e^{-\frac{R}{L}t + C_1} = e^{-\frac{R}{L}t}e^{C_1} = Ce^{-\frac{R}{L}t} \qquad (2)$$

Where:

$C = e^{C_1}$

As :

$$i(0) = I_0 \qquad (3)$$

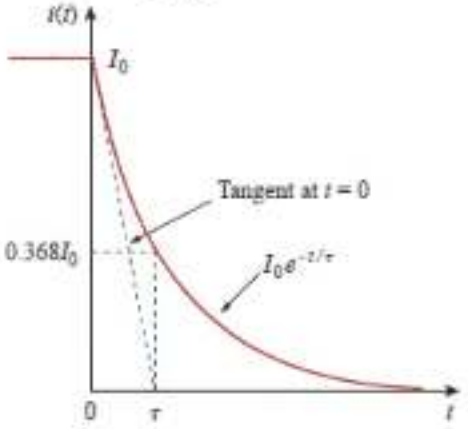
Thus from (5) and (6), we find:

$$I_0 = C \qquad (4)$$

Substituting (4) in (2), we find:

$$i(t) = I_0 e^{-\frac{R}{L}t} \qquad (5)$$

This shows that the current of the source free RL circuit is an exponential decay of the initial current. Since the response is due to the initial energy stored and the physical characteristics of the circuit and not due to some external voltage or current source, it is called the natural response of the circuit, and is shown in the following figure.



Note that, as t increases, the current decreases toward zero. The rapidity with which the current decreases is expressed in terms of the time constant, denoted by τ.

The time constant of a circuit is defined as the time required for the response to decay to a factor of $\frac{1}{e}$ or 36.8 percent of its initial value. Or from (5):

$$\frac{1}{e} = e^{-1} = e^{-\frac{t}{RC}}, \text{ or } \tau = RC$$

Thus equation (5) can be written as:

$$i(t) = I_0 e^{-\frac{t}{\tau}} \qquad (6)$$

Energy conservation:

From (8). the current in the resistor is:

$$i(t) = I_0 e^{-\frac{R}{L}t}$$

The power dissipated on the resistor is:

$$p(t) = Ri^2 = R(I_0)^2 e^{-\frac{2R}{L}t}$$

The energy dissipated on the resistor is:

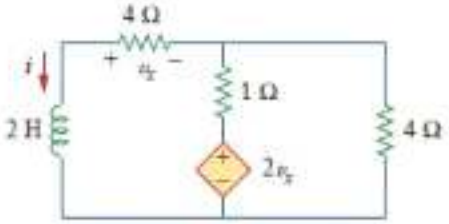
$$w(t) = \int_0^t p(t)dt = \int_0^t R(I_0)^2 e^{-\frac{2R}{L}t}dt = R(I_0)^2 \int_0^t e^{-\frac{2R}{L}t}dt = -\frac{LR(I_0)^2}{2R}\int_0^t e^{-\frac{2R}{L}t}d\left(\frac{-2R}{L}t\right)$$

$$= -\frac{LR(I_0)^2}{2R}e^{-\frac{2R}{L}t}\Big|_0^t = -\frac{L(I_0)^2}{2}\left(e^{-\frac{2R}{L}t} - 1\right)$$

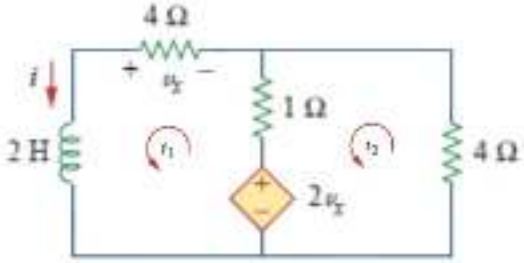
When $t \to \infty$: $w = \frac{1}{2} L(I_o)^2$ which is the energy previously stored in the inductor.

Example 1:

Find i and v_x in the following circuit, let $i(0) = 5A$



Solution:



Method 1:

For mesh 1:

$$4i_1 + 2\frac{di_1}{dt} + (i_1 - i_2) = 2v_x$$

And:

$$v_x = -4i_1$$

Thus:

$$4i_1 + 2\frac{di_1}{dt} + (i_1 - i_2) = -8i_1$$

Or:

$$12i_1 + 2\frac{di_1}{dt} + (i_1 - i_2) = 0$$

$$13i_1 + 2\frac{di_1}{dt} - i_2 = 0 \qquad (1)$$

For mesh 2:

$$4i_2 + (i_2 - i_1) = -2v_x$$

Or:

$$4i_2 + (i_2 - i_1) = 8i_1$$

$$5i_2 = 9i_1$$

Or:

$$i_2 = \frac{9}{5} i_1 \qquad (2)$$

Substituting (2) in (1), we find that:

$$13i_1 + 2\frac{di_1}{dt} - \frac{9}{5} i_1 = 0$$

Or:

$$56i_1 + 10\frac{di_1}{dt} = 0$$

$$\frac{di_1}{dt} = -5.6i_1$$

$$\frac{di_1}{i_1} = -5.6dt$$

$$\ln i_1 = -5.6t + C_1$$

$$i_1(t) = e^{-5.6t+C_1} = e^{C_1} e^{-5.6t}$$

$$i_1(0) = e^{C_1} = 5$$

Thus:

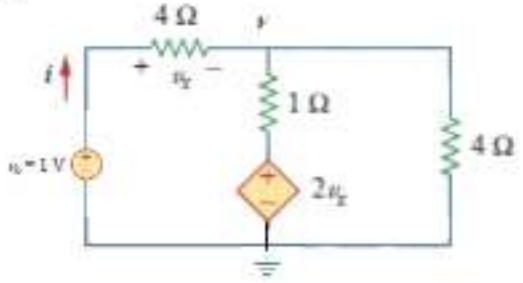
$$i(t) = i_1(t) = 5e^{-5.6t}$$

And:

$$v_x(t) = -4i(t) = -20e^{-5.6t}$$

Method 2:

The Thevenin's equivalent of the circuit as seen from the inductor terminals can be found as usual, by applying a voltage source and finding the current, (or by applying a current source and finding the voltage) as below.



Using nodal analysis:

$$\frac{V-1}{4} + \frac{V-2v_x}{1} + \frac{V}{4} = 0$$

Or:

$$V - 1 + 4V - 8v_x + V = 0 \qquad (1)$$

But:

$$v_x = 1 - V$$

Thus:

$$V-1+4V-8(1-V)+V-0$$

Or:

$$14V=9$$

Or:

$$V=\frac{9}{14}V$$

And:

$$i=\frac{1-V}{4}=\frac{5}{56}$$

And:

$$R_{eq}=\frac{1}{i}=\frac{56}{5}\Omega$$

$$\tau=\frac{L}{R_{eq}}=\frac{10}{56}s$$

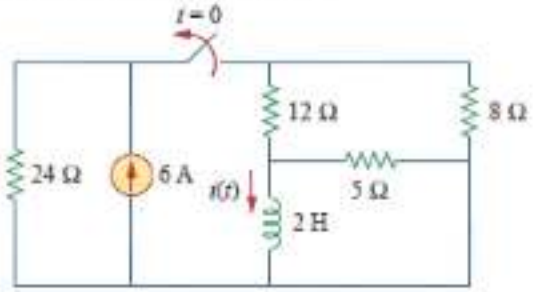
$$i(t)=I_0e^{-5.6t}=5e^{-\frac{t}{\tau}}=5e^{-5.6t}A$$

And:

$$v_x(t)=-4i(t)=-20e^{-5.6t}$$

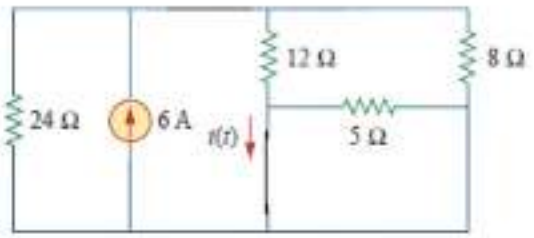
Example 2:

For the circuit in the figure below, find $i(t)$ for $t>0$.

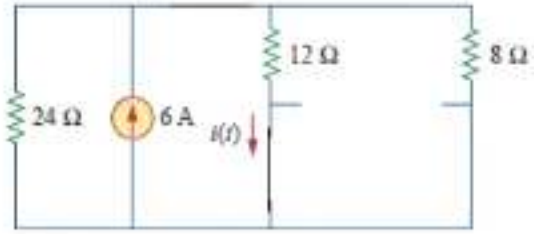


Solution:

For $t<0$, the switch was closed and the current acts as a short circuit. the circuit is as follows:



The 5Ω resistor is short circuited and the circuit is:

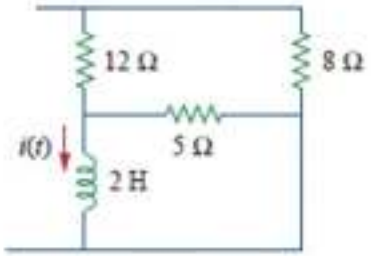


The 3 resistors are in parallel, the equivalent resistance is:

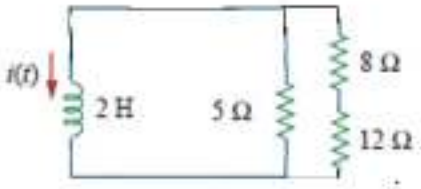
$$\frac{1}{R_{eq}} = \frac{1}{24} + \frac{1}{12} + \frac{1}{8} = \frac{6}{24} = \frac{1}{4}$$

$R_{eq} = 4\Omega$ and $V = 6x4 = 24V$ and $I_o = \frac{24}{12} = 2A$

When $t > 0$, the switch is open and the voltage source is disconnected. We now have the source-free RL circuit in the figure below.



And it is equivalent to:



From which:

$$R_{eq} = \frac{5x20}{5+20} = 4\Omega$$

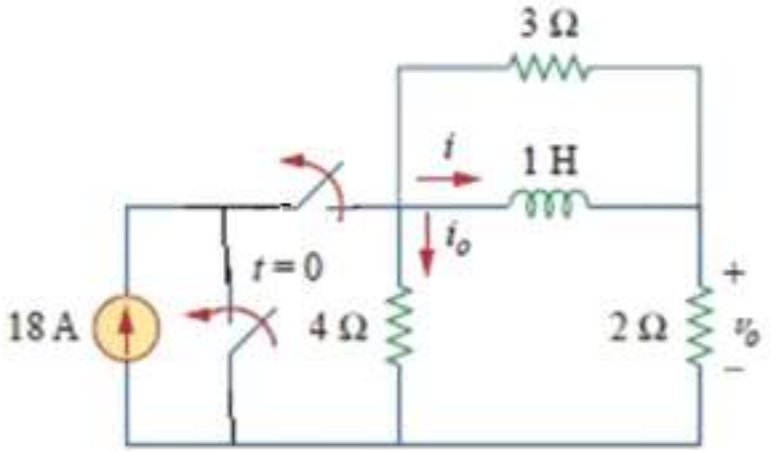
And:

$$\tau = \frac{L}{R_{eq}} = \frac{2}{4} = 0.5s$$

$$i(t) = I_0 e^{-\frac{t}{\tau}} = 2e^{-2t} A$$

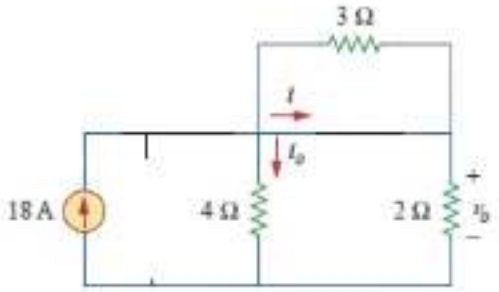
Example 3:

Determine i, i_0 and v_o for all t in the following circuit.



Solution:

When the switch was open the inductor acts as a short circuit and the equivalent circuit is as follows.



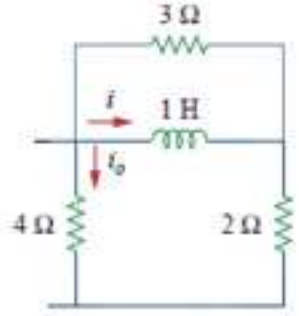
In this circuit, the 3Ω resistor is short-circuited and the 4Ω and the 2Ω resistors act as a current divider, thus:

$$i_o(t \le 0) = (18)\frac{2}{2+4} = 6A$$

$$i(t \le 0) = 18 - 6 = 12A$$

$$v_0(t \le 0) = 12x2 = 24V$$

When the switch is closed the current source is short circuited and the circuit acts as a source free, is as follows.



In this circuit:

$$R_{eq} = \frac{3(2+4)}{3+2+4} = 2\Omega$$

$$\tau = \frac{L}{R_{eq}} = \frac{1}{2} = 0.5s$$

Thus:

$$i(t) = \begin{cases} 12A & for\ t < 0 \\ 12e^{-\frac{t}{\tau}} = 12e^{-2t}A & for\ t > 0 \end{cases}$$

The voltage across the inductor is:

$$v(t) = L\frac{di}{dt} = \begin{cases} 0V & for \quad t < 0 \\ -24e^{-2t}V & for \quad t > 0 \end{cases}$$

And:

$$i_o(t) = \frac{v(t)}{6} = -4e^{-2t} \quad \text{for} \quad t > 0$$

And:

$$i_o(t) = \begin{cases} 6A & for\ t < 0 \\ -4e^{-2t}A & for\ t > 0 \end{cases}$$

And:

$$v_0(t) = -2i_o(t) = 8e^{-2t} \quad \text{for} \quad t > 0$$

And:

$$v_0(t) = \begin{cases} 24V & for\ t < 0 \\ 8e^{-2t}V & for\ t > 0 \end{cases}$$

Singularity Functions

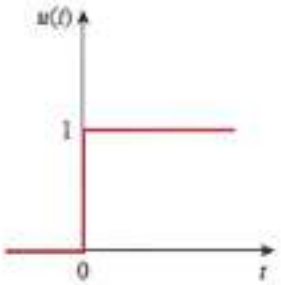
Singularity functions (also called switching functions) serve as good approximations to the switching signals that arise in circuits with switching operations.

By definition, singularity functions are functions that either are discontinuous or have discontinuous derivatives.

The three most widely used singularity functions are the unit step, the unit impulse, and the unit ramp functions.

The unit step function $u(t)$: The unit step function u(t) is 0 for negative values of t and 1 for positive values of t, or:

$$u(t) = \begin{cases} 0, & for \quad t < 0 \\ 1, & for \quad t > 0 \end{cases}$$, see the figure below.

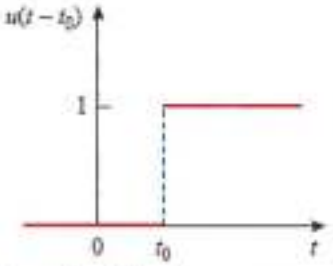


The unit step function is undefined at $t = 0$, where it changes abruptly from 0 to 1.

If the abrupt change occurs at $t = t_0$ (where $t_0 > 0$) instead of $t = 0$, the unit step function becomes:

$$u(t - t_0) = \begin{cases} 0, & for \quad t < t_0 \\ 1, & for \quad t > t_0 \end{cases}$$

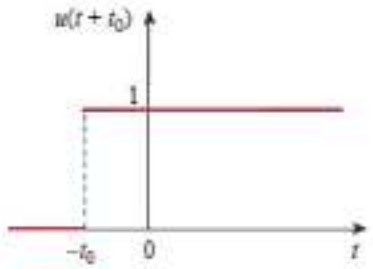
which is the same as saying that $u(t)$ is delayed by t_0 seconds, as shown in the following figure.



If the change is at $t = -t_0$, the unit step function becomes:

$$u(t + t_0) = \begin{cases} 0, & for \quad t < -t_0 \\ 1, & for \quad t > -t_0 \end{cases}$$

meaning that is advanced by t_0 seconds, as shown in the following figure.



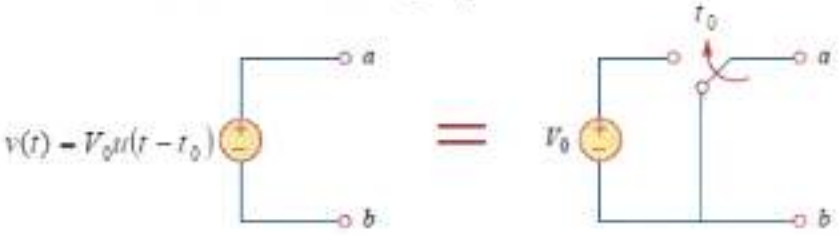
The step function can be used to represent an abrupt change in voltage or current. For example, the voltage:

$$v(t) = \begin{cases} 0, & for \quad t < t_0 \\ V_0, & for \quad t > t_0 \end{cases}$$

Can be expressed as:

$v(t) = V_0 u(t - t_0)$

And the equivalent circuit is as in the following figure:



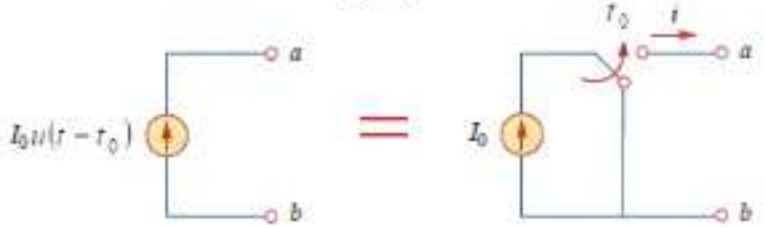
Similarly, the current:

$$i(t) = \begin{cases} 0, & for \quad t < t_0 \\ I_0, & for \quad t > t_0 \end{cases}$$

Can be expressed as:

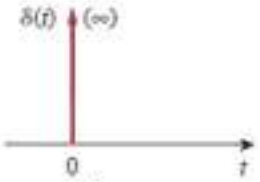
$i(t) = I_0 u(t - t_0)$

And the equivalent circuit is as in the following figure:



The unit impulse function, (delta function), $\delta(t)$: The unit impulse function $\delta(t)$ is the derivative of the unit step function $u(t)$, or:

$$\delta(t) = \frac{d}{dt} u(t) = \begin{cases} 0, & t < 0 \\ undefined, & t = 0 \\ 0, & t > 0 \end{cases}$$, see the following figure.

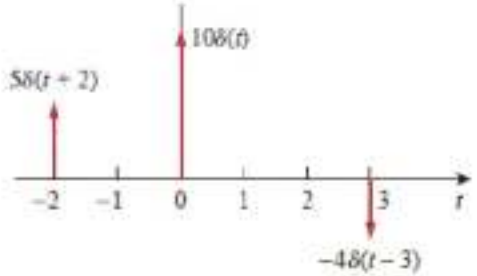


The unit impulse function $\delta(t)$ is zero everywhere except at $t = 0$, where it is undefined.

For the unit impulse function:

$$\int_{0^-}^{0^+} \delta(t).dt = 1$$

The following figure shows the impulse functions $5\delta(t+2)$, $10\delta(t)$ and $-4\delta(t-3)$.



To illustrate how the impulse function affects other functions, let us evaluate the integral

$\int_a^b f(t).\delta(t-t_0)dt$, *where* $a < t_0 < b$, thus:

$$\int_a^b f(t).\delta(t-t_0)dt = \int_a^b f(t_0).\delta(t-t_0)dt = f(t_0)\int_a^b \delta(t-t_0)dt = f(t_0)$$

Or:

$$\int_a^b f(t).\delta(t-t_0)dt = f(t_0)$$

This shows that when a function is integrated with the impulse function, we obtain the value of the function at the point where the impulse occurs. This is a highly useful property of the impulse function known as the sampling or sifting property. The special case is for $t_0 = 0$, or:

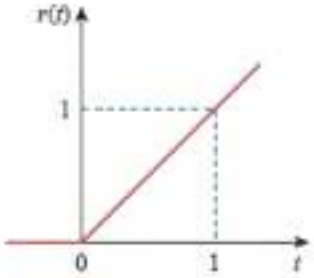
$$\int_{0^-}^{0^+} f(t).\delta(t)dt = f(0)$$

The unit ramp function, $r(t)$: the unit ramp function is the integral of the unit step function $u(t)$, or:

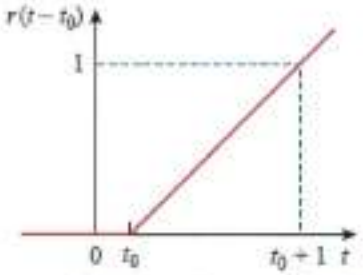
$r(t) = \int_{-\infty}^{t} u(t)dt = tu(t)$, or:

$$r(t) = \begin{cases} 0, & t \le 0 \\ t, & t \ge 0 \end{cases}$$

Or, the unit ramp function is zero for negative values of t and has a unit slope for positive values of t, see the following figure.

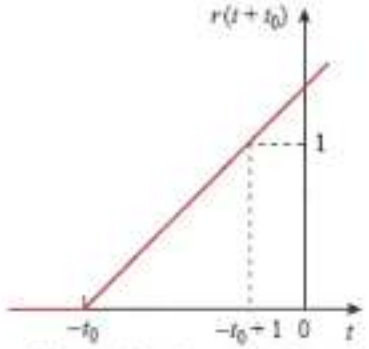


The unit ramp function may be delayed or advanced as shown in the following figure.



delayed unit ramp function.

$$r(t-t_0)=\begin{cases}0, & t\le t_0\\ t-t_0, & t\ge t_0\end{cases}$$



advanced unit ramp function

$$r(t+t_0)=\begin{cases}0, & t\le -t_0\\ t+t_0, & t\ge -t_0\end{cases}$$

It has to be kept in mind that:

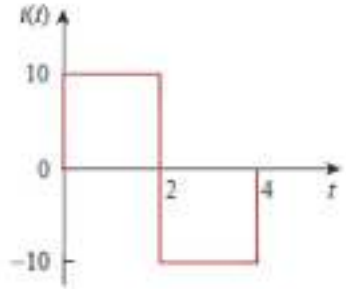
$$\delta(t)=\frac{d}{dt}u(t) \text{ and } u(t)=\frac{d}{dt}r(t)$$

Also:

$$u(t)=\int_{-\infty}^{t}\delta(t)dt \text{ and } r(t)=\int_{-\infty}^{t}u(t)dt$$

Example 1:

Express the current pulse in the figure below in terms of the unit step. Find its and its integral and sketch them.



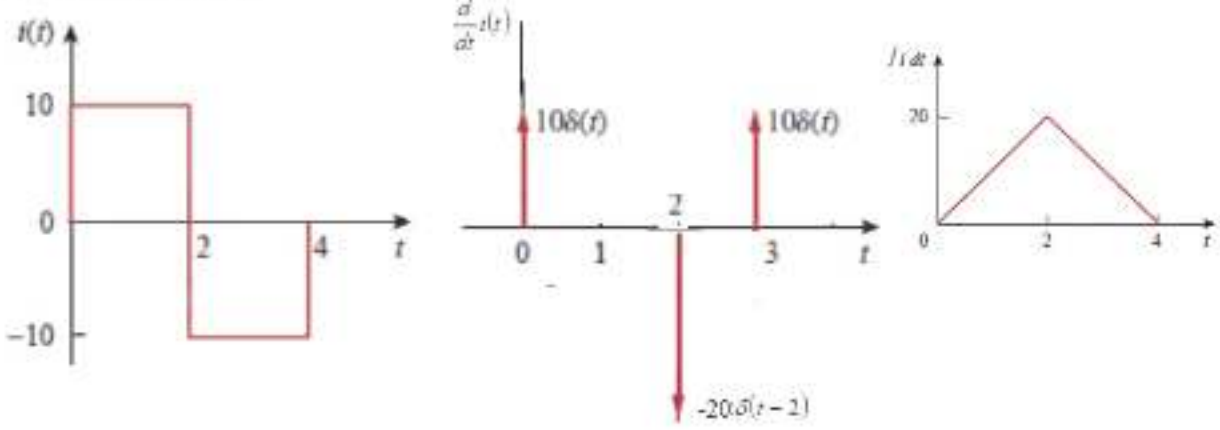
Solution:

$$i(t) = 10u(t) - 20u(t-2) + 10u(t-4) = 10[u(t) - 2u(t-2) + u(t-4)]$$

$$\frac{d}{dt}i(t) = 10[\delta(t) - 2\delta(t-2) + \delta(t-4)]$$

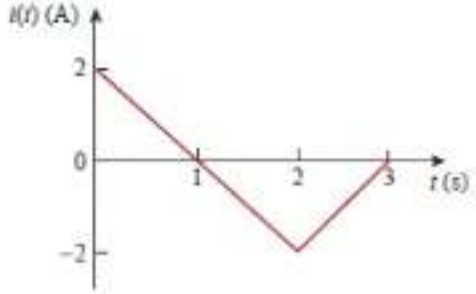
$$\int_{-\infty}^{t} i(t)dt = 10[r(t) - 2r(t-2) + r(t-4)]$$

See the figure below.



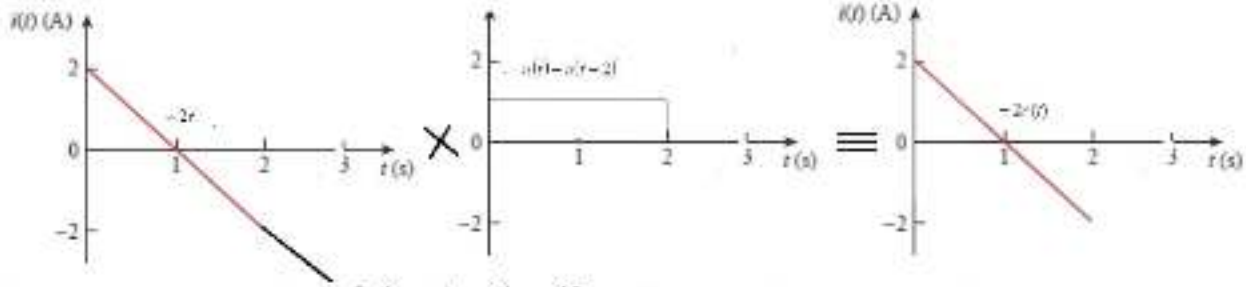
Example 2:

Express the function shown in terms of singularity functions.

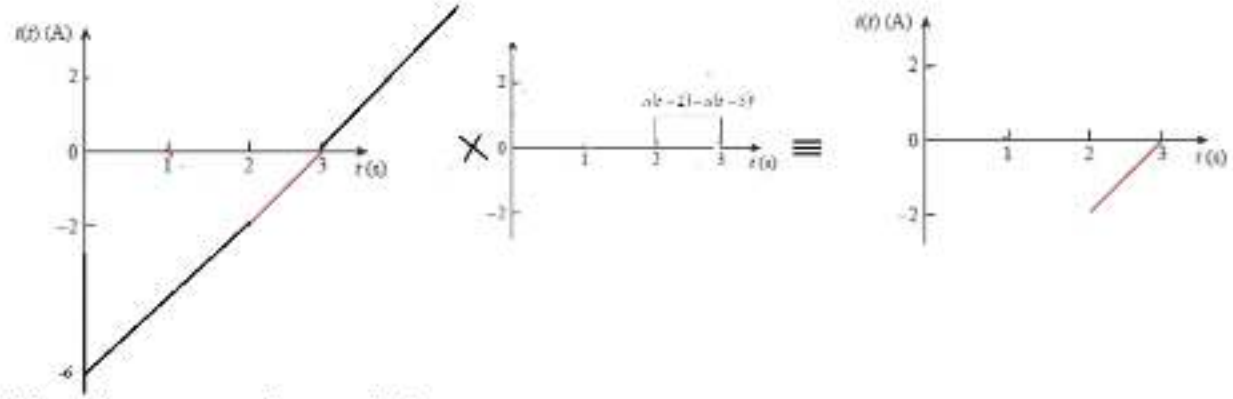


Solution:

The first slope can be expressed as: $-2t+2$. As it is between 0 and 2, thus we can write it as $(2-2t)[u(t)-u(t-2)]$. This is clear from the following figure.



The second slope is $(2t-6)[u(t-2)-u(t-3)]$ and is shown in the figure below.



Adding these two results, we find:

$$
\begin{aligned}
i(t) &= (2-2t)[u(t)-u(t-2)]-(2t-6)[u(t-2)-u(t-3)] \\
&= 2u(t)-2u(t-2)-2tu(t)+2tu(t-2)+2tu(t-2)-6u(t-2)-2tu(t-3)+6u(t-3) \\
&= 2u(t)-8u(t-2)-2tu(t)+4tu(t-2)-2tu(t-3)+6u(t-3) \\
&= 2u(t)-8u(t-2)-2tu(t)+4(t-2+2)u(t-2)-2(t-3-3)u(t-3)+6u(t-3) \\
&= 2u(t)-8u(t-2)-2tu(t)+4(t-2)u(t-2)+8u(t-2)-2(t-3)u(t-3)-6u(t-3)+6u(t-3) \\
&= 2u(t)-2tu(t)+4(t-2)u(t-2)-2(t-3)u(t-3) \\
&= 2u(t)-2r(t)+4r(t-2)-2r(t-3)
\end{aligned}
$$

Example 3:

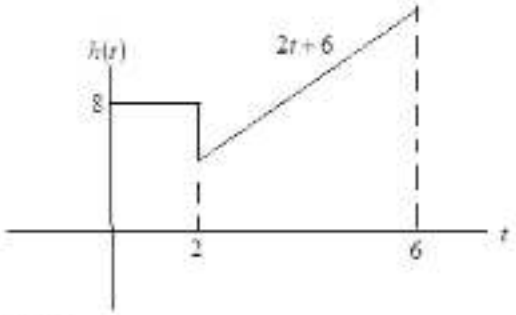
If:

$$
h(t)=\begin{cases} 0, & t<0 \\ 8, & 0<t<2 \\ 2t+6, & 2<t<6 \\ 0, & t>6 \end{cases}
$$

express $h(t)$ in terms of the singularity functions.

Solution:

This function is shown in the figure below.



This function can be expressed as:

$$h(t)=8[u(t)-u(t-2)]+(2t+6)[u(t-2)-u(t-6)]$$

$$
\begin{aligned}
h(t) &= 8u(t) - 8u(t-2) + 2tu(t-2) + 6u(t-2) - 2tu(t-6) - 6u(t-6) \\
&= 8u(t) - 8u(t-2) + 2(t-2+2)u(t-2) + 6u(t-2) - 2(t-6-6)u(t-6) - 6u(t-6) \\
&= 8u(t) - 8u(t-2) + 2(t-2)u(t-2) + 4u(t-2) + 6u(t-2) - 2(t-6)u(t-6) - 12u(t-6) - 6u(t-6) \\
&= 8u(t) + 2u(t-2) + 2(t-2)u(t-2) - 2(t-6)u(t-6) - 18u(t-6) \\
&= 8u(t) + 2u(t-2) - 18u(t-6) + 2r(t-2) - 2r(t-6)
\end{aligned}
$$

Example 4:

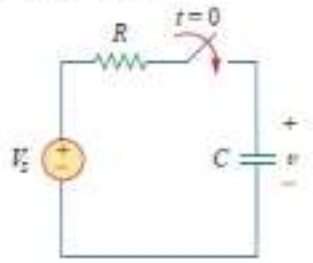
Evaluate the following integrals: $\int_{-\infty}^{\infty}(t^3+5t^2+10)\delta(t+3)dt$, $\int_{0}^{10}\delta(t-\pi)\cos 3t\,dt$

$$\int_{-\infty}^{\infty}(t^3+5t^2+10)\delta(t+3)dt = (t^3+5t^2+10)\Big|_{t=-3} = -27+45+10 = 28$$

$$\int_{0}^{10}\delta(t-\pi)\cos 3t\,dt = \cos 3t\Big|_{t=\pi} = -1$$

Step Response of an RC Circuit

The step response is the response of the circuit due to a sudden application of a dc voltage or current source. Consider the RC circuit shown.



For this circuit:

$$Ri + v_C = V_S \qquad (1)$$

And:

$$i = C\frac{dv_C}{dt} \qquad (2)$$

From (1) and (2):

$$RC\frac{dv_C}{dt} + v_C = V_S \qquad (3)$$

$$RC\frac{dv_C}{dt} = V_S - v_C$$

$$\frac{dv_C}{V_S - v_C} = \frac{1}{RC}dt$$

$$\frac{d(V_S - v_C)}{V_S - v_C} = -\frac{1}{RC}dt$$

$$\ln(V_S - v_C) = -\frac{1}{RC}t + C_1$$

$$V_S - v_C = e^{-\frac{1}{RC}t - C_1} = e^{C_1}e^{-\frac{1}{RC}t}$$

$$v_C = V_S - e^{C_1}e^{-\frac{1}{RC}t} \qquad (4)$$

Assume that $v_C(0) = V_0$

Thus:

$$V_0 = V_S - e^{C_1}$$

Or:

$$e^{C_1} = V_S - V_0 \qquad (5)$$

Substituting (5) in (4),

$$v_C(t) = V_S - (V_S - V_0)e^{-\frac{1}{RC}t} = V_S + (V_0 - V_S)e^{-\frac{1}{RC}t} = V_S + (V_0 - V_S)e^{-\frac{t}{\tau}} \qquad (6)$$

Note that, from this equation:

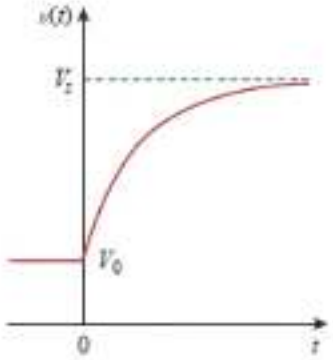
$v_c(0) = V_0$

And for $t \geq 0$, $v_C(t) = V_S + (V_0 - V_S)e^{-\frac{t}{\tau}} = V_0 e^{-\frac{t}{\tau}} - V_S\left(1 - e^{-\frac{t}{\tau}}\right)$

Or:

$$v_C(t) = \begin{cases} V_0 & t < 0 \\ V_0 e^{-\frac{t}{\tau}} - V_S\left(1 - e^{-\frac{t}{\tau}}\right) & t > 0 \end{cases}$$

And is shown in the figure below.



The previous equation, shows that the complete response of this circuit has two components $v_c(t) = v_n + v_f$

Where:

- v_n is the free response caused by the initial conditions, $v_n(t) = V_0 e^{-\frac{t}{\tau}}$
- v_f, the forced response caused by the supply, $v_f(t) = V_S\left(1 - e^{-\frac{t}{\tau}}\right)$

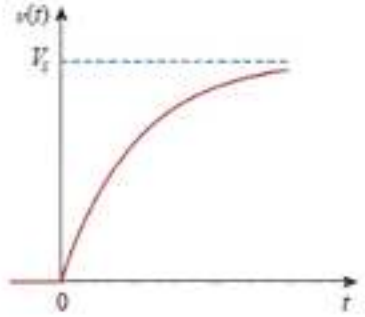
If the capacitor is initially uncharged, or if $v_c(0) = 0$, thus:

$$v_c(t) = V_S\left(1 - e^{-\frac{t}{\tau}}\right)$$

Or:

$$v_C(t) = V_S\left(1 - e^{-\frac{t}{\tau}}\right)u(t)$$

And is shown in the figure below.



Another way of looking at the complete response is to break into two components—one temporary and the other permanent, i.e.

for $t > 0$, $v_c(t) = V_0 e^{-\frac{t}{\tau}} + V_s\left(1 - e^{-\frac{t}{\tau}}\right) = (V_0 - V_s)e^{-\frac{t}{\tau}} + V_s$

or:

Complete response = transient response (*temporary part*) + steady-state response (*permanent part*).

Or:

$v_C(t) = v_t + v_{ss}$

Where:

$v_t = (V_0 - V_s)e^{-\frac{t}{\tau}}$, and

$v_{ss} = V_s$

Note that, the transient response is temporary and decays to zero as time approaches infinity. The steady-state response is the portion of the complete response that remains after the transient response has died out. Thus, the steady-state response is the behavior of the circuit a long time after an external excitation is applied.

Note also that, the first decomposition of the complete response is in terms of the source of the responses, while the second decomposition is in terms of the permanency of the responses.

Note also that the capacitor voltage can also be written as:

$v_c(t) = v(\infty) + [v(0) - v(\infty)]e^{-\frac{t}{\tau}}$

Where $v(0)$ is the initial voltage at $t = 0^+$ and $v(\infty)$ is the final or steady state value.

Note that if the switch changes position at time $t = t_0$ instead of at $t = 0$, there is a time delay in the response so that the above equation becomes:

$v_c(t) = v(\infty) + [v(t_0) - v(\infty)]e^{-\frac{(t-t_0)}{\tau}}$

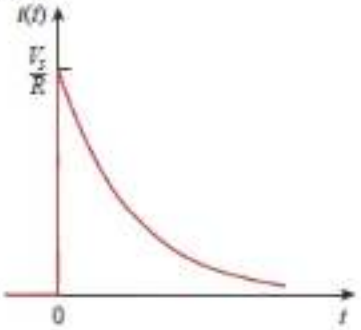
The current through the capacitor is:

$$i(t) = C\frac{dv_C}{dt} = C\frac{d}{dt}\left(V_S - V_S e^{-\frac{t}{\tau}}\right) = \frac{C}{\tau}V_S e^{-\frac{t}{\tau}} = \frac{C}{RC}V_S e^{-\frac{t}{\tau}} = \frac{V_S}{R}e^{-\frac{t}{\tau}}$$

Or:

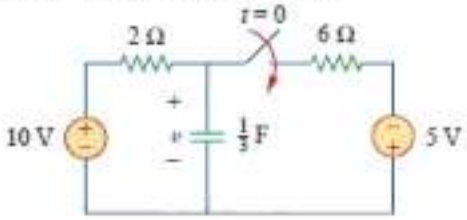
$$i(t) = \frac{V_S}{R}e^{-\frac{t}{\tau}}u(t)$$

And is shown in the figure below.



Example 1:

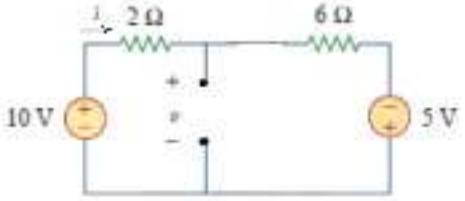
For the circuit of the figure below, find $v(t)$ for $t > 0$. Assume the switch has been open for a long time and is closed at $t = 0$. Calculate $v(t)$ at $t = 0.5s$



Solution:

In this circuit, when the was open $v_C(t < 0) = 10V$ and $v_C(0^-) = v_C(0^+) = 10V$

When the switch is closed, $v(\infty)$ is found from the following circuit.



From this circuit:

$$i = \frac{10+5}{2+6} = \frac{15}{8}A$$

$$v(\infty) = 10 - 2i = 10 - \frac{30}{8} = 6.25V$$

$R_{eq} = \frac{6x2}{6+2} = 1.5\Omega$ and $\tau = RC = 1.5\left(\frac{1}{3}\right) = 0.5s$, thus, for $t > 0$

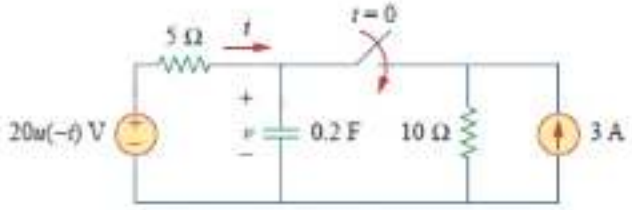
$v_c(t) = v(\infty) + [v(0) - v(\infty)]e^{-\frac{t}{\tau}} = 6.25 + (10 - 6.25)e^{-2t} = 6.25 + 3.75e^{-(1/\tau)t}V$

For $t = 0.5s$, $v_c(0.5) = 6.25 + 3.75e^{-0.5(0.5)}V = 7.63V$

Example 2:

The switch in the figure below is closed at $t = 0$. Find $i(t)$ and $v(t)$ for all time.

Note that $u(-t) = 1$ for $t < 0$ and 0 for $t > 0$. Also, $u(-t) = 1 - u(t)$

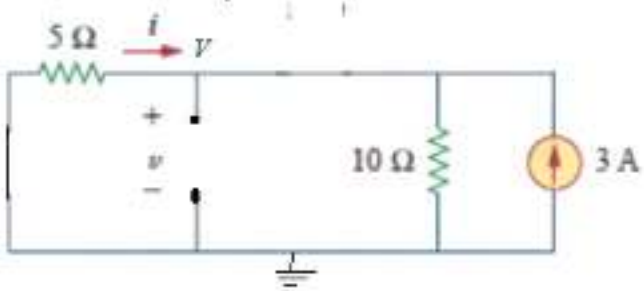


Solution:

For $t < 0$, the switch is open and $v = 20V$. Thus $v(0^+) = v(0^-) = 20V$.

And $i = 0$

When the switch is closed, and at steady state. the circuit is



For this circuit and by nodal method:

$\frac{V}{5} + \frac{V}{10} = 3$

From which, $V = 10V$, or $v(\infty) = 10V$

$R_{eq} = \frac{5x10}{5+10} = \frac{50}{15} = \frac{10}{3}\Omega$

$\tau = R_{eq}.C = \frac{10}{3}x0.2 = \frac{2}{3}s$

Thus:

$v_c(t) = v(\infty) + [v(0) - v(\infty)]e^{-\frac{t}{\tau}} = 10 + (20 - 10)e^{-1.5t} = 10 + 10e^{-1.5t}$

And $i(t) = -\frac{v(t)}{5} = -2 - 2e^{-1.5t}$

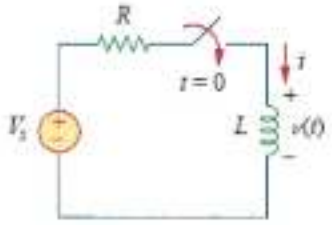
Thus:

$$v_c(t) = \begin{cases} 20V & t < 0 \\ 10 + 10e^{-1.5t} & t > 0 \end{cases}$$

$$i(t) = \begin{cases} 0 & t < 0 \\ -2 - 2e^{-1.5t} & t > 0 \end{cases}$$

Step Response of an RL Circuit

Consider the following circuit.



For this circuit:

$$Ri + L\frac{di}{dt} = V_s \qquad (1)$$

$$\frac{di}{dt} = \frac{V_s - Ri}{L}$$

$$\frac{di}{V_s - Ri} = \frac{1}{L}dt$$

$$\int \frac{di}{V_s - Ri} = \int \frac{1}{L}dt$$

$$-\frac{1}{R}\int \frac{d(V_s - Ri)}{V_s - Ri} = \int \frac{1}{L}dt$$

$$-\frac{1}{R}\ln(V_s - Ri) = \frac{1}{L}t + C_1$$

$$\ln(V_s - Ri) = -\frac{R}{L}t - RC_1 = -\frac{R}{L}t + C_2$$

$$V_s - Ri = e^{-\frac{R}{L}t + C_2} = e^{C_2}.e^{-\frac{R}{L}t}$$

$$i = \frac{V_s}{R} - \frac{e^{C_2}}{R}.e^{-\frac{R}{L}t} = \frac{V_s}{R} - C_3.e^{-\frac{R}{L}t} \qquad (2)$$

Let $i(0^-) = i(0^+) = I_0$

Thus:

$$I_0 = \frac{V_s}{R} - C_3$$

Or:

$$C_3 = \frac{V_s}{R} - I_0$$

Substituting this in (2), we find:

$$i(t) = \frac{V_s}{R} - \left(\frac{V_s}{R} - I_0\right)e^{-\frac{R}{L}t} = \frac{V_s}{R} + \left(I_0 - \frac{V_s}{R}\right)e^{-\frac{R}{L}t}$$

This result can also be found from:

$$i(t) = i(\infty) + [i(0) - i(\infty)]e^{-\frac{t}{\tau}}$$

Where:

$$i(\infty) = \frac{V_S}{R}, \quad i(0) = I_0, \qquad \tau = \frac{L}{R}.$$

Note that, if the switching takes place at $t = t_0$ instead of $t = 0$, thus:

$$i(t) = \begin{cases} 0, & t < t_0 \\ \frac{V_S}{R}\left(1 - e^{-\frac{t}{\tau}}\right) & t > t_0 \end{cases}$$

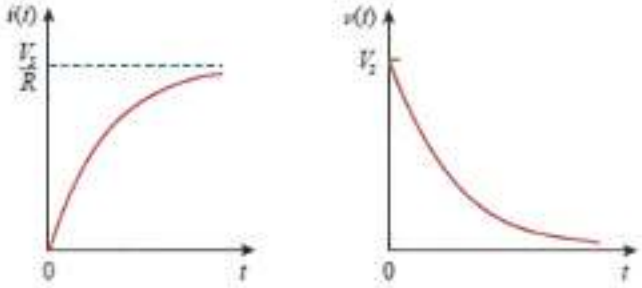
Or:

$$i(t) = \frac{V_S}{R}\left(1 - e^{-\frac{t}{\tau}}\right)u(t)$$

The voltage across the inductor is obtained from:

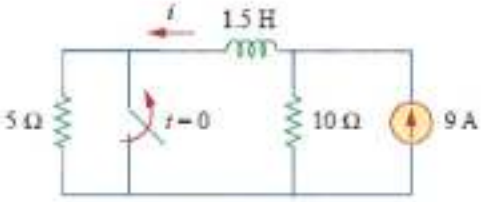
$$v(t) = L\frac{di}{dt} = \frac{L}{\tau}\frac{V_S}{R}e^{-\frac{t}{\tau}} = V_S e^{-\frac{t}{\tau}}u(t)$$

The plot of $i(t)$ and $v(t)$ are shown below.



Example 1:

The switch in the figure below has been closed for a long time. It opens at $t = 0$. Find $i(t)$ for $t > 0$.



Solution:

For $t < 0$, the 5Ω resistor is short circuited and the inductor acts as a short circuit, thus:

$$i(0^+) = i(0^-) = 9A$$

When the switch is opened, the current at infinity will be divided between the two branches, and:

$$i(\infty) = (9)\frac{10}{10+5} = 6A$$

$R_{Th} = 5 + 10 = 15\Omega$ and $\tau = \dfrac{L}{R_{Th}} = \dfrac{1.5}{15} = 0.1s$

Thus, for $t > 0$:

$$i(t) = i(\infty) + [i(0) - i(\infty)]e^{-\frac{t}{\tau}}$$

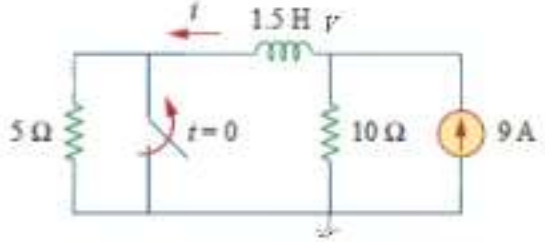
Or:

$$i(t) = 6 - [9 - 6]e^{-\frac{t}{0.1}}$$

$$i(t) = 6 + 3e^{-10t}A$$

Another method:

For $t > 0$, the switch is opened an the following equation can be written for node V.



$$\frac{V - L\dfrac{di}{dt}}{5} + \frac{V}{10} = 9$$

Or:

$$3V - 2L\frac{di}{dt} = 90 \quad (1)$$

Also:

$$V = 5i + L\frac{di}{dt} \qquad (2)$$

Substituting (2) in (1):

$$3\left(5i + L\frac{di}{dt}\right) - 2L\frac{di}{dt} = 90$$

$$15i + 1.5\frac{di}{dt} = 90$$

$$\frac{di}{dt} = \frac{90 - 15i}{1.5}$$

$$\int \frac{di}{90 - 15i} = \int \frac{1}{1.5}dt$$

$$-\frac{1}{15}\int \frac{d(90 - 15i)}{90 - 15i} = \int \frac{1}{1.5}dt$$

$$\ln(90 - 15i) = -10t + C_2$$

$90 - 15i = e^{-10t - C_3} = e^{C_3} e^{-10t}$

$i = 6 - C_3 e^{-10t}$

But $i(0^+) = i(0^-) = 9A$

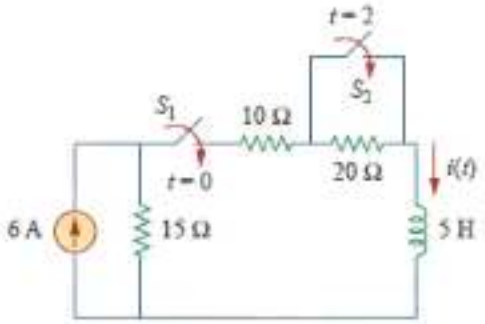
Thus:

$9 = 6 - C_3$ or $C_3 = -3$ and $i(t) = 6 + 3e^{-10t}$ which is the same result as before.

Note the differential equation can also be written using mesh analysis.

Example 2:

Switch S_1 in the following figure is closed at $t = 0$ and switch S_2 is closed at $t = 2s$. Calculate $i(t)$ for all t. Find $i(1)$ and $i(3)$.



Solution:

For $t < 0$, the two switches are open and $i(t) = 0$.

$i(0^+) = i(0^-) = 0$

When S_1 is closed $i(\infty) = 6\dfrac{15}{15+10+20} = 2A$

$R_{Th} = 20 + 10 + 15 = 45\Omega$ and $\tau = \dfrac{L}{R_{Th}} = \dfrac{5}{45} = \dfrac{1}{9}s$

Thus for $0 \le t \le 2$, $i(t) = i(\infty) + [i(0) - i(\infty)]e^{-\frac{t}{\tau}}$, or:

$i(t) = 2 - 2e^{-9t}A = 2(1 - e^{-9t})A$

At $t = 2s$, $i(2) = 2(1 - e^{-18})A \approx 2A$

$t \ge 2$, When S_2 is closed $i(2^+) = i(2) = 2A$ and:

$i(\infty) = 6\dfrac{15}{15+10} = 3.6A$

$R_{Th} = 10 + 15 = 25\Omega$ and $\tau = \dfrac{L}{R_{Th}} = \dfrac{5}{25} = \dfrac{1}{5}s$ and $i(t) = i(\infty) + [i(0) - i(\infty)]e^{-\frac{t}{\tau}}$

$i(t) = 3.6 + [2 - 3.6]e^{-5(t-2)}A$

$i(t) = 3.6 - 1.6e^{-5(t-2)}A$

Thus:

$$i(t) = \begin{cases} 0, & t \le 0 \\ 2\left(1 - e^{-0.5t}\right)A & 0 \le t \le 2 \\ 3.6 - 1.6e^{-0.5(t-2)}A & 0 \ge 2 \end{cases}$$

$$i(1) = 2\left(1 - e^{-0.5}\right) = 1.9997A$$

$$i(3) = 3.6 - 1.6e^{-0.5(3-2)} = 3.589A$$

Second-Order Circuits

second-order circuits are those which responses are described by differential equations that contain second derivatives. These circuits have two storage elements.

Typical examples of second-order circuits are RLC circuits, in which the three kinds of passive elements are present. Examples of such circuits are series and parallel RLC circuits. Other examples are RL and RC circuits, provided elements of the same type cannot be represented by an equivalent single element.

Finding Initial and Final Values

Finding the initial and the final values of the voltages of the capacitors and the currents of the inductors and their derivatives, are the kay points for finding the solutions of the second-order circuits.

Two key points have to be kept in mind:

a) the capacitor voltage is always continuous so that:

$$v_c(0^+) = v_c(0^-)$$

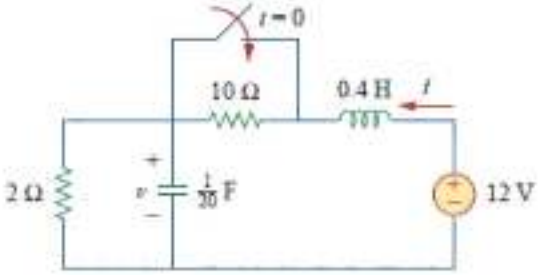
b) the inductor current is always continuous so that:

$$i_L(0^-) = i_L(0^+)$$

where $t(0^-)$ denotes the time just before a switching event and $t(0^+)$ is the time just after the switching event, assuming that the switching event takes place at $t = 0$.

Example 1:

The switch in the following figure was open for a long time but closed at $t = 0$.



Determine:

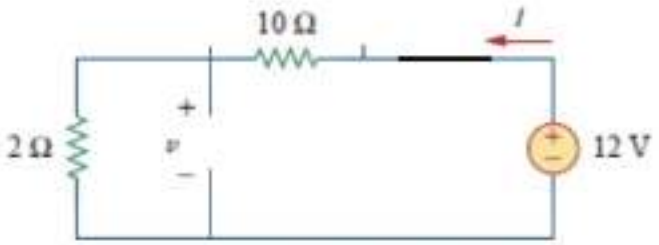
(a) $i(0^-)$, $v(0^+)$

(b) $\frac{di(0^+)}{dt}$, $\frac{dv(0^+)}{dt}$

(c) $i(\infty)$, $v(\infty)$

Solution

For $t = 0^-$, the equivalent circuit is as in the following figure.



From this figure:

$$i(0^-) = \frac{12}{2+10} = 1A$$

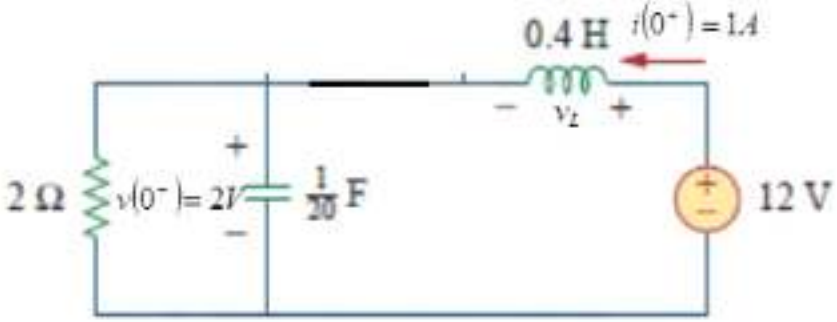
$$v(0^-) = \frac{12(2)}{2+10} = 2V$$

Thus:

$$i(0^+) = i(0^-) = 1A$$

$$v(0^+) = v(0^-) = 2V$$

For $t = 0^+$, the equivalent circuit is as in the following figure.



From this figure:

$$v_L = 12 - 2 = 10V$$

But:

$$v_L = L\frac{di}{dt}, \text{ or :}$$

$$\frac{di(0^+)}{dt} = \frac{v_L(0^+)}{L} = \frac{10}{0.4} = 25A/s$$

Also:

$$i_c(0^-) = i(0^-) - \frac{v_c(0^+)}{2} = 1 - \frac{2}{2} = 0A$$

But:

$$i_c(0^-) = C\frac{dv(0^-)}{dt}, \text{ or :}$$

$$\frac{dv(0^-)}{dt} = \frac{i_C(0^+)}{C} = 0V/s$$

At $t \to \infty$, the equivalent circuit is:



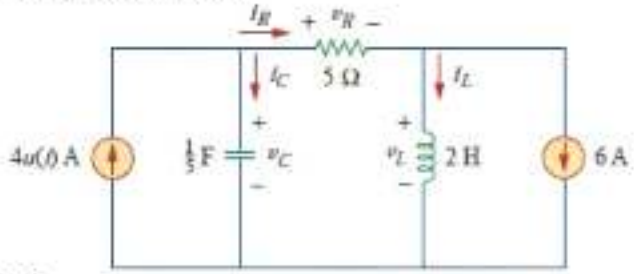
Thus:

$$i(\infty) = \frac{12}{2} = 6A$$

$$v(\infty) = 12V$$

Example 2:

For the circuit in the following figure, find:



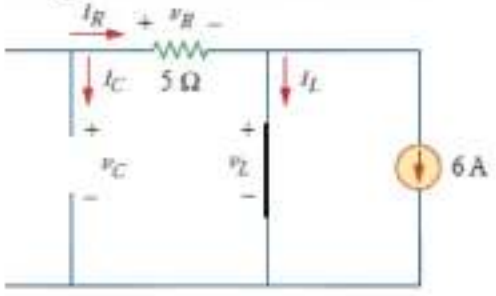
(a) $i_L(0^-)$, $v_C(0^-)$, $v_R(0^-)$

(b) $\frac{di_L(0^-)}{dt}$, $\frac{dv_C(0^+)}{dt}$, $\frac{dv_R(0^+)}{dt}$

(c) $i_L(\infty)$, $v_C(\infty)$, $v_R(\infty)$

Solution:

a) For $t = 0^-$, $u(0^-) = 0$ and the equivalent circuit is as below.



From this circuit:

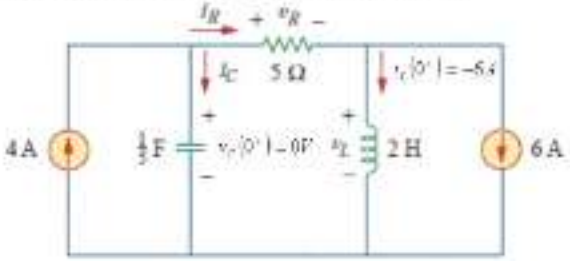
$i_L(0^-) = -6A$, $v_c(0^-) = 0V$, thus:

$i_L(0^+) = i_L(0^-) = -6A$

$v_c(0^+) = v_c(0^-) = 0V$

As $i_L(0^-) = -6A$, thus $i_x(0^+) = 0A$ and $v_x(0^-) = 0V$

b) For $t = 0^+$, $u(0^+) = 1$ and the equivalent circuit is as below.



For this circuit:

As $v_c(0^+) = 0V$, thus the 4A source is shorted by the capacitor and:

$4 = C\dfrac{dv_c(0^-)}{dt}$ and $\dfrac{dv_c(0^+)}{dt} = \dfrac{4}{C} = 20V/s$

For the middle loop:

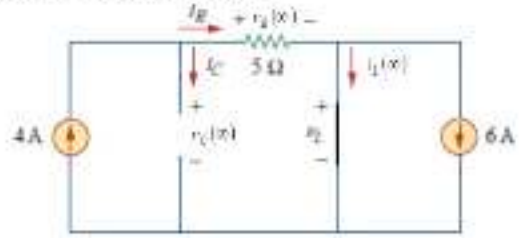
$v_c(0^+) - 5i_x - v_L = 0$ or $0 - 5(0) - v_L = 0$, or $v_L = 0$, or $L\dfrac{di_L(0^-)}{dt} = 0$, or $\dfrac{di_L(0^-)}{dt} = 0$.

$i_R(0^-) = i_L(0^-) + 6$

Thus:

$$\frac{di_R(0^+)}{dt} = \frac{di_L(0^+)}{dt} = 0$$

c) For $t \to \infty$, the equivalent circuit is as below.



Thus:

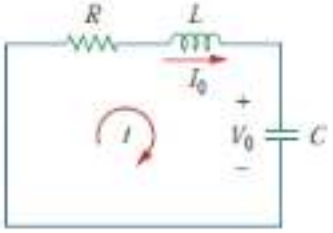
$i_L(\infty) = -6 + 4 = -2A$

$v_c(\infty) - 5(4) = 0$, or $v_c(\infty) = 20V$

$v_x(\infty) = 5(4) = 20V$

The Source-Free Series RLC Circuit

Consider the series RLC circuit shown below. The circuit is being excited by the energy initially stored in the capacitor and the inductor. The energy is represented by the initial capacitor voltage V_0 and initial inductor current I_0.



Applying KVL around the loop:

$$Ri + L\frac{di}{dt} + \frac{1}{C}\int_{-\infty}^{t} i\,dt = 0 \qquad (1)$$

Differentiating with respect to t:

$$R\frac{di}{dt} + L\frac{d^2i}{dt^2} + \frac{1}{C}i = 0$$

Or:

$$\frac{d^2i}{dt^2} + \frac{R}{L}\frac{di}{dt} + \frac{1}{LC}i = 0 \qquad (2)$$

The characteristic equation of this second-order differential equation is:

$$s^2 + \frac{R}{L}s + \frac{1}{LC} = 0 \qquad (3)$$

The roots of this characteristic equation are:

$$s_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = -\frac{R}{2L} \pm \frac{\sqrt{\left(\frac{R}{L}\right)^2 - \frac{4}{LC}}}{2} = -\frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$

Or:

$$s_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}} \qquad (4)$$

$$s_2 = -\frac{R}{2L} - \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}} \qquad (5)$$

Assuming:

$$\alpha = \frac{R}{2L} \quad \text{and} \quad \omega = \frac{1}{\sqrt{LC}}$$

The two roots can be expressed as:

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} \qquad (6)$$

$$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} \qquad (7)$$

These two roots are usually called the natural frequencies of the circuit and are measured in nepers per second (Np/s), because they are associated with the natural response of the circuit.

$\alpha = \dfrac{R}{2L}$ is called the damping factor (or the neper frequency) and is expressed in nepers per second.

$\omega_0 = \dfrac{1}{\sqrt{LC}}$ is known as the resonant frequency (or the undamped natural frequency) and is expressed in radians per second (rad/s).

The solution of the differential equation is:

$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$

The constants A_1 and A_2 are determined from the initial values $i(0)$ and $\dfrac{di(0)}{dt}$.

Note that, as stated before, $i(0) = I_0$; and $\dfrac{di(0)}{dt}$ can be found from equation (1), re-written below.

$$Ri + L\frac{di}{dt} + \frac{1}{C}\int_{-\infty}^{t} i\,dt = 0$$

From this equation, re-written at $t = 0$:

$$RI_0 + L\frac{di(0)}{dt} + V_0 = 0$$

Or:

$$\frac{di(0)}{dt} = -\frac{1}{L}(RI_0 + V_0)$$

The roots of the characteristic equation described by (6) and (7), implies that there are three types of solutions of the differential equation, these are:

a) overdamped solution, for which $\alpha > \omega_0$ and the two roots are real.

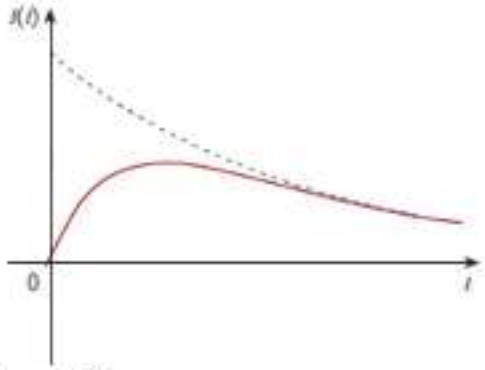
b) critically damped solution, for which $\alpha = \omega_0$ and the two roots are real and equal.

a) underdamped solution, for which $\alpha < \omega_0$ and the two roots are complex.

These three cases are discussed below.

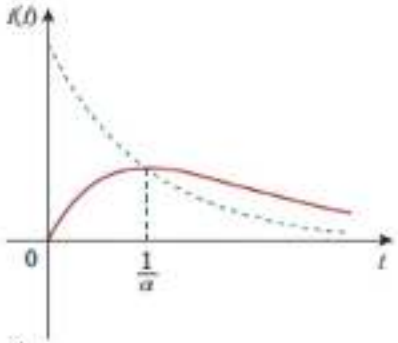
Overdamped solution, $(\alpha > \omega_0)$

For this case $\alpha > \omega_0$, or $\dfrac{R}{2L} > \dfrac{1}{\sqrt{LC}}$, or $\dfrac{R}{2\sqrt{L}} > \dfrac{1}{\sqrt{C}}$, or $\dfrac{1}{\sqrt{C}} < \dfrac{R}{2\sqrt{L}}$ $C > \dfrac{4L}{R^2}$. For this case the two roots are real and negative and the response is $i(t) = A_1 e^{s_1 t} - A_2 e^{s_2 t}$ which decays and approaches zero as t increases, as shown below.



Critically damped solution, $(\alpha = \omega_0)$

or $\frac{R}{2L} = \frac{1}{\sqrt{LC}}$, or $\frac{R}{2\sqrt{L}} = \frac{1}{\sqrt{C}}$, or $\frac{1}{\sqrt{C}} = \frac{R}{2\sqrt{L}}$ $C = \frac{4L}{R^2}$. For this case the two roots are real and equal, or $s_1 = s_2 = -\alpha = -\frac{R}{2L}$ and the response is $i(t) = (A_1 + A_2 t)e^{-\alpha t}$ and is shown in the figure below.



Underdamped solution, $(\alpha < \omega_0)$

For this case $\alpha < \omega_0$, or $\frac{R}{2L} < \frac{1}{\sqrt{LC}}$, or $\frac{R}{2\sqrt{L}} < \frac{1}{\sqrt{C}}$, or $\frac{1}{\sqrt{C}} > \frac{R}{2\sqrt{L}}$ $C < \frac{4L}{R^2}$. For this case the two roots are complex, and:

$s_{1,2} = -\alpha \pm \sqrt{-(\omega_0^2 - \alpha^2)} = -\alpha \pm j\omega_d$

Where:

$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$ and is called the damping frequency.

The solution is:

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$
$$= A_1 e^{(-\alpha + j\omega_d)t} + A_2 e^{(-\alpha - j\omega_d)t} = A_1 e^{-\alpha t} e^{j\omega_d t} + A_2 e^{-\alpha t} e^{-j\omega_d t}$$
$$= e^{-\alpha t}\left(A_1 e^{j\omega_d t} + A_2 e^{-j\omega_d t}\right) = e^{-\alpha t}\left[A_1(\cos\omega_d t + j\sin\omega_d t) + A_2(\cos\omega_d t - j\sin\omega_d t)\right]$$
$$= e^{-\alpha t}\left[(A_1 + A_2)\cos\omega_d t + j(A_1 - A_2)\sin\omega_d t\right]$$
$$= e^{-\alpha t}\left[B_1\cos\omega_d t + B_2\sin\omega_d t\right]$$

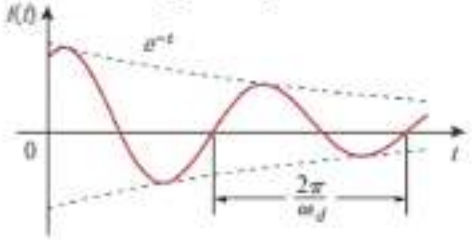
Where:

$B_1 = A_1 + A_2$ and $B_2 = j(A_1 - A_2)$

it is clear that the natural response for this case is exponentially damped and oscillatory in nature.

The response has a time constant of $\frac{1}{\alpha}$ and a period of $T = \frac{2\pi}{\omega_d}$.

The figure below depicts a typical underdamped response.



Important Notes:

1) Damping which is continuous decrease in the amplitude of the response is caused by the gradual loss of the initial stored energy. It is caused by the presence of resistance R. The damping factor α determines the rate at which the response is damped. If $R = 0$ then $\alpha = 0$ and we have an LC circuit with $\frac{1}{\sqrt{LC}}$ as the undamped natural frequency. Since $\alpha < \omega_0$ in this case, the response is not only undamped but also oscillatory. The circuit is said to be loss-less, because the dissipating or damping element (R) is absent. By adjusting the value of R, the response may be made undamped, overdamped, critically damped, or underdamped.

2) Oscillatory response is possible due to the presence of the two types of storage elements. Having both L and C allows the flow of energy back and forth between the two.

3) In general, it is difficult to tell from the waveforms the difference between the overdamped and critically damped responses. The critically damped case is the borderline between the underdamped and overdamped cases and it decays the fastest. With the same initial conditions, the overdamped case has the longest settling time, because it takes the longest time to dissipate the initial stored energy. If we desire the response that approaches the final value most rapidly without oscillation or ringing, the critically damped circuit is the right choice.

Example 1:

For the series RLC circuit, If $R = 10\Omega$, $L = 5H$ and $C = 2mF$, find α, ω_0, s_1 and s_2. What type of natural response will the circuit have?

$$\alpha = \frac{R}{2L} = \frac{10}{2(5)} = 1s^{-1}$$

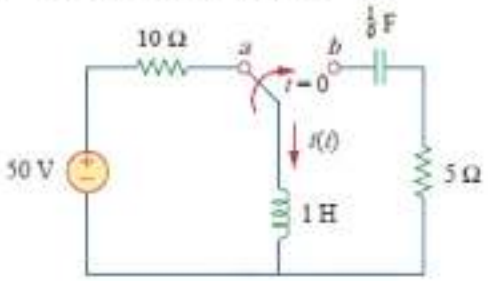
$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{5(2x10^{-3})}} = 10rad/s$$

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -1 \pm \sqrt{1^2 - 10^2} = -1 \pm j9.95$$

The natural response is underdamped.

Example 2:

The circuit in the following figure has reached steady state at $t = 0^-$. If the make-before-break switch moves to position b at $t = 0$, calculate $i(t)$ for $t > 0$.



Solution:

For $t < 0$, the switch was in position a and the inductor acts as a short-circuit, the current through the inductor is:

$$i(0^-) = \frac{50}{10} = 5A$$

For $t > 0$, the switch is in position b with $i(0^+) = i(0^-) = 5A$

$$\alpha = \frac{R}{2L} = \frac{5}{2(1)} = 2.5$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1\left(\frac{1}{9}\right)}} = 3$$

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -2.5 \pm \sqrt{(2.5)^2 - 3^2} = -2.5 \pm j1.6583$$

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} = A_1 e^{(-2.5 + j1.6583)t} + A_2 e^{(-2.5 - j1.6583)t}$$

$$i(t) = A_1 e^{-2.5t} e^{j1.6853t} + A_2 e^{-2.5t} e^{-j1.6853t}$$

$$i(t) = A_1 e^{-2.5t}(\cos 1.6853t + j \sin 1.6853t) + A_2 e^{-2.5t}(\cos 1.6853t - j \sin 1.6853t)$$

$$i(t) = (A_1 + A_2) e^{-2.5t} \cos 1.6853t + j(A_1 - A_2) e^{-2.5t} \sin 1.6853t$$

$$i(t) = B_1 e^{-2.5t} \cos 1.6853t + B_2 e^{-2.5t} \sin 1.6853t \qquad (1)$$

B_1 and B_2 are found from the initial conditions:

$$i(0^+) = i(0^-) = 5A \qquad (2)$$

$\frac{di(0^+)}{dt}$ is found from the right-hand loop formed when the switch is in position b, or:

$$I.\frac{di(0^-)}{dt} + v_C(0^-) + 5i(0^-) = 0$$

But: $v_C(0^+) = v_C(0^-) = 0$, thus:

$$I.\frac{di(0^-)}{dt} + 5(5) = 0$$

Or:

$$\frac{di(0^-)}{dt} = -\frac{25}{1} = -25A/s \qquad (2)$$

Substituting conditions (2) and (3) in equation (1), we find:

$$5 = B_1$$

$$\frac{di(t)}{dt} = B_1\left(-1.6853e^{-2.5t}\sin 1.6853t - 2.5\cos 1.6853t.e^{-2.5t}\right) + B_2\left(1.6853e^{-2.5t}\cos 1.6853t - 2.5\sin 1.6853t.e^{-2.5t}\right)$$

$$-25 = B_1(-2.5) + B_2(1.6853)$$

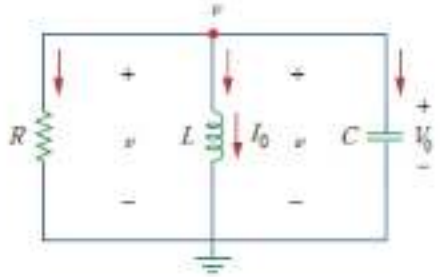
$$B_2 = \frac{-25 + 5(2.5)}{1.6853} = -7.417$$

Thus:

$$i(t) = 5e^{-2.5t}\cos 1.6853t - 7.417e^{-2.5t}\sin 1.6853t \quad A$$

The Source-Free Parallel RLC Circuit

Consider the parallel RLC circuit shown below. The circuit is being excited by the energy initially stored in the capacitor and the inductor. The energy is represented by the initial capacitor voltage V_0 and initial inductor current I_0.



Applying KCL at the top node:

$$\frac{v}{R} + \frac{1}{L}\int_{-\infty}^{t} v\,dt + C\frac{dv}{dt} = 0 \qquad (1)$$

Differentiating with respect to t:

$$\frac{1}{R}\frac{dv}{dt} + \frac{1}{L}v + C\frac{d^2v}{dt^2} = 0$$

Or:

$$\frac{d^2v}{dt^2} + \frac{1}{RC}\frac{dv}{dt} + \frac{1}{LC}v = 0 \qquad (2)$$

The characteristic equation of this second-order differential equation is:

$$s^2 + \frac{1}{RC}s + \frac{1}{LC} = 0 \qquad (3)$$

The roots of this characteristic equation are:

$$s_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = -\frac{1}{2RC} \pm \frac{\sqrt{\left(\frac{1}{RC}\right)^2 - \frac{4}{LC}}}{2} = -\frac{1}{2RC} \pm \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}}$$

Or:

$$s_1 = -\frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}} \qquad (4)$$

$$s_2 = -\frac{1}{2RC} - \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}} \qquad (5)$$

Assuming:

$$\alpha = \frac{1}{2RC} \quad \text{and} \quad \omega = \frac{1}{\sqrt{LC}}$$

The two roots can be expressed as:

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} \qquad (6)$$

$$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} \qquad (7)$$

These two roots are usually called the natural frequencies of the circuit and are measured in nepers per second (Np/s), because they are associated with the natural response of the circuit.

$\alpha = \dfrac{1}{2RC}$ is called the damping factor (or the neper frequency) and is expressed in nepers per second.

$\omega_0 = \dfrac{1}{\sqrt{LC}}$ is known as the resonant frequency (or the undamped natural frequency) and is expressed in radians per second (rad/s).

The solution of the differential equation is:

$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$

The constants A_1 and A_2 are determined from the initial values $v(0)$ and $\dfrac{dv(0)}{dt}$.

Note that, as stated before, $v(0) = V_0$; and $\dfrac{dv(0)}{dt}$ can be found from equation (1), re-written below.

$$\frac{v}{R} + \frac{1}{L}\int_{-\infty}^{t} v.dt + C\frac{dv}{dt} = 0$$

From this equation, re-written at $t = 0$:

$$\frac{V_0}{R} + I_0 + C\frac{dv(0)}{dt} = 0$$

Or:

$$\frac{dv(0)}{dt} = -\frac{1}{C}\left(\frac{V_0}{R} + I_0\right)$$

The roots of the characteristic equation described by (6) and (7), implies that there are three types of solutions of the differential equation, these are:

a) overdamped solution, for which $\alpha > \omega_0$, and the two roots are real.

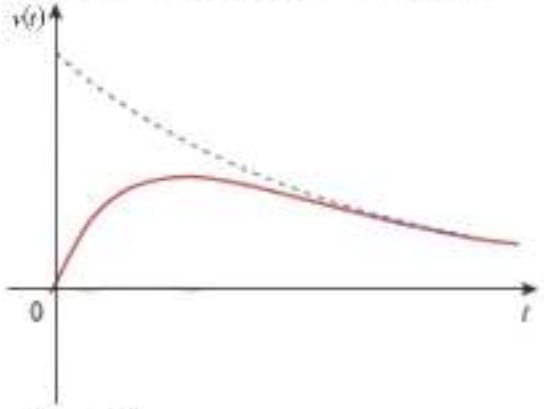
b) critically damped solution, for which $\alpha = \omega_0$, and the two roots are real and equal.

a) underdamped solution, for which $\alpha < \omega_0$, and the two roots are complex.

These three cases are discussed below.

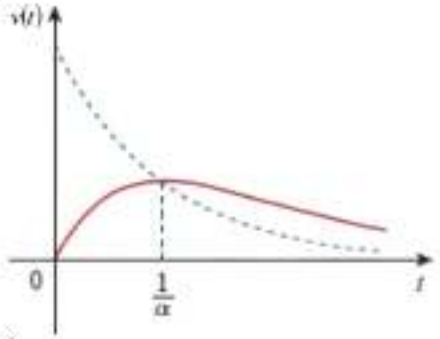
Overdamped solution, $(\alpha > \omega_0)$

For this case $\alpha > \omega_0$, or $\frac{1}{2RC} > \frac{1}{\sqrt{LC}}$, or $\frac{1}{2R\sqrt{C}} > \frac{1}{\sqrt{L}}$, or $\sqrt{L} > 2R\sqrt{C}$ or $L > 4R^2C$. For this case the two roots are real and negative and the response is $v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$ which decays and approaches zero as t increases, as shown below.



Critically damped solution, $(\alpha = \omega_0)$

or $\frac{1}{2RC} = \frac{1}{\sqrt{LC}}$, or $\frac{1}{2R\sqrt{C}} = \frac{1}{\sqrt{L}}$, or $\sqrt{L} = 2R\sqrt{C}$, or $L = 4R^2C$. For this case the two roots are real and equal, or $s_1 = s_2 = -\alpha = -\frac{1}{2RC}$ and the response is $v(t) = (A_1 + A_2 t)e^{-\alpha t}$ and is shown in the figure below.



Underdamped solution, $(\alpha < \omega_0)$

For this case $\alpha < \omega_0$, or $\frac{1}{2RC} < \frac{1}{\sqrt{LC}}$, or $\frac{1}{2R\sqrt{C}} < \frac{1}{\sqrt{L}}$, or $\sqrt{L} < 2R\sqrt{C}$ or $L < 4R^2C$. For this case the two roots are complex, and:

$s_{1,2} = -\alpha \pm \sqrt{-\left(\omega_0^2 - \alpha^2\right)} = -\alpha \pm j\omega_d$

Where:

$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$ and is called the damping frequency.

The solution is:

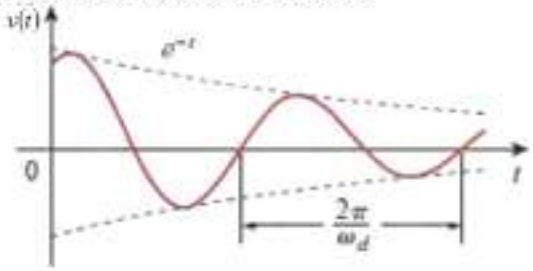
$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$
$$= A_1 e^{(-\alpha + j\omega_d)t} + A_2 e^{(-\alpha - j\omega_d)t} = A_1 e^{-\alpha t} e^{j\omega_d t} - A_2 e^{-\alpha t} e^{-j\omega_d t}$$
$$= e^{-\alpha t}\left(A_1 e^{j\omega_d t} + A_2 e^{-j\omega_d t}\right) = e^{-\alpha t}\left[A_1(\cos\omega_d t + j\sin\omega_d t) + A_2(\cos\omega_d t - j\sin\omega_d t)\right]$$
$$= e^{-\alpha t}\left[(A_1 + A_2)\cos\omega_d t + j(A_1 - A_2)\sin\omega_d t\right]$$
$$= e^{-\alpha t}\left[B_1\cos\omega_d t + B_2\sin\omega_d t\right]$$

Where:

$B_1 = A_1 + A_2$ and $B_2 = j(A_1 - A_2)$

it is clear that the natural response for this case is exponentially damped and oscillatory in nature. The response has a time constant of $\frac{1}{\alpha}$ and a period of $T = \frac{2\pi}{\omega_d}$.

The figure below depicts a typical underdamped response.



Important Notes:

1) Damping which is continuous decrease in the amplitude of the response is caused by the gradual loss of the initial stored energy. It is caused by the presence of resistance R. The damping factor α determines the rate at which the response is damped. If $R = 0$ then $\alpha = 0$ and we have an LC circuit with $\frac{1}{\sqrt{LC}}$ as the undamped natural frequency. Since $\alpha < \omega_0$ in this case, the response is not only undamped but also oscillatory. The circuit is said to be loss-less, because the dissipating or damping element (R) is absent. By adjusting the value of R, the response may be made undamped, overdamped, critically damped, or underdamped.

2) Oscillatory response is possible due to the presence of the two types of storage elements. Having both L and C allows the flow of energy back and forth between the two.

3) In general, it is difficult to tell from the waveforms the difference between the overdamped and critically damped responses. The critically damped case is the borderline between the underdamped and overdamped cases and it decays the fastest. With the same initial conditions, the overdamped case has the longest settling time, because it takes the longest time to dissipate the initial stored energy. If we desire the response that approaches the final value most rapidly without oscillation or ringing, the critically damped circuit is the right choice.

Example 1:

For the parallel RLC circuit, If $R = 2\Omega$, $L = 0.4H$, $C = 25mF$, $v(0) = 0$, $i(0) = 10mA$. find $v(t)$ $v(0) = 0$ for $t > 0$.

Solution:

$$\alpha = \frac{1}{2RC} = \frac{1}{2(2)(25x10^{-3})} = 10s^{-1}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{0.4(25x10^{-3})}} = 10rad/s$$

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -10 \pm \sqrt{10^2 - 10^2} = -10$$

The natural response is critically damped.

$$v(t) = (A_1 + A_2 t)e^{-\alpha t} = (A_1 + A_2 t)e^{-10t}$$

From $v(0) = 0$ we find $A_1 = 0$ or $v(t) = A_2 te^{-10t}$

From this:

$$\frac{dv(t)}{dt} = A_2(-10te^{-10t} + e^{-10t}) \text{ and } \frac{dv(0)}{dt} = A_2$$

But:

$$\frac{dv(0)}{dt} = -\frac{1}{C}\left(\frac{V_0}{R} + I_0\right) = -\frac{1}{25x10^{-3}}\left(\frac{0}{2} + 10x10^{-3}\right) = 0.4$$

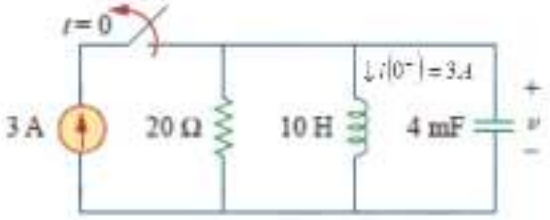
Thus:

$A_2 = 0.4$ and:

$v(t) = 0.4te^{-10t}$ V

Example 2:

For the circuit in the following figure, find $v(t)$ for $t > 0$.



Solution:

For $t < 0$, the switch was closed and the inductor acts as a short-circuit, thus $v(0^-) = 0$ and $i(0^-) = 3A$.

For $t > 0$, the switch is opened, thus: $v(0^+) = v(0^-) = 0$ and $i(0^+) = i(0^-) = 3A$

$$\alpha = \frac{1}{2RC} = \frac{1}{2(20)(4x10^{-3})} = 6.25$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10x4.x10^{-3}}} = 5$$

$\alpha > \omega_0$, thus the process is overdamped and:

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} = -6.25 + \sqrt{(6.25)^2 - 5^2} = -2.5$$

$$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} = -6.25 - \sqrt{(6.25)^2 - 5^2} = -10$$

$$v(t) = A_1 e^{s_1 t} - A_2 e^{s_2 t} = A_1 e^{-2.5t} + A_2 e^{-10t} \qquad (1)$$

B_1 and B_2 are found from the initial conditions:

$v(0^+) = v(0^-) = 0V$

Or:

$$A_1 + A_2 = 0 \qquad (2)$$

From (1):

$$\frac{dv(t)}{dt} = -2.5A_1 e^{-2.5t} - 10A_2 e^{-10t}$$

And:

$$\frac{dv(0)}{dt} = -2.5A_1 - 10A_2 \qquad (3)$$

$$\frac{dv(0)}{dt} = -\frac{1}{C}\left(\frac{V_0}{R} + I_0\right) = -\frac{1}{4x10^{-3}}\left(\frac{0}{20} + 3\right) = -750 \qquad (4)$$

Thus:

$$2.5A_1 + 10A_2 = 750 \qquad (5)$$

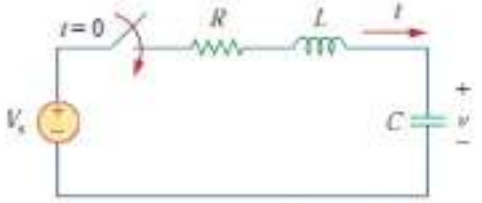
From (2) and (5):

$A_1 = -100$ and $A_2 = 100$

$$v(t) = -100e^{-2.5t} + 100e^{-10t} = 100\left(e^{-10t} - e^{-2.5t}\right)V$$

Step Response of a Series RLC Circuit

The step response is obtained by the sudden application of a dc source. Consider the series *RLC* circuit shown in the figure below.



Applying KVL around the loop for $t > 0$:

$$Ri - L\frac{di}{dt} + v = V_s \qquad (1)$$

But:

$$i = C\frac{dv}{dt} \qquad (2)$$

Substituting (2) in (1), we have:

$$RC\frac{dv}{dt} + LC\frac{d^2v}{dt^2} + v = V_s$$

Or:

$$\frac{d^2v}{dt^2} + \frac{R}{L}\frac{dv}{dt} + \frac{1}{LC}v = \frac{V_s}{LC}$$

The solution to this equation has two components: the transient response $v_t(t)$ and the steady-state response $v_{ss}(t)$ that is,

$$v(t) = v_t(t) + v_{ss}(t)$$

The form of the transient response is the same as the form of the solution obtained for the source-free circuit. Therefore, the transient response for the overdamped, underdamped, and critically damped cases are:

- $v_t(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$ for overdamped case
- $v_t(t) = (A_1 - A_2 t)e^{-\alpha t}$ for critically damped case
- $v_t(t) = (A_1 \cos \omega_d t - A_2 \sin \omega_d t)e^{-\alpha t}$ for underdamped case

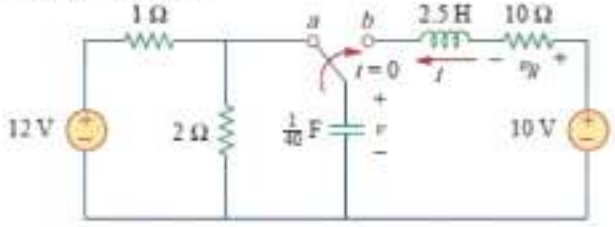
The steady-state response is the final value of $v(t)$, which is the source voltage V_s. Hence

- $v(t) = V_s + A_1 e^{s_1 t} + A_2 e^{s_2 t}$ for overdamped case
- $v(t) = V_s + (A_1 + A_2 t)e^{-\alpha t}$ for critically damped case
- $v(t) = V_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t)e^{-\alpha t}$ for underdamped case

The values of the constants A_1 and A_2 are obtained from the initial conditions: $v(0)$ and $\frac{dv(0)}{dt}$.

Example:

Having been in position a for a long time, the switch in the following figure is moved to position b at $t = 0$. Find $v(t)$ and $v_R(t)$ for $t > 0$.



Solution:

For $t < 0$, the switch is in position a, thus:

$$v(0^-) = (12)\frac{2}{1+2} = 8V, \quad i(0^-) = 0A$$

For $t > 0$, the switch is in position b, thus:

$$v(0^+) = v(0^-) = 8V, \quad i(0^+) = i(0^-) = 0A$$

$$v(\infty) = 10V, \; i(\infty) = 0A$$

$$\alpha = \frac{R}{2L} = \frac{10}{2x2.5} = 2s^{-1}, \qquad \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{2.5x\frac{1}{40}}} = 4s^{-1}$$

$\alpha < \omega_0$, the case is underdamped, thus:

$$v(t) = v(\infty) + (A_1 \cos\omega_d t + A_2 \sin\omega_d t)e^{-\alpha t},$$

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2} = \sqrt{12} = 3.464$$

And:

$$v(t) = 10 + (A_1 \cos 3.464t + A_2 \sin 3.464t)e^{-2t} \qquad (1)$$

$v(0^+) = v(0^-) = 8V$, thus:

$8 = 10 + A_1$, or

$$A_1 = -2 \qquad (2)$$

or

$$i(0^+) = C\frac{dv(0^+)}{dt} = 0, \text{ thus } \frac{dv(0^+)}{dt} = 0$$

From (1):

$$\frac{dv(t)}{dt} = -2(A_1 \cos 3.464t - A_2 \sin 3.464t)e^{-2t} + e^{-2t}(-3.464A_1 \sin 3.464t + 3.464A_2 \cos 3.464t)$$

$$\frac{dv(0)}{dt} = -2(A_1) + (3.464A_2)$$

Or:

$$-2A_1 + 3.464A_2 = 0 \qquad (3)$$

Substituting (2) in (3),

$3.464A_2 = -4$, or $A_2 = -1.1547$

Or:

$v(t) = 10 + (-2\cos 3.464t - 1.1547\sin 3.464t)e^{-2t}$

$v(t) = 10 - (2\cos 3.464t + 1.1547\sin 3.464t)e^{-2t}$

For the current:

$i(t) = i(\infty) + (B_1 \cos \omega_d t + B_2 \sin \omega_d t)e^{-\alpha t}$,

But $i(\infty) = 0A$, $\alpha = 2s^{-1}$ and $\omega_d = 3.464s^{-1}$, thus:

$i(t) = (B_1 \cos 3.464t + B_2 \sin 3.464t)e^{-2t}$

Also $i(0^-) = i(0^-) = 0A$, thus:

$B_1 = 0$ and:

$i(t) = B_2 \sin 3.464te^{-2t}$ (4)

When the switch is in position b,

$$10 - 10i(0^-) - L\frac{di(0^-)}{dt} - v(0^-) = 0$$

Or:

$$10 - 10(0) - L\frac{di(0^-)}{dt} - 8 = 0$$

$$\frac{di(0^+)}{dt} = \frac{2}{L} = \frac{2}{2.5} = 0.8A/s \quad (5)$$

From (4):

$$\frac{di(t)}{dt} = B_2\left(-2\sin 3.464te^{-2t} + 3.464\cos 3.464te^{-2t}\right)$$

$\frac{di(0^+)}{dt} = B_2(3.464) = 0.8$, or:

$$B_2 = \frac{0.8}{3.464} = 0.231$$

And:

$i(t) = 0.231\sin 3.464te^{-2t}$

And:

$v_o(t) = 10i = 2.31\sin 3.464te^{-2t}V$

Step Response of a Parallel RLC Circuit

The step response is obtained by the sudden application of a dc source. Consider the parallel *RLC* circuit shown in the figure below.



Applying KVL around the loop for $t > 0$:

$$\frac{v}{R} + i + C\frac{dv}{dt} = I_s \qquad (1)$$

But:

$$v = L\frac{di}{dt} \qquad (2)$$

Substituting (2) in (1), we have:

$$\frac{L}{R}\frac{di}{dt} + i + LC\frac{d^2i}{dt^2} = I_s$$

Or:

$$\frac{d^2i}{dt^2} + \frac{1}{RC}\frac{di}{dt} + \frac{1}{LC}i = \frac{I_s}{LC}$$

The solution to this equation has two components: the transient response $i_t(t)$ and the steady-state response $i_{ss}(t)$ that is,

$$i(t) = i_t(t) + i_{ss}(t)$$

The form of the transient response is the same as the form of the solution obtained for the source-free circuit.

The characteristic equation of this second-order differential equation is:

$$s^2 + \frac{1}{RC}s + \frac{1}{LC} = 0$$

The roots of this characteristic equation are:

$$s_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = -\frac{1}{2RC} \pm \frac{\sqrt{\left(\frac{1}{RC}\right)^2 - \frac{4}{LC}}}{2} = -\frac{1}{2RC} \pm \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}}$$

Or:

$$s_1 = -\frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}} \qquad (4)$$

$$s_2 = -\frac{1}{2RC} - \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}} \qquad (5)$$

Assuming:

$$\alpha = \frac{1}{2RC} \quad \text{and} \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

The two roots can be expressed as:

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} \qquad (6)$$

$$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} \qquad (7)$$

Therefore, the transient response for the overdamped, underdamped, and critically damped cases are:

- $i_t(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$ for overdamped case
- $i_t(t) = (A_1 + A_2 t)e^{-\alpha t}$ for critically damped case
- $i_t(t) = (A_1 \cos \omega_d t + A_2 \sin \omega_d t)e^{-\alpha t}$ for underdamped case

The steady-state response is the final value of $v(t)$, which is the source voltage V_s. Hence

- $i(t) = I_s + A_1 e^{s_1 t} + A_2 e^{s_2 t}$ for overdamped case
- $i(t) = I_s + (A_1 + A_2 t)e^{-\alpha t}$ for critically damped case
- $i(t) = I_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t)e^{-\alpha t}$ for underdamped case

The values of the constants A_1 and A_2 are obtained from the initial conditions: $i(0)$ and $\frac{di(0)}{dt}$.

Example:

Find $i(t)$ and $v(t)$ in following circuit.



Solution:

Method (1):

For $t < 0$, $u(t) = 0$, and: $v(0^-) = 0$ and $i(0^-) = 0$.

Thus:

$v(0^+) = v(0^-) = 0$ and $i(0^+) = i(0^-) = 0$.

For $t > 0$, $u(t) = 1$, and:

$$C\frac{dv}{dt} + i = 12 \qquad (1)$$

But:

$$v = L\frac{di}{dt} \qquad (2)$$

Substituting (2) in (1):

$$LC\frac{d^2i}{dt^2} + i = 12$$

Or:

$$\frac{d^2i}{dt^2} + i = 12$$

$$i(t) = i_t(t) + i_{ss}(t)$$

The component $i_t(t)$ is the solution of the homogeneous equation, whose characteristic equation is:

$s^2 + 1 = 0$, from which $s_{1,2} = \pm i$, thus:

$$i_t(t) = A_1e^{it} + A_2e^{-it} = A_1(\cos t + i\sin t) + A_2(\cos t - i\sin t) = (A_1 + A_2)\cos t + i(A_1 - A_2)\sin t$$

$$i_t(t) = B_1\cos t + B_2\sin t \qquad (3)$$

$$i_{ss} = 12A$$

And:

$$i(t) = 12 + B_1\cos t + B_2\sin t$$

But $i(0^+) = 0$, thus $0 = 12 + B_1$, or $B_1 = -12$:

$$i(t) = 12 - 12\cos t + B_2\sin t \qquad (4)$$

Also:

$$v(0^+) = 0 = L\frac{di(0^-)}{dt}, \text{ or } \frac{di(0^-)}{dt} = 0$$

From (4):

$$\frac{di}{dt} = 12\sin t + B_2\cos t$$

Thus:

$$B_2 = 0$$

Or:

$$i(t) = 12 - 12\cos t = 12(1 - \cos t)$$

And:

$$v(t) = L\frac{di}{dt} = 5(12)\sin t = 60\sin t$$

Method (2):

This circuit can be treated as a parallel RLC with $R \to \infty$, thus:

$$\alpha = \frac{1}{2RC} = 0,\ \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{5(0.2)}} = 1rad/s$$

As $\alpha < \omega_0$, the process is underdamped and:

$$i(t) = I_S + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t}$$

$$i(t) = 12 + (A_1 \cos \omega_d t + A_2 \sin \omega_d t)$$

But $i(0^+) = 0$, thus $0 = 12 + A_1$, or $A_1 = -12$:

$$i(t) = 12 - 12\cos t + B_2 \sin t \qquad (5)$$

Also:

$$v(0^+) = 0 = L\frac{di(0^+)}{dt}, \text{ or } \frac{di(0^+)}{dt} = 0$$

From (5):

$$\frac{di}{dt} = 12\sin t + B_2 \cos t$$

Thus:

$$B_2 = 0$$

Or:

$$i(t) = 12 - 12\cos t = 12(1 - \cos t)$$

And:

$$v(t) = L\frac{di}{dt} = 5(12)\sin t = 60\sin t$$

General Second-Order Circuits

Given a second-order circuit, we determine its step response $x(t)$ (which may be voltage or current) by taking the following four steps:

1) We first determine the initial conditions $x(0)$ and $\frac{dx(0)}{dt}$ and the final value $x(\infty)$.

2) We turn off the independent sources and find the form of the transient response $x_t(t)$ by applying KCL and KVL. Once a second-order differential equation is obtained, we determine its characteristic roots. Depending on whether the response is overdamped, critically damped, or underdamped, we obtain $x_t(t)$ with two unknown constants.

3) We obtain the steady-state response as $x_{ss}(t) = x(\infty)$.

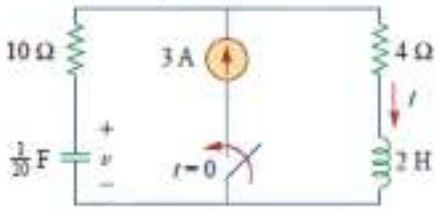
4) The total response is now found as the sum of the transient response and steady-state response

$$x(t) = x_t(t) + x_{ss}(t)$$

We finally determine the constants associated with the transient response by imposing the initial conditions $x(0)$ and $\frac{dx(0)}{dt}$ determined in step 1.

Example 1:

Determine v and i for in the circuit of the figure below.

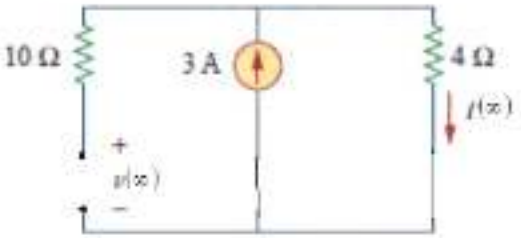


Solution:

For $t < 0$, the switch is open, thus:

$v(0^-) = 0V$ and $i(0^-) = 0A$

For $t \to \infty$, the switch is closed, the equivalent circuit is as below.

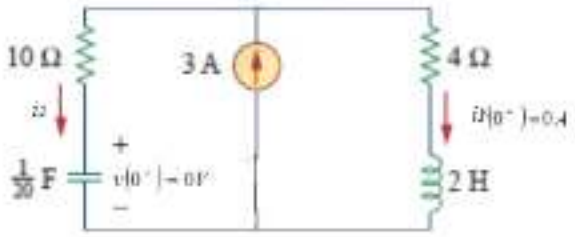


From this circuit:

$i(\infty) = 3A$, $v(\infty) = 3x4 = 12V$

For $t > 0$, the switch is closed, and:

$v(0^+) = v(0^-) = 0V$ and $i(0^-) = i(0^-) = 0A$, the equivalent circuit is:



From this circuit:

$i_1(0^+)+i_2(0^+)=3$

Or:

$0+i_2(0^-)=3$

And:

$i_2(0^+)=3$

Or:

$C\dfrac{dv(0^+)}{dt}=3$ and $\dfrac{dv(0^+)}{dt}=\dfrac{3}{C}=60V/s$

From the circuit:

$i_1+i_2=3$ (1)

Also,

$$4i_1+L\frac{di_1}{dt}=10i_2+\frac{1}{C}\int_0^t i_2\,dt$$

Or:

$$4i_1+2\frac{di_1}{dt}=10i_2+20\int_0^t i_2\,dt \qquad (2)$$

Deriving this equation:

$$4\frac{di_1}{dt}+2\frac{d^2i_1}{dt^2}=10\frac{di_2}{dt}+20i_2 \qquad (3)$$

From (1):

$i_2=3-i_1$ and: $\dfrac{di_2}{dt}=-\dfrac{di_1}{dt}$ (4)

Substituting (4) in (3):

$$4\frac{di_1}{dt}+2\frac{d^2i_1}{dt^2}=-10\frac{di_1}{dt}+20(3-i_1)$$

Or:

$$2\frac{d^2i_1}{dt^2}+14\frac{di_1}{dt}+20i_1=60$$

$$\frac{d^2i_1}{dt^2}+7\frac{di_1}{dt}+10i_1=30$$

The characteristic equation is:

$s^2+7s+10=0$

$$s_{1,2} = \frac{-7 \pm \sqrt{49 - 4x10}}{2} = \frac{-7 \pm 3}{2} = -5, -2$$

$$i_0(t) = A_1 e^{-5t} + A_2 e^{-2t}$$

$$i_{1\infty} = i_1(\infty) = 3$$

Thus:

$$i(t) = i_1(t) = i_0(t) + i_\infty(t) = 3 + A_1 e^{-5t} + A_2 e^{-2t} \quad (5)$$

But:

$$i(0^+) = i(0^-) = 0A$$

Or:

$$3 + A_1 + A_2 = 0$$

Or:

$$A_1 + A_2 = -3 \qquad (6)$$

Equation (2), for $t = 0^+$ can be written as:

$$4i_1(0^+) + 2\frac{di_1(0^+)}{dt} = 10i_2(0^+) - v(0^+)$$

Or:

$$4(0) + 2\frac{di_1(0^+)}{dt} = 10(3) + 0$$

Or:

$$\frac{di(0^-)}{dt} = \frac{di_1(0^+)}{dt} = 15$$

From (5)

$$\frac{di(0^-)}{dt} = -5A_1 - 2A_2$$

Or:

$$-5A_1 - 2A_2 = 15 \qquad (7)$$

From (6) and (7), we find:

$A_1 = -3$ and $A_2 = 0$

Thus:

$$i(t) = 3 - 3e^{-5t} = 3(1 - e^{-5t})A \qquad (8)$$

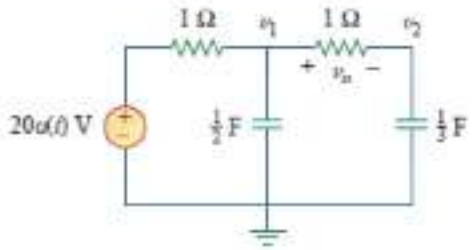
From (1) and (8):

$$i_2(t) = 3 - i(t) = 3e^{-5t}$$

$$v(t) = \frac{1}{C}\int_0^t i_c\,dt = 20\int_0^t 3e^{-5t}\,dt = -12\int_0^t e^{-5t}\,d(-5t) = -12e^{-5t}\Big|_0^t = -12(e^{-5t} - 1) = 12(1 - e^{-5t})V$$

Example 2:

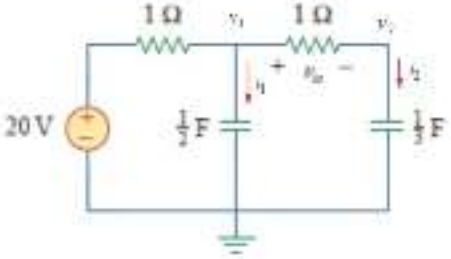
For $t > 0$ obtain $v_0(t)$ in the following circuit.



Solution

For $t < 0$, $u(t) = 0$, thus: $v_1(0^-) = 0$ and $v_2(0^-) = 0$

For $t > 0$, $u(t) = 1$, the equivalent circuit is as below.



For $t = 0^-$, $v_1(0^-) = v_1(0^-) = 0$ and $v_2(0^-) = v_2(0^-) = 0$

$$i_1(0^+) = C_1 \frac{dv_1(0^-)}{dt} = \frac{20}{1} = 20A$$

$$\frac{dv_1(0^-)}{dt} = \frac{20}{C_1} = 40V/s$$

$$i_2(0^+) = C_2 \frac{dv_2(0^+)}{dt} = 0$$

$$\frac{dv_2(0^-)}{dt} = \frac{0}{C_2} = 0V/s$$

Using nodal analysis:

$$\frac{v_1 - 20}{1} + C_1 \frac{dv_1}{dt} - C_2 \frac{dv_2}{dt} = 0 \qquad (1)$$

$$\frac{v_1 - v_2}{1} = C_2 \frac{dv_2}{dt} \qquad (2)$$

From (2):

$$v_1 - v_2 = C_2 \frac{dv_2}{dt} \qquad (3)$$

Substituting (3) in (1),

$$v_2 + C_2 \frac{dv_2}{dt} - 20 + C_1 \frac{d}{dt}\left(v_2 + C_2 \frac{dv_2}{dt}\right) + C_2 \frac{dv_2}{dt} = 0$$

Or:

$$v_2 + (2C_2 + C_1)\frac{dv_2}{dt} - 20 + C_1 C_2 \frac{d^2 v_2}{dt^2} = 0$$

$$C_1C_2\frac{d^2v_2}{dt^2}+(2C_2+C_1)\frac{dv_2}{dt}+v_2=20$$

$$\frac{1}{6}\frac{d^2v_2}{dt^2}+\frac{7}{6}\frac{dv_2}{dt}+v_2=20$$

$$\frac{d^2v_2}{dt^2}+7\frac{dv_2}{dt}+6v_2=120$$

The characteristic equation is:

$$s^2+7s+6=0$$

$$(s+6)(s+1)=0$$

And:

$$s_1=-6,\ s_2=-1$$

$$v_{2n}(t)=A_1e^{-t}+A_2e^{-6t}$$

$$v_{2p}(t)=v_2(\infty)=20V$$

Thus:

$$v_2(t)=v_{2p}(t)+v_{2n}(t)=20+A_1e^{-t}+A_2e^{-6t}$$

$$v_2(0^+)=0=20+A_1+A_2$$

Or:

$$A_1+A_2=-20 \qquad (4)$$

Also:

$$\frac{dv_2(0^+)}{dt}=0=-A_1-6A_2$$

$$A_1+6A_2=0 \qquad (5)$$

From (4) and (5):

$$A_1=-24,\ A_2=4$$

Or:

$$v_2(t)=20-24e^{-t}+4e^{-6t}$$

From (3):

$$v_1=v_2+C_2\frac{dv_2}{dt}=20-24e^{-t}+4e^{-6t}+\frac{1}{3}\left(24e^{-t}-24e^{-6t}\right)$$

$$=20-16e^{-t}-4e^{-6t}$$

And:

$$v_0(t)=v_1-v_2=8e^{-t}-8e^{-6t}=8\left(e^{-t}-e^{-6t}\right)V$$